DIPLOMARBEIT

In-Situ Measurement of the Resistivity during Annealing of Neutron Irradiated High-Temperature Superconductors

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Dedication

Dedicated to my uncle, Dipl.-Ing. Davor Špoljarić, a devoted chemist and fighter, whose courage during difficult times inspires this effort. May this work honor his strength and perseverance in the face of severe illness.

Kurzfassung

Die Kernfusion wird seit Jahrzehnten als potenzielle Energiequelle der Zukunft erforscht. Für die in Fusionsreaktoren erforderlichen hohen Magnetfelder bieten Hochtemperatur-Supraleiter (HTS) gegenüber Tieftemperatur-Supraleitern Vorteile, da sie kompaktere Magnetdesigns ermöglichen. Eine zentrale Herausforderung ist die Degradation von HTS unter Neutronenbestrahlung. Frühere Studien von Unterrainer, Fischer, Lorenz, *et al.* haben gezeigt, dass die Degradation von neutronenbestrahltem Seltenerd-Barium-Kupferoxid (REBCO) HTS durch thermische Behandlungen teilweise rückgängig gemacht werden kann. Dabei wurde eine lineare Wiederherstellung von T_c sowie eine nicht-monotone Erholung von j_c mit zunehmender Annealingtemperatur beobachtet [1]. Jedoch bleiben die zugrunde liegenden Defektwiederherstellungsprozesse und charakteristischen Temperaturen unklar. Es wird erwartet, dass dieselben Defekte, die die Suprafluiddichte (ρ_s) beeinflussen, auch zum spezifischen Widerstand im Normalzustand (ρ_n) beitragen, wie durch die Matthiessen-Regel und das Homes-Gesetz beschrieben.

Ein chemisches Ätzverfahren sowie zwei Generationen von maßgeschneiderten Probenhaltern mit integrierter Messtechnik wurden für die in-situ Wärmebehandlung von REBCO-Supraleitern entwickelt, getestet und erfolgreich eingesetzt. Die Wärmebehandlungen wurden in reiner O_2-Umgebung bei Temperaturen zwischen 50 °C und 450 °C durchgeführt. Unbestrahlte Bruker (YBCO)-Proben zeigten weitgehend das erwartete Widerstandsverhalten, während SuperPower (GdBCO)-Proben zunächst anomale Messkurven aufgrund eines Kurzschlusses zwischen der Silberschicht und dem Substrat aufwiesen, welcher durch eine Anpassung des Ätzmusters behoben werden konnte.

Der thermische und schnelle Neutronenfluss wurde im zentralen Bestrahlungsrohr (ZBR) und in einer Bestrahlungsposition außerhalb des Reflektors bei einer Reaktorleistung von 250 kW mittels Neutronenaktivierungsanalyse und γ-Spektroskopie bestimmt. Die berechneten Flusswerte wurden bezüglich der Bestrahlungszeit korrigiert. Für den thermischen Fluss wurde der Neutroneneinfangquerschnitt für die Neutronentemperatur angepasst und eine Positionskorrektur durchgeführt. Dies ergab thermische Neutronenflüsse von $\Phi_{th} = 9{,}51 \times 10^{16}\ \mathrm{m^{-2}s^{-1}}$ (ZBR) und $3{,}73 \times 10^{15}\ \mathrm{m^{-2}s^{-1}}$ (Position außerhalb des Reflektors). Die schnellen Flusswerte wurden mithilfe der Methode der effektiven Schwellenenergie berechnet, wobei

erhebliche Unsicherheiten durch die Wahl der Neutronenabsorptionswirkungsquerschnitte und Schwellenenergien bestehen.

Diese Arbeit stellt eine validierte Messplattform bereit und liefert bestmögliche Flussdaten, wodurch die Grundlage für zukünftige quantitative Untersuchungen zur Defektwiederherstellung in bestrahlten REBCO-Supraleitern geschaffen wird.

Abstract

Nuclear fusion has been researched for decades as a potential energy source of the future. For the high magnetic fields required in fusion reactors, high-temperature superconductors (HTS) offer advantages over low-temperature superconductors, enabling more compact magnet designs. A key challenge is the degradation of HTS under neutron irradiation. Previous studies conducted by Unterrainer, Fischer, Lorenz, *et al.* demonstrated that the degradation of neutron-irradiated rare-earth barium copper oxide (REBCO) HTS can be counteracted by thermal treatments. These HTS showed partial linear recovery of T_c and non-monotonic recovery of j_c with increasing annealing temperature [1], but the underlying defect recovery processes and characteristic temperature windows remain unclear. In this work, in-situ resistance measurements during annealing are used to identify these recovery temperatures, based on the expectation that the same defects affecting superfluid density (ρ_s) also contribute to normal-state resistivity (ρ_n) as described by Matthiessen's rule and Homes' law.

A chemical etching process and two generations of custom sample holders with integrated instrumentation were developed, validated and successfully deployed for in-situ thermal treatment of REBCO tapes. The annealing runs were performed in pure O_2 atmosphere, using temperature profiles between 50 °C and 450 °C. Pristine Bruker (YBCO) samples largely showed the expected resistance behavior, while SuperPower (GdBCO) samples initially exhibited anomalous curves due to a short circuit between the silver layer and the substrate, which was resolved by adapting the etching pattern.

Thermal and fast neutron fluxes were determined at the central irradiation position (CIP) and irradiation tube located outside the graphite reflector at a reactor power of 250 kW using neutron activation analysis and γ-spectroscopy. The computed flux values were corrected for irradiation time. For the thermal flux, the neutron absorption cross-section was corrected for the neutron temperature, and a position correction was applied. This yielded thermal neutron fluxes of $\Phi_{th} = 9.51 \times 10^{16}\,\mathrm{m^{-2}s^{-1}}$ at the CIP and $3.73 \times 10^{15}\,\mathrm{m^{-2}s^{-1}}$ at the irradiation tube outside the graphite reflector. Fast flux was estimated using effective threshold energies, with substantial uncertainties due to neutron absorption cross-section and threshold energy selection.

This thesis establishes a validated measurement platform and provides best-effort flux data, laying the foundation for future quantitative studies of defect recovery in irradiated REBCO conductors.

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List of Abbreviations

2G	second-generation
BCS	Bardeen-Cooper-Schrieffer
CIP	central irradiation position
CPS	counts per second
EMF	electromotive force
GdBCO	gadolinium barium copper oxide
GL	Ginzburg-Landau
HPGe	high purity germanium
HTS	high-temperature superconductor
LTS	low-temperature superconductor
MR	moderating ratio
NAA	neutron activation analysis
PCB	printed circuit board
REBCO	rare-earth barium copper oxide
SMU	source measure unit
SP	SuperPower
TRIGA	Training, Research, Isotopes, General Atomic
VTI	variable temperature insert
YBCO	yttrium barium copper oxide

1. Introduction

There is a clear need for clean, low-carbon, and abundant energy sources to meet the rising energy demand of our world. Renewable energy sources, primarily solar and wind, are becoming increasingly present in our energy mix, but remain too volatile without a suitable storage solution to reliably power modern daily life. Nuclear power by fission is one of the most important low-carbon energy sources globally. In the European Union, nuclear power has remained the largest low-carbon energy source in recent years, accounting for approximately 25 % of total annual electricity production. In search of an alternative energy source with similar energy density and production capability, nuclear fusion is the main contender. Fusion technology has been researched for decades and has yielded numerous significant achievements. Nevertheless, the technological challenges associated with harnessing fusion energy remain enormous and require ongoing, substantial research and development. The goal of this thesis is to make a relatively small, yet significant, contribution to the understanding of defect annealing in high-temperature superconductor (HTS), which are required for achieving high magnetic fields in compact fusion devices.

1.1. Aim of this Thesis

This master thesis extends previous research on the recoverability of the superconducting properties of HTS, specifically gadolinium barium copper oxide (GdBCO), after neutron irradiation by thermal treatments. It has been shown that the critical temperature decreases linearly with neutron fluence and increases linearly with annealing temperature, while the critical current density recovers non-monotonically [1]. In order to improve our understanding of the underlying physics, the evolution of the resistance under high-temperature thermal annealing in a pure oxygen environment will be measured. For this purpose, a new sample holder will be designed and manufactured. It is expected that the collected data on the change in resistance at different temperatures will provide information about which physical processes lead to the recovery of the critical temperature and the closely related critical current density. For this thesis, the qualitative form of the curves is of greater importance than the exact values of the resistivity. The critical temperature will be measured before and after neutron irradiation in a

17 T cryostat. The neutron irradiation will be performed at the central irradiation position (CIP) of the Training, Research, Isotopes, General Atomic (TRIGA) Mark II reactor at a thermal power of 250 kW.

Additionally, the fast and thermal neutron flux will be determined both at the CIP and the central irradiation tube located outside the graphite reflector, termed Weber tube, of the TRIGA Mark II reactor at 250 kW by neutron activation measurements. This is done to accurately determine the defect density in the irradiated superconductors, which is proportional to the neutron flux. This is especially relevant for GdBCO conductors, as the degradation of the critical temperature is 14 to 15 times stronger compared to thermally shielded samples due to neutron-gamma capture reactions of gadolinium. As the neutron field of the TRIGA Mark II reactor differs from that of a fusion reactor, an accurate description of the neutron flux is essential to evaluate the significance of our observations and assess their limitations.

As pointed out in the doctoral thesis "Universal Degradation of Superconducting Properties of REBCO Tapes in Radiation Environments" [2] by Unterrainer, past and present publications on neutron flux values deviate by as much as a factor of two. The aim of the neutron flux measurement part of this thesis is to formulate a reliable flux measurement workflow that can be easily reproduced. In the course of this, possible causes for the deviation in different flux measurements will be analyzed and discussed.

1.2. Methodology

The methodology of this diploma thesis consists of both a theoretical and an experimental part. For the theoretical section, a comprehensive literature review was conducted, focusing particularly on neutron flux measurements. As similar work has been carried out previously, the method of neutron activation analysis is well covered in the literature. The selection of activation materials and target isotopes was largely based on experience from previous studies. Additionally, a radiation protection concept was developed, as substantial activation was expected upon irradiation of the samples at full thermal reactor power (250 kW). To support this, a Python script was created, which will be described in detail later. The formula for thermal and fast neutron flux calculation was derived by the author using existing literature, and tailored to the measurement principle of γ-spectroscopy and the specific circumstances of the irradiation experiment.

For the resistivity measurement during annealing, theoretical analysis addressed several sub-projects: the photolithography process for van der Pauw structures, chemical etching of different structures, and the design and instrumentation for resistivity measurement, all of which required analysis of various publications and manuals. During the design phase of the first-generation sample holder, commercial products for similar applications and relevant publications were examined to find a suitable approach for contacting the samples. The second-generation contacting arrangement was developed iteratively to fit the adjusted design goals, with input from both the research group and the workshop. Additionally, extensive literature research was conducted to investigate the unexpected behavior of SuperPower HTS.

The experimental work was divided into two main areas: neutron flux measurement and resistivity measurement during annealing. For neutron flux measurement, samples were selected, prepared, and encapsulated in quartz-glass capsules. Irradiation was performed at two different positions for varying durations. Sample insertion was carried out by reactor personnel, while the author monitored the process and measured the irradiation time.

After sufficient decay (cooling) time, samples were unpacked and measured. This process was first practiced without radioactive material to minimize exposure and adhere to the radiation protection principle "as low as reasonably achievable", aiming for the lowest practical radiation dose. Neutron flux was initially calculated using Excel, and later, Python scripts were developed to correct for the factors of irradiation time and position, yielding the most accurate flux values. After further decay, the residual radioactivity of irradiated samples was measured. If deemed safe in consultation with radiation protection officers, samples were returned to Mario Villa, the reactor operations manager.

The experimental section for resistivity measurement during annealing involved the preparation of the superconductor samples. Contacting the probes required both chemical etching of the samples and the design of a custom sample holder and contact arrangement. An iterative approach was used for both chemical etching and sample holder design, resulting in two generations of sample holders and various contact arrangements. The different sample holder configurations were tested extensively with pristine superconductor samples.

The technical drawings produced in the course of this work (see chapter section A.1 and section B.2) were created using Autodesk Inventor 2024 and 2025.

The superconductor samples were also encapsulated in quartz-glass capsules and irradiated at the CIP in the TRIGA reactor. Their critical temperature was determined using a 17 T cryostat at several stages: before irradiation, after irradiation, and after annealing. After irradiation, samples were unpacked and their resistivity was measured during annealing in a tube furnace under a pure oxygen environment. Critical temperature and resistance data were subsequently analyzed and processed using custom Python scripts.

The Python script codes written for this thesis were developed in the integrated development environment Visual Studio Code. Visual Studio Code features, together with GitHub Copilot, the possibility to write code with the help of different generative artificial intelligence models, which can be selected. The Python scripts described in the following section were written over the course of several months with the help of the Anthropic model Claude Sonnet (versions 3.5–4.0). The author of this thesis reviewed the relevant code sections, edited them if needed, and takes full responsibility for the final version used for calculations. The scripts are available on the author's GitHub page[1].

- **Radioactive decay and γ-dose rate calculator**: Computes the remaining activity and γ-dose rate of selected radionuclides after an arbitrary, user-defined time interval.
- **Neutron flux calculator**: Calculates the neutron flux at different positions based on irradiation data (see section B.1) and performs uncertainty analysis using the uncertainties package. Additionally, both the irradiation time (see subsection 2.5.3 and subsection 3.1.5) and irradiation position (see subsection 3.1.6) were corrected and visualized with dedicated Python scripts.
- **Four-wire measurement automation and data logging**: Automates the annealing measurement process by controlling the Keithley source measure unit (SMU), acquiring voltage drop data, synchronizing it with PT100 temperature sensor readings and saving the combined data in a .dat file for subsequent analysis.
- **Annealing data analysis**: Interpolates and merges voltage, temperature and time data from the .dat file, applies data smoothing to remove outliers and visualizes voltage and resistance as functions of both temperature and time during thermal annealing experiments.
- **Critical temperature data analysis**: Analyzes measurements from the 17 T cryostat, calculates two characteristic critical temperature values and visualizes the superconducting phase transition.

For the citation of references, the IEEE style was adopted. As a reference guide, the IEEE Reference Guide [3] was used. For citations placed at the end of sentences, the convention described in [4, p. 17] was followed. If the citation is placed before the period (full stop), it refers solely to the content of that sentence. If the citation is positioned after the period, it refers to the entire preceding paragraph or all the content since the previous citation.

[1] https://github.com/danielhackl-tu/

2. Theoretical Foundations

2.1. Neutron and Neutron Temperature

The atomic nucleus generally consists of neutrons and protons. The discovery of the neutron in 1932 by Chadwick laid the foundation for further discoveries in nuclear physics. [5] In order to distinguish different energy groups of neutrons, a nomenclature convention was introduced, which is not precise and can deviate depending on the literature. In Table 2.1, the different energy groups and their corresponding names can be seen. If a neutron is in equilibrium with its surrounding medium, the neutron can be described by a Maxwell-Boltzmann-distribution with a mean kinetic energy of $k_B T$. [6] This mean kinetic energy can be calculated according to:

$$\langle E_{kin} \rangle = \frac{f}{2} \cdot k_B T \tag{2.1}$$

$$\begin{aligned} \text{where} \quad \langle E_{kin} \rangle &: \text{average (mean) kinetic energy (eV)} \\ f &: \text{degrees of freedom; } f = 3 \text{ for purely translational motion} \\ k_B &: \text{Boltzmann constant } (8.617333 \times 10^{-5}\ \text{eV/K}) \\ T &: \text{absolute temperature (K)} \end{aligned}$$

For neutrons in thermal equilibrium, which is defined at room temperature of $20\,^{\circ}\mathrm{C}$ ($T_0 = 293.15\,\mathrm{K}$), this value is:

$$\langle E_{kin} \rangle \approx 0.0379 \text{ eV} \tag{2.2}$$

In literature, the simplified term for calculating the nominal energy value at $T_0 = 293.15\,\mathrm{K}$ is generally used as a reference for thermal neutrons:

$$E = k_B T_0 \approx 0.0253 \text{ eV} \tag{2.3}$$

Energy Range	Name	Velocity (km/s)
0 – 0.025 eV	Cold	0 – 2.19
0.025 eV	Thermal	≈ 2.19
0.025 – 0.4 eV	Epithermal	2.19 – 8.75
0.4 – 0.6 eV	Cadmium	8.75 – 10.71
0.6 – 1.0 eV	Epicadmium	10.71 – 13.83
1 – 10 eV	Slow	13.83 – 43.74
10 – 300 eV	Resonance	43.74 – 239.56
300 eV – 1 MeV	Intermediate	239.56 – 13 831
1 – 20 MeV	Fast	13 831 – 61 853
> 20 MeV	Relativistic	> 61 853

Table 2.1.: Classification of neutron energy ranges (adapted from [6, p. 308]) and corresponding velocities. Velocities were calculated with the relation $v = \sqrt{\frac{2E}{m}}$ and are given in km/s.

2.2. Nuclear Fission

Nuclear fission was discovered in 1938 by the chemists Otto Hahn and Fritz Strassmann and the physicists Lise Meitner and Otto Robert Frisch. A neutron is absorbed by a ^{235}U atom (see Figure 2.1a). ^{236}U is formed as a intermediate nucleus. According to the liquid drop model, the nucleus starts to vibrate like a liquid which absorbed a drop. If enough energy is supplied with the absorption of the neutron, the nucleus splits into two lighter fission fragments. The required energy for this process depends on the isotope. For the isotopes ^{233}U, ^{235}U, ^{239}Pu and ^{241}Pu, the necessary kinetic neutron energy is 0 MeV, leading to their classification as fissile material. Around 200 MeV of binding energy is released per fission reaction, resulting from the mass difference between parent and daughter nuclide. [5] This is described by Einsteins equation describing the equivalence of mass and energy:

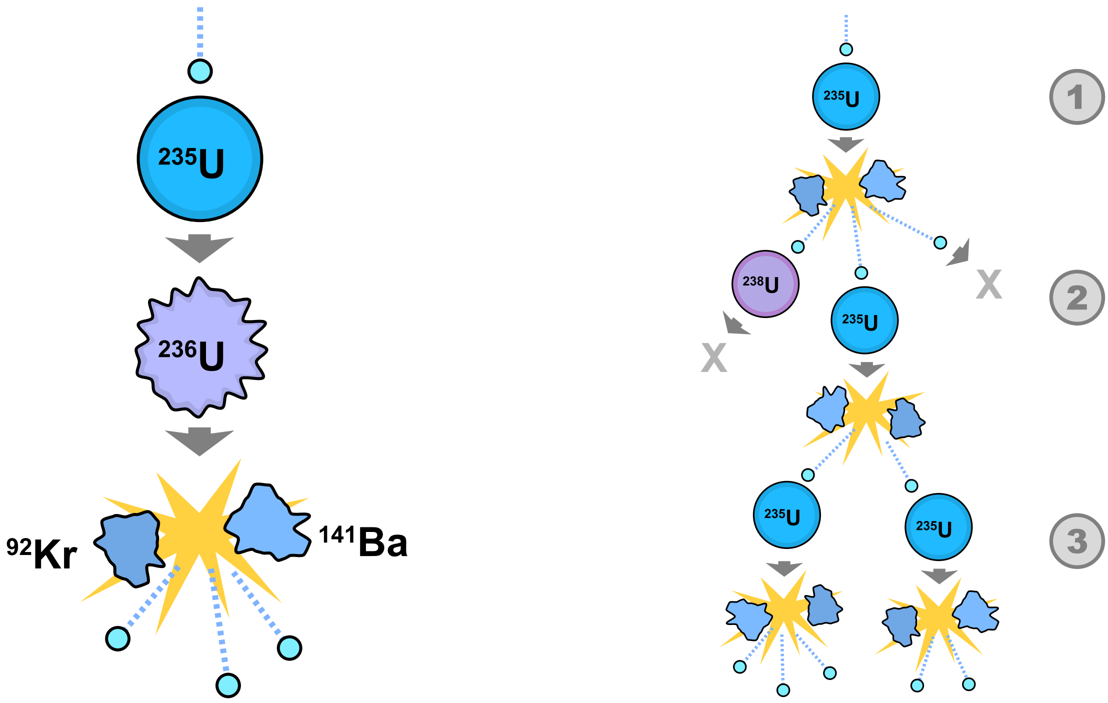
$$E_\mathrm{B} = \gamma \, \Delta m \cdot c^2 \tag{2.4}$$

where E_B : binding energy
γ : Lorentz factor
Δm : mass defect
c : speed of light in vacuum (2.99792458×10^8 m $\cdot$ s^{-1})

As a result of the fission process, prompt neutrons, gamma rays from the fission fragments and further decay products as well as delayed neutrons are emitted. As more neutrons are released than absorbed, a chain reaction can be achieved.

This is portrayed in Figure 2.1b, where fission occurs at step one. During step two, not every emitted neutron is absorbed by ^{235}U, as the more abundant ^{238}U absorbs it, leading to no fission as the probability of fission by a thermal neutron is much lower. Another neutron is not absorbed by any uranium nucleus, while the third leads to a second fission process. In step three, two ^{235}U nuclei undergo fission. If fissile material is configured in a geometrical way that the probability of neutron absorption is high, a sustained chain reaction can be achieved and the released energy can be harnessed. [5]

The first controlled nuclear chain reaction was performed in December 1942 at the Chicago Pile 1 reactor, an experimental graphite pile constructed by Enrico Fermi. This new source of energy of enormous scale was utilized subsequently for non-peaceful applications in form of nuclear and thermonuclear bombs, as well as peaceful purposes in nuclear power, isotope production and research reactors. [7] At the TRIGA Mark II research reactor in Vienna, which was used for the experiments in this thesis, the fissile material is also ^{235}U.



(a) Nuclear fission [8]

(b) Fission chain reaction [9]

Figure 2.1.: Simple diagram of nuclear fission and a schematic of a nuclear fission chain reaction.

2.3. Radioactive Decay

Radioactivity is the spontaneous transformation of an unstable nucleus into another while emitting high-energy (ionizing) radiation. Ionizing radiation is a type of radiation that can indirectly or directly ionize the atoms or molecules present in matter when it penetrates it, i.e. it generates electrically charged atoms or molecules. [10] With ionizing radiation, a distinction can be made between the most relevant radiation sources:

- **Particle radiation**: α, β and neutron radiation
- **Electromagnetic radiation**: γ and X-ray radiation

The activity of a radioactive substance with N number of nuclei is defined by the number of radioactive decays (including isomeric transitions [11]) per unit of time [5, p. 28]:

$$A(t) = -\frac{dN(t)}{dt} = \lambda N \quad [A] = 1\ Bq \equiv s^{-1} \tag{2.5}$$

where N : initial number of nuclei
$A(t)$: remaining activity after time t (Bq)
λ : nuclide-specific decay constant (s^{-1})
t : time (s)

During radioactive decay, the original atomic nuclei are transformed into other nuclei, resulting in a decrease (negative sign) in the original number of N atomic nuclei. The activity is equivalent to the number of nuclear decays per second. The number of nuclei decaying per unit time is proportional to the number of nuclei still present. Each radioactive isotope has a half-life specific to the isotope. In one half-life, the number of atoms and therefore the activity of a radioactive substance has decreased by half. The decay constant λ indicates the probability of decay and is related to half-life via the following relationship [5, p. 28]:

$$T_{1/2} = \frac{1}{\tau} = \frac{ln(2)}{\lambda} \tag{2.6}$$

where $T_{1/2}$: half-life (s)
τ : mean lifetime (s)
λ : nuclide-specific decay constant (s^{-1})

Solving the differential equation of Equation 2.5, we get the decay law which describes the exponential decrease in the number of nuclei with time [5, p. 28]:

$$N(t) = N_0 \cdot e^{-\lambda \cdot t} \tag{2.7}$$

or with Equation 2.5 we can calculate the remaining activity at any arbitrary time:

$$A(t) = \lambda N_0 \cdot e^{-\lambda \cdot t} = A_0 \cdot e^{-\lambda \cdot t} \tag{2.8}$$

where N_0 : initial number of nuclei (at time $t = 0$)
A_0 : initial activity (activity at time $t = 0$, Bq)
t : time (s)

2.4. Gamma Radiation

After α- and β-decay, the atomic nucleus often remains in an excited state. Subsequently, competing with internal conversion[1], this energy can be emitted as γ-radiation from the nucleus to ultimately transition to the ground state. This also occurs when particle emission is energetically impossible, leaving only gamma transformation as an option. The γ-transition represents an isomeric transition since the mass number remains unchanged. This process is also called gamma decay and is included in the definition of activity, i.e., the activity of a hypothetical pure γ-emitter indicates the number of gamma rays emitted per second. [11]

The emitted γ-quantum can be described as either a photon or an electromagnetic wave according to wave-particle duality. A photon has no mass, but can be assigned an energy:

$$E = h \cdot \nu = \frac{hc}{\lambda} \tag{2.9}$$

Furthermore, the photon can also be associated with a momentum p:

$$E = p \cdot c \tag{2.10}$$

[1] Internal conversion describes the nuclear decay process where an excited nucleus transfers its energy electromagnetically to an electron from the atomic shell. This electron is subsequently emitted.

where E : energy of the photon (J)
h : Planck constant ($6.62607015 \times 10^{-34}$ J · s)
ν : frequency of the photon (Hz)
c : vacuum speed of light (2.99792458×10^{8} m · s^{-1})
λ : wavelength of the photon (m)
p : momentum of the photon (kg · m · s^{-1})

2.4.1. Gamma Radiation Dose Rate

The gamma radiation dose rate of a radioactive source can be calculated by assuming that the radiation source is an ideal point-source emitting constant electromagnetic radiation. As depicted in Figure 2.2, a source S emits flux lines (e.g., gamma rays), which are proportional to the strength of the source. In our case, this strength corresponds to the activity. The flux lines penetrate a certain area A at a radial distance r. If we double the distance ($2r$), the same number of flux lines now penetrates an area which is four times its original size A at distance r.

This phenomenon is described by the inverse square law, which states that the density of flux lines is inversely proportional to the radial distance from the source. In order to calculate the correct gamma dose rate, we have to multiply the activity with a nuclide-specific gamma dose rate constant, which can be found in the literature. For this thesis, values from [12] were utilized. The gamma dose rate can be calculated for any arbitrary time using Equation 2.8 and [13, p. 77]:

$$\dot{D}(t,r) = \frac{A(t) \cdot \Gamma}{r^2} \tag{2.11}$$

where $\dot{D}(t,r)$: gamma dose rate at time t and distance r (mSv · h^{-1})
$A(t)$: activity at time t (MBq)
Γ : gamma dose rate constant (mSv · m^2 · h^{-1} · MBq^{-1})
r : distance from the point-source (m)

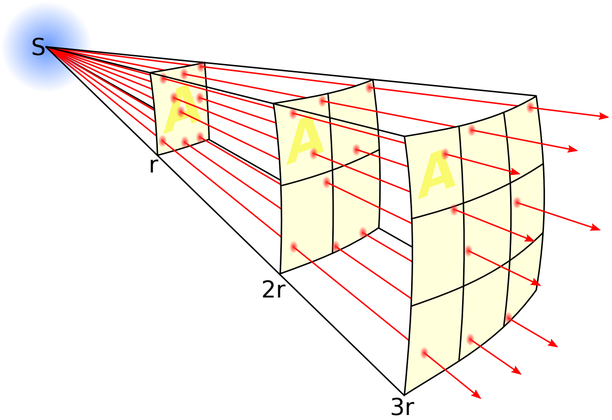


Figure 2.2.: Illustration of the inverse square law [14]

2.5. Neutron Flux

A mono-energetic neutron beam with the intensity I can be described by the product of the number of neutrons per volume and their velocity v [15, eq. (3.1)]:

$$I = n \cdot v \tag{2.12}$$

If this neutron beam now strikes a thin target, the number of collisions F per volume and time increment can be described by [15, eq. (3.16)]:

$$F = \Sigma_t \cdot n \cdot v = \Sigma_t \cdot I \tag{2.13}$$

The product $n \cdot v$ is called neutron flux or neutron flux density [15, eq. (3.20)]:

$$\Phi = n \cdot v \tag{2.14}$$

$$\begin{aligned} \text{where} \quad \Sigma_t &: \text{macroscopic total cross-section } (\mathrm{cm}^{-1}) \\ n &: \text{neutron number density } (\mathrm{cm}^{-3}) \\ v &: \text{velocity of the neutrons } (\mathrm{cm\,s^{-1}}) \\ I &: \text{neutron beam intensity } (\mathrm{s}^{-1}) \\ \Phi &: \text{neutron flux } (\mathrm{cm^{-2}\,s^{-1}}) \end{aligned}$$

If we generalize our approach to include not only mono-energetic neutrons, but neutrons with energies between E and dE, we can describe the differential interaction rate with [15, eq. (5.2)]:

$$dF = \Sigma_t \cdot n(E)dE \cdot v(E) \tag{2.15}$$

Integrating this equation, we get the total interaction rate [15, eq. (5.3)]:

$$F = \int_0^\infty \Sigma_t(E)\, n(E)\, v(E)\, dE = \int_0^\infty \Sigma_t(E)\, \Phi(E)\, dE, \tag{2.16}$$

with the energy-dependent neutron flux, also called neutron flux per unit energy [15, eq. (5.4)]:

$$\Phi(E) = n(E) \cdot v(E) \tag{2.17}$$

The neutron flux determination of reactors, in particular TRIGA research reactors, is a well covered field in nuclear physics. The approach to determine the neutron flux of the TRIGA Mark II reactor was shaped by academic studies done at the reactor of the university, in particular [16], [17] and a more recent PhD thesis [18]. Similar experiments are also covered in the course "Practical Exercises at the Reactor", described in [19], which the author previously completed and provided valuable insights into the approach.

The selected method is called **neutron activation analysis (NAA)**: An element is selected as a neutron target. This target is referred to as activation probe or flux monitor. It is exposed to the neutron flux, subsequently leading through interaction with the neutrons to the formation of radioactive isotopes. These isotopes undergo radioactive decay and emit gamma radiation, which can be measured by detectors. With knowledge of all the relevant parameters, the neutron flux can then be calculated.

As the neutron flux is generally a function of energy ($\Phi = \Phi(E)$), it is common to determine the flux for different energy groups. For this purpose, different materials are suitable for activation at different energies. Based on the proposed materials in [19, Tab. 2.1], four different elements were selected initially for NAA: titanium, cobalt, nickel, indium and gold (see Table 2.3 and Table 2.7).

In subsection 2.5.1 and subsection 2.5.2, the physical parameters of the materials required for NAA are explained in detail, while the underlying physics is explained in the following section. As the neutron flux for every irradiation experiment can deviate slightly, a reference monitor had to be selected, which was added to every irradiation capsule. For this purpose, nickel was selected, as it is abundant and was used previously by the low temperature and superconductivity physics group as a reference and fast flux monitor for irradiation of superconductor samples.

The activation of an infinitely thin sample by a mono-energetic neutron beam can be described by:

$$A_\infty = \lambda \cdot N_\infty = N \cdot \sigma_a \cdot \Phi \tag{2.18}$$

With the total number of target nuclei being:

$$N = n_{\text{target}} \cdot V \tag{2.19}$$

And the relation between the macroscopic and microscopic cross-section [15, p. 56]:

$$\Sigma_a = n_{\text{target}} \cdot \sigma_a \tag{2.20}$$

where
A_∞ : saturation activity (Bq)
N : total number of target nuclei in the sample (dimensionless)
n_{target} : number density of target atoms (cm^{-3})
V : irradiated sample volume (cm^3)
σ_a : microscopic absorption cross-section (cm^2)
Σ_a : macroscopic absorption cross-section (cm^{-1})

Starting with the rate equation for activated nuclei N, which describes the rate of change of the radioactive nuclei as the difference between induced activity and the decay of these nuclei:

$$\frac{dN(t)}{dt} = A_\infty - \lambda \cdot N(t) \tag{2.21}$$

With integration over the irradiation time we get the expression for the activity at the end of irradiation:

$$A(t_{\text{irr}}) = A_\infty(1 - e^{-\lambda t_{\text{irr}}}) = A_\infty \cdot S = N \cdot \sigma_a \cdot \Phi \cdot S \tag{2.22}$$

with the saturation factor S:

$$S = 1 - e^{-\lambda t_{\text{irr}}} \tag{2.23}$$

For extremely long irradiation times $t_{\text{irr}} \rightarrow \infty$ we receive the saturation activity. The saturation activity (A_∞) describes the maximum activity achievable by irradiating N nuclei. At the constant value A_∞, the rate of formation of radioactive nuclei equals the rate of their decay. After irradiation and between the measurement of the activated material, time t_{cool} passes. The isotope undergoes radioactive decay during t_{cool}, which is considered with an additional term:

$$A(t_{\text{irr}}, t_{\text{cool}}) = A_\infty \cdot S \cdot e^{-\lambda \cdot t_{\text{cool}}} = A_\infty \cdot S \cdot C \tag{2.24}$$

Defining the cooling factor C:

$$C = e^{-\lambda \cdot t_{\text{cool}}} \tag{2.25}$$

The activity at measurement time is equal to Equation 2.24:

$$A_{\text{meas}} = A_\infty \cdot S \cdot C \tag{2.26}$$

Leading to the relation between the measured activity and the neutron flux:

$$A_{\text{meas}} = N \cdot \sigma_a \cdot \Phi \cdot S \cdot C \tag{2.27}$$

Rearranging to solve for the neutron flux we get:

$$\Phi = \frac{A_{\text{meas}}}{N \cdot \sigma_a \cdot S \cdot C} \tag{2.28}$$

2.5.1. Thermal Neutron Flux

As described in *Neutronenphysikalisches Praktikum I: Physik und Technik der Aktivierungssonden* by Bensch and Fleck [20], thermal and epithermal neutrons cause exothermic (n,γ) and (n,f)[2] nuclear reactions in exposed activation probes. These reactions differ decisively from reactions caused by fast neutrons, which are often endothermic. Exothermic reactions release energy during the reaction process, whereas endothermic reactions require energy input from the environment. Mathematically, this is described by the Q-value [15, p. 27]:

$$Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}} \right) c^2 \tag{2.29}$$

where Q : energy released or absorbed (MeV)
m : mass of reactants or products (u)
c : speed of light in vacuum ($2.99792458 \times 10^8\ \text{m} \cdot \text{s}^{-1}$)

[2] neutron-induced fission reactions

- **Exothermic reaction:** $Q > 0$ — energy is released.
- **Endothermic reaction:** $Q < 0$ — energy must be supplied.

For low neutron energies up to few eV, the neutron absorption cross-section can be described by [5]:

$$\sigma_a \propto \frac{1}{v} \tag{2.30}$$

with v being the velocity of the neutron (see Figure 2.3). This relation implies that the probability of absorption increases, if the energy and therefore the velocity of the neutron decreases. This can be explained by the fact that a slower neutron remains in the vicinity of the nucleus for a longer period of time. [5]

This region is followed by the resonance region, which can extend from a few eV up to 100 keV, depending on the specific reaction. [5] This region is dominated by sharp increases and decreases of the absorption cross-section in short energy intervals. A nucleus can assume discrete energies similar to electrons of the atomic shell, which leads to higher and lower probabilities of neutron absorption at different energies. In simplified terms, it is more and less energetically ideal for the energy state of the core to absorb the particle. This is described by quantum mechanics.

The resonance region is followed by the fast region, which is characterized by further decreasing neutron cross-sections. In this region, the cross-sections converge toward the geometric nuclear cross-section value. Inelastic scattering becomes relevant at energies above 10 keV. [5]

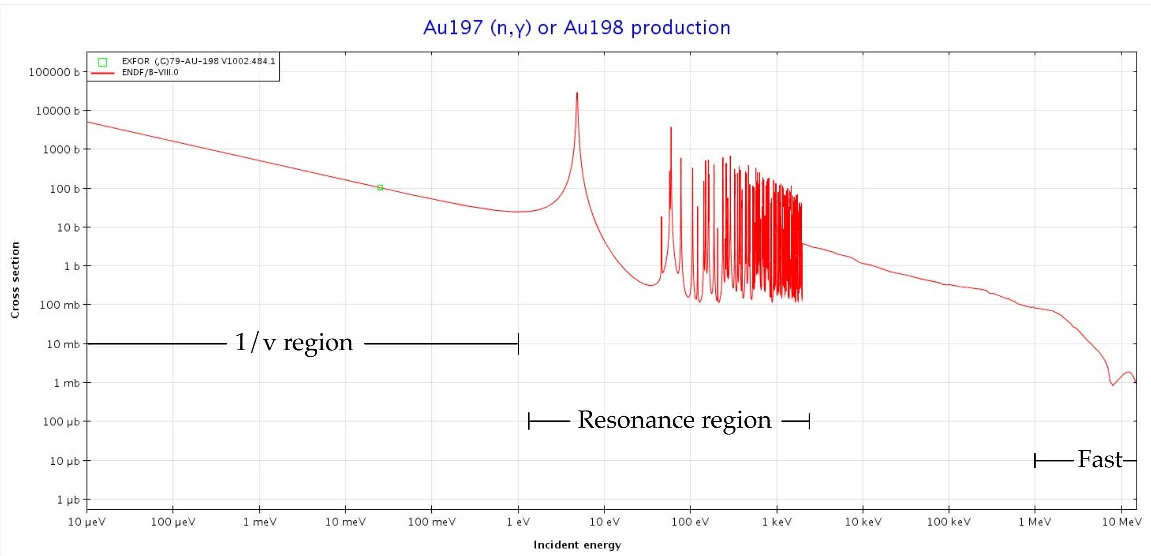


Figure 2.3.: Different characteristic regions of the neutron absorption cross-section of the ^{197}Au(n, γ)^{198}Au reaction. Original plot from [21] with added annotation, marking the different regions.

As (n,γ)-reactions are mainly caused by both thermal and epithermal neutrons, leading to the total saturation activity in an unshielded sample [20, eq. (6.1.1)]:

$$A_{\mathrm{t}} = A_{\mathrm{th}} + A_{\mathrm{epi}} \tag{2.31}$$

In order to separate the activation by thermal and epithermal neutrons, the cadmium difference method can be applied. Cadmium-shielding with adequate thickness absorbs practically all thermal neutrons while absorbing almost no epithermal neutrons as the absorption cross-section drops by several orders of magnitude at around 0.5 eV. Adequate thickness of a cadmium filter is defined in [20, p. 167] as:

- **Collimated Neutron Beam**: $d \geq 1\,\mathrm{mm}$
- **Isotropic Neutron Field**: $d \geq 0.5\,\mathrm{mm}$

The effective cut-off energy of cadmium (E_{Cd}) is not a perfect step function and therefore somewhere between 0.4 eV and 0.7 eV. The energy threshold (E_{ET}) between the thermal and epithermal neutron spectrum is at around 0.1 eV ($3\,k_{\mathrm{B}}T$ and $5\,k_{\mathrm{B}}T$). As E_{ET} is lower than E_{Cd}, a fraction of epithermal neutrons is absorbed by the cadmium filter. This is corrected with the cadmium correction factor F_{Cd} [20, eq. (6.1.2)]:

$$A_{\mathrm{ep}} = F_{\mathrm{Cd}} \cdot A_{\mathrm{Cd}} \tag{2.32}$$

With:

$$F_{\mathrm{Cd}} > 1 \tag{2.33}$$

Resulting in the expression for thermal activation by means of the difference method [20, eq. (6.1.3a)]:

$$A_{\mathrm{th}} = A_{\mathrm{t}} - F_{\mathrm{Cd}} \cdot A_{\mathrm{Cd}} \tag{2.34}$$

An overview of the cadmium correction factors for the two relevant isotopes of gold and indium are presented in Table 2.2. F_{Cd} varies depending on the literature source. For the thermal flux measurement of this work, cadmium shielding with a thickness of 0.65 mm to 1.0 mm was used. For gold, the cadmium correction factor of 1.15 [20, p. 179] was selected for subsequent flux calculations, as Bensch and Fleck [20] is the most detailed source and the origin of the data is comprehensible.

For the thermal flux determination, three elements were selected based on previous experiments and existing literature, including [19], [22], [23]: gold, cobalt and indium (Table 2.3). The ideal activation probe material should offer a (n,γ)-reaction with a high thermal cross-section and a γ-line that can easily be attributed to the activated isotope. The half-life should be in the magnitude of hours to days.

Isotope	d_{Cd} (mm)	F_{Cd}	Reference
^{198}Au	0.5	1.14	[20, p. 179]
^{198}Au	1.0	1.15	[20, p. 179]
^{198}Au	unknown	1.22	[19, p. 28]
^{115}In	0.5	1.32	[20, p. 179]
^{115}In	1.0	1.40	[20, p. 179]
^{115}In	unknown	1.31	[22, p. 2]

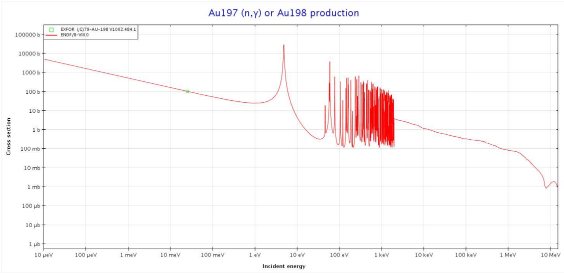
Table 2.2.: Different cadmium correction factors for ^{198}Au and ^{115}In, depending on the thickness d_{Cd} and reference.

Cobalt irradiation produces the radioactive isotope ^{60}Co, which has a half-life of over five years. For our experiment, this implies substantial activation of material with a decay time of over a decade until safe activity is reached. This lead to the elimination of cobalt as a thermal activation material in order to minimize radioactive waste. For indium, the reaction ^{113}In(n, γ)^{114m1}In was selected as the cross-section is the highest of all considered reactions and the decay time is longer than of ^{198}Au.

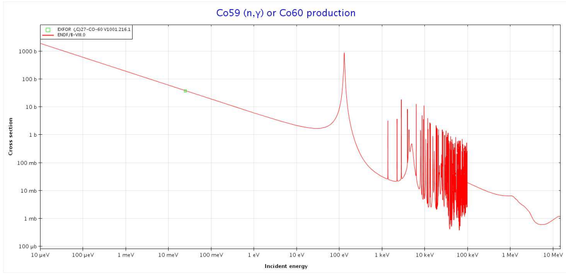
Due to irradiation time constraints, all activation probes were irradiated simultaneously at the CIP, which is also termed the central thimble in the literature. This proved to be a challenge, as the decay time has to be similar for all irradiated isotopes. This demand stems from the fact that the cool-off time between irradiation and handling is determined by the most active isotope and highest dose rate (^{198}Au). This lead to the elimination of the ^{115}In(n, γ)^{116m1}In reaction. Virtually all thermal measurements with indium utilize the latter reaction path. The cadmium shielding factor for the reaction ^{113}In(n, γ)^{114m1}In is therefore not readily accessible. To determine this factor, a separate experiment would be necessary, which was beyond the scope of this thesis.

Element	Reaction	$T_{1/2}$ [24]	$\sigma_{a,th}$ (b) [25]
Gold	^{197}Au(n, γ)^{198}Au	2.6941(2) d	98.70259 ± 0.19519
Cobalt	^{59}Co(n, γ)^{60}Co	5.27(4) y	37.18373
Indium	^{113}In(n, γ)^{114m1}In	49.51(1) d	12.13492
Indium	^{115}In(n, γ)^{116m1}In	54.29(17) min	202.275

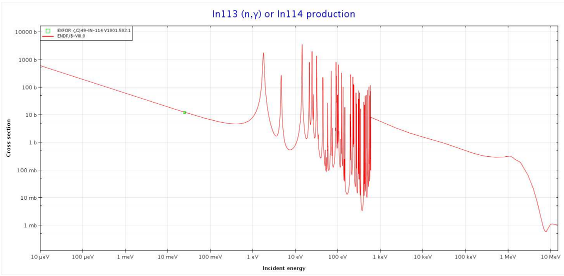
Table 2.3.: Materials selected for the neutron activation analysis for the thermal neutron flux. Uncertainties in $T_{1/2}$ refer to the last digit in parentheses. [19], [24], [25]



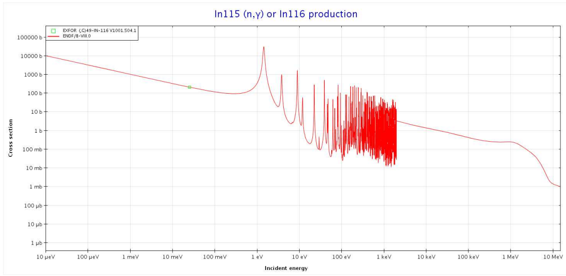
(a) Cross-section of ^{197}Au(n, γ)^{198}Au reaction [21]



(b) Cross-section of ^{59}Co(n, γ)^{60}Co reaction [21]



(c) Cross-section of ^{113}In(n, γ)^{114m1}In reaction [21]



(d) Cross-section of ^{115}In(n, γ)^{116m1}In reaction [21]

Figure 2.4.: Nuclear reaction cross-sections for neutron activation analysis. Comparison of cross-sections for different isotopes showing thermal and resonance regions.

Correction of the Thermal Cross-Section

In order to calculate the correct thermal neutron cross-section of the thermal activation probes, the neutron temperature has to be determined at the irradiation position. This temperature is increasing with reactor power and is by first approximation equal to the temperature of the moderator. As most materials have a $1/v$–distribution for their neutron absorption cross-section, including the common moderators light water and graphite, slower neutrons are disproportionally absorbed at lower energies compared to higher ones. This leads to a hardening of the neutron spectrum, which can be approximated by a Maxwell-Boltzmann-distribution with the effective neutron temperature (T_n) [5, eq. (4.42)]:

$$T_\mathrm{n} = T_\mathrm{m}\left(1 + c_\mathrm{T}\frac{\Sigma_\mathrm{a}(k_\mathrm{B}T)}{\xi\Sigma_\mathrm{s}}\right) \tag{2.35}$$

$$\frac{\Sigma_\mathrm{a}(k_\mathrm{B}T)}{\xi\Sigma_\mathrm{s}} < 0.1 \tag{2.36}$$

Equation 2.35 comprises the inverse moderating ratio (MR), which originally describes how effectively a moderator slows down neutrons compared to how much it absorbs them and is defined by [5, p. 82]:

$$\text{Moderating Ratio (MR)} = \frac{\xi \Sigma_s}{\Sigma_a} \tag{2.37}$$

where T_n : neutron temperature (K)
T_m : moderator temperature (K)
c_T : characteristic value (dimensionless)
$\Sigma_a(k_B T)$: macroscopic absorption cross section (cm^{-1})
ξ : average logarithmic energy decrement (dimensionless)
Σ_s : macroscopic scattering cross section (cm^{-1})

By first approximation, T_n is equal to the temperature of the moderator. The second term is the inverse of the MR, which is equal to the ratio between the macroscopic absorption cross-section (Σ_a) and the stopping power ($\xi\Sigma_s$) multiplied by a moderator-dependent value c_T. This value ranges according to the literature depending on the moderation material from $c_T = 1$ for hydrogen, $c_T \approx 4$ [5, p. 72] to $c_T \approx 1.2 - 1.8$ for light-water moderated reactors [26, p. 381]. The condition of Equation 2.36 results from the postulate, that the real spectral distribution does not deviate too strong from a Maxwell-Boltzmann distribution. If this is condition is not met, the spectral distribution cannot be described by an effective neutron temperature. The neutron moderation in the TRIGA Mark II reactor occurs through two separate mechanisms:

- **Self-moderation via hydrogen in the fuel** (Uranium and Zirconium Hydride-alloy) [27, p. 8]. According to [28, p. 10], around 44 % of the total neutron moderation occurs in the fuel, while the majority occurs in the surrounding water. As shown in [29, Tab. 9], this ratio remains approximately constant for H_2O and U–ZrH temperatures during steady operation up to $P_{th} = 250$ kW.
- **Moderation via light water** (H_2O)

As the moderation characteristics of the TRIGA reactor are complex functions of the hydride, water and uranium ratios [27, p. 8], simplifications had to be done to be able to calculate the approximate effective neutron temperature: For the neutrons irradiating the CIP, it was assumed that 80 % of the moderation is performed by the fuel ($ZrH_{1.6}$), while only 20 % of the neutrons are slowed down in light water. This assumption is supported by the geometry of the core (see Figure 3.1c), which surrounds the central irradiation tube in close proximity, leaving only a miniscule space filled with water. Additionally, the irradiation capsules are filled with water, leading to our estimation of 20 %. For the neutrons reaching the Weber tube positioned outside the graphite reflector, only an estimated fraction of 10 % is expected to be thermalized by ZrH, while the majority is moderated by water and graphite. In order to simplify further calculation, it was assumed that 100 % of the neutrons in the Weber tube are slowed down by light water. In reality, the Weber tube is positioned just outside the graphite. Inside the Weber tube, virtually no moderation occurs, as the tube is dry. Therefore, for this location, the majority of neutron moderation is most likely provided by the graphite, whose temperature is unknown. As described in [18, p. 130], the graphite reflector is clad in aluminum. There is also an air gap between the graphite and aluminum. The water temperatures used for further calculations were estimated based on the expertise of reactor personnel, whereas the fuel temperature was determined previously:

- Fuel temperature at 250 kW: 200 °C [30, p. 4]
- Water temperature in the CIP at 250 kW: 60 – 70 °C
- Moderator temperature surrounding the Weber tube at 250 kW: 30 – 40 °C

Calculating T_n, further estimations had to be done: The characteristic value c_T has been selected as an average of 1.5 based on the literature values for light-water reactors. The inverse of the MR is expected to decrease with increased neutron energies and was estimated based on the MRs of 70 [31, p. 123] and 68 [5, p. 83] for light water.

While the MR of $ZrH_{1.6}$ is not available, an approximation can be done based on the MR of ZrH_2, which is 55 at room temperature according to [32, Tab. 4]. As $ZrH_{1.6}$ has a smaller hydrogen content ratio compared to ZrH_2 by factor 0.8, we can expect a MR of 44 for $ZrH_{1.6}$ at room temperature. The MR will decrease with increased fuel temperature due to the cell effect, which is extensively described in [33]. In order to determine the exact decrease of the MR, a complex calculation is necessary with data, that was not available at the time of the creation of this thesis. As a result, a MR of 44 was utilized for further calculations. The estimated values of c_T and $\frac{\Sigma_\mathrm{a}}{\xi\Sigma_\mathrm{s}}$ as well as the calculated values of T_n are depicted in Table 2.4.

Position	Material	T_m (K)	c_T	MR	$\frac{\Sigma_\mathrm{a}}{\xi\Sigma_\mathrm{s}}$	T_n (K)
Weber	H_2O	313.15	1.5	70	0.0143	319.8604
CIP	H_2O	343.15	1.5	68	0.0147	350.7195
CIP	$ZrH_{1.6}$	473.15	1.5	44	0.0227	489.2801

Table 2.4.: Characteristic values for the calculation of the effective neutron temperature [31, p. 123], [5, p. 83], [34, Tab. 1]

The corrected thermal absorption cross-section at the neutron-temperature T_n can be calculated according to [5, eq. (4.43)] with the relation:

$$\bar{\sigma}_\mathrm{A}(T_\mathrm{n}) = \frac{\sqrt{\pi}}{2} g_a(T_\mathrm{n}) \sqrt{\frac{T_0}{T_\mathrm{n}}} \sigma_\mathrm{a,th}(k_B T_0) \tag{2.38}$$

where $\bar{\sigma}_{\mathrm{A}}(T_{\mathrm{n}})$: effective absorption cross section at T_{n} (barn)
$g_a(T_{\mathrm{n}})$: Westcott factor (dimensionless)
T_{n} : neutron temperature (K)
$T_0 = 293.15\,\mathrm{K}$: reference temperature (room temperature)
$\sigma_{\mathrm{a,th}}(k_{\mathrm{B}}T_0)$: microscopic absorption cross section at thermal energy $k_{\mathrm{B}}T_0$ (barn)
k_{B} : Boltzmann constant ($1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$)

The Westcott factor $g_a(T_{\mathrm{n}})$ describes the deviation of the neutron absorption cross-section of a nuclide from the $1/v$-distribution [5, p. 72]. The tabulated values were extracted from [35, Tab. 4] for ^{197}Au and are depicted in Table 2.5. The Westcott factor at 343.15 K was calculated as the arithmetic mean of $g_a\,(333.15\,\mathrm{K}) = 1.0086$ and $g_a\,(353.15\,\mathrm{K}) = 1.0096$ and is denoted with an asterisk.

Isotope	g_a (313.5 K)	g_a (343.15 K)	g_a (473.15 K)
^{197}Au	1.0076	1.0091*	1.0156
Position (Moderator)	Weber (H_2O)	CIP (H_2O)	CIP (ZrH)

Table 2.5.: Westcott factors of ^{197}Au at different neutron temperatures [35, Tab. 4]

Position	Isotope	Moderator	T_0 (K)	$\sigma_{\mathrm{a,th}}(k_{\mathrm{B}}T_0)$ (b)	$\bar{\sigma}_A(T_{\mathrm{n}})$ (b)
Weber	^{197}Au	H_2O	293.15	98.70259	84.37745
CIP	^{197}Au	H_2O	293.15	98.70259	80.69984
CIP	^{197}Au	$ZrH_{1.6}$	293.15	98.70259	68.76419

Table 2.6.: Corrected thermal absorption cross-section for the CIP and the Weber tube outside the graphite reflector

For the CIP, a temperature-corrected and weighted absorption cross-section can be calculated:

$$\bar{\sigma}_{\mathrm{A,weighted}} \approx 0.8 \cdot \bar{\sigma}_{\mathrm{A,ZrH_{1.6}}}(T_{\mathrm{n}}) + 0.2 \cdot \bar{\sigma}_{\mathrm{A,H_2O}}(T_{\mathrm{n}}) \approx 71.1513\ \mathrm{b} \tag{2.39}$$

Combining Equation 2.28 and Equation 2.34 we receive the general formula for calculating the neutron flux, which has to be adapted further:

- Shielded and unshielded samples generally have different datasets leading to the need of separate calculation of the different terms.

- The corrected thermal absorptions cross-section $\bar{\sigma}_A$ needs to be utilized instead of σ_a.
- As samples are not infinitely thin, self-shielding has to be considered with a self-shielding factor (G), which is described with the expression [22, p. 2]:

$$G = \frac{\bar{\Phi}}{\Phi_0} = \frac{1 - e^{-\mu_a \delta}}{\mu_a \delta} \tag{2.40}$$

$$\delta = \frac{m}{A_{\text{area}}} \tag{2.41}$$

The thermal neutron flux can be calculated using the following relation, which is an adaptation of the flux equations presented in [19], [22], [23]:

$$\Phi_{\text{th}} = \frac{A_{\text{bare}} \cdot e^{\lambda \cdot t_{\text{cool,bare}}}}{N_{\text{bare}} \cdot \bar{\sigma}_A(T_n) \cdot S_{\text{bare}} \cdot G} - F_{\text{Cd}} \cdot \frac{A_{\text{Cd}} \cdot e^{\lambda \cdot t_{\text{cool,Cd}}}}{N_{\text{Cd}} \cdot \bar{\sigma}_A(T_n) \cdot S_{\text{Cd}} \cdot G} \tag{2.42}$$

$$N = \frac{m \cdot N_A}{M} \cdot \theta \tag{2.43}$$

According to [22, p. 2], the mass absorption coefficient is $0.3018\,\text{cm}^2\text{g}^{-1}$ for gold and $1.02\,\text{cm}^2\text{g}^{-1}$ for indium. For our specific activation samples, the neutron self-shielding factor was calculated and yielded $G \approx 1$. This is a result of the extremely thin, light and small samples that were used, implying that self-shielding can be neglected for this specific experiment. This yields the formula relevant for our thermal flux determination:

$$\Phi_{\text{th}} \approx \frac{A_{\text{bare}} \cdot e^{\lambda \cdot t_{\text{cool,bare}}}}{N_{\text{bare}} \cdot \bar{\sigma}_A(T_n) \cdot S_{\text{bare}}} - F_{\text{Cd}} \cdot \frac{A_{\text{Cd}} \cdot e^{\lambda \cdot t_{\text{cool,Cd}}}}{N_{\text{Cd}} \cdot \bar{\sigma}_A(T_n) \cdot S_{\text{Cd}}} \tag{2.44}$$

where Φ_{th} : thermal neutron flux ($\text{cm}^{-2}\text{s}^{-1}$)
Φ_{fast} : fast neutron flux ($\text{cm}^{-2}\text{s}^{-1}$)
A_{bare} : activity of the unshielded activation foil (Bq)
A_{Cd} : activity of the Cd-shielded activation foil (Bq)
A : activity of the activation foil for threshold reactions (Bq)
F_{Cd} : cadmium correction factor (dimensionless)
λ : decay constant of the produced radionuclide (s^{-1})
N : number of target nuclei in the activation foil (dimensionless)
$\bar{\sigma}_{\text{A}}(T_{\text{n}})$: Maxwellian-averaged absorption cross section at T_{n} (cm^2)
σ_{th} : thermal neutron cross-section (cm^2)
σ_{eff} : effective cross-section for fast neutrons (cm^2)
S : saturation factor (dimensionless)
C : cooling factor (dimensionless)
G : neutron self-shielding factor (dimensionless)
μ_{a} : mass absorption coefficient (cm^2g^{-1})
δ : surface density ($\text{g}\,\text{cm}^{-2}$)
m : mass of the activation foil (g)
A_{area} : area of the activation foil (cm^2)
N_A : Avogadro's number ($6.02214 \times 10^{23}\ \text{mol}^{-1}$)
M : atomic mass of the target element ($\text{g}\,\text{mol}^{-1}$)
θ : isotopic abundance of the target nuclide (dimensionless)
t_{irr} : irradiation time (s)
t_{cool} : cooling time between end of irradiation and start of measurement (s)
$t_{\text{cool,bare}}$: cooling time for bare foil (s)
$t_{\text{cool,Cd}}$: cooling time for Cd-shielded foil (s)
$T_{1/2}$: half-life of the produced radionuclide (s)

2.5.2. Fast Neutron Flux

For fast neutrons (> 1 MeV), the radiative capture cross-section drops significantly to a few millibarns, requiring alternative reaction paths for a reliable detection.

Threshold reactions offer the ideal attributes, as they are strong endothermic reactions with cross-sections greater than zero from certain energies. These threshold energies are reaction specific and the cross-sections are generally energy-dependent. Different isotopes and reactions can thus be selected to cover different energy groups. The different endothermic reactions comprise (n,p), (n,α), (n,n′), (n,2n) and more rarely (n,f)-reactions, which have resonance character at different energies and generally are sensitive to energies between 1 MeV and 20 MeV. [20]

If both the energy-dependent reaction cross-section and flux spectrum are known, the method of effective energy threshold can be applied. As described in [20], this method was developed first by Hughes and published in [36]. It enables the simple flux calculation with threshold detectors. The flux spectrum has to be known - this is normally given for measurements inside the core of a thermal or epithermal reactor, where the flux can be described by a fission or moderation spectrum. For measurements outside the core, the spectrum generally deviates from the in-core spectrum, as the number of scattered neutrons is greater than fission neutrons. [20] This information is crucial to assess the accuracy of our measurements, which will be conducted at two positions: At the CIP, which is inside the core, and the Weber tube, which is located outside both the core and graphite reflector.

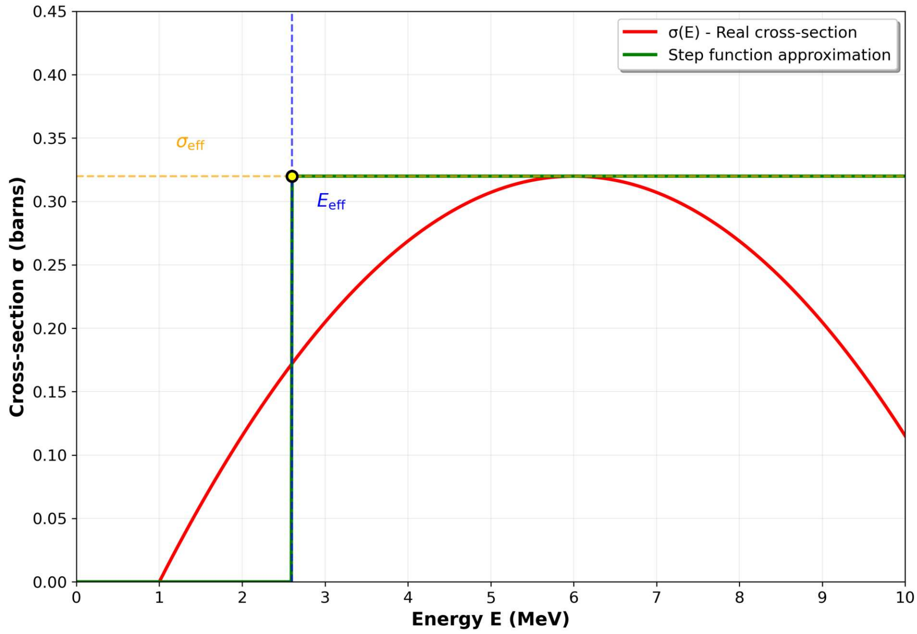


Figure 2.5.: Illustration of the method of effective threshold energies showing the real energy-dependent cross-section $\sigma(E)$ and its step function approximation with the pair of values σ_{eff}, E_{eff}.

The method of effective threshold energies essentially substitutes the energy-dependent cross-section with an effective, constant cross-section, which is described by a step-function (see Figure 2.5): Below the effective energy, the cross-section is zero. Above the threshold E_{eff}, the cross-section is constant with the value σ_{eff}. This is described by the relation [20, eq. (7.1.2)]:

$$\int_0^\infty \sigma(E)\Phi(E)\, dE = \sigma_{\text{eff}} \int_{E_{\text{eff}}}^\infty \Phi(E)\, dE \tag{2.45}$$

For the activation of a thin threshold detector per volume ($N\sigma(E)d \ll 1$), the following equation is valid [20, eq. (7.1.3)]:

$$A = N \cdot \sigma_{\text{eff}} \int_{E_{\text{eff}}}^\infty \Phi(E)\, dE \tag{2.46}$$

or [20, eq. (7.1.4)]:

$$A = N \cdot \sigma_{\text{eff}} \cdot \Phi_{\text{eff}} \tag{2.47}$$

The neutron flux integrated over the interval (E_{eff}, ∞) is described by [20, eq. (7.1.5)]:

$$\Phi_{\text{eff}} = \int_{E_{\text{eff}}}^\infty \Phi(E)\, dE \tag{2.48}$$

If the spectrum corresponds to a fission spectrum the subsequent relation is valid [20, eq. (7.1.6)]:

$$\int_{E_{\text{eff}}}^\infty \Phi(E)\, dE = \frac{A}{N \cdot \sigma_{\text{eff}}} \tag{2.49}$$

The final equation used for calculating the fast flux is obtained by adding the two correction terms for cooling ($C = e^{\lambda \cdot t_{\text{cool}}}$) and saturation ($S$):

$$\Phi_{\text{fast}} = \frac{A \cdot e^{\lambda \cdot t_{\text{cool}}}}{N \cdot \sigma_{\text{eff}} \cdot S} \tag{2.50}$$

The different threshold detector elements were selected both based on previously publicized neutron flux determinations, namely [16] and [17], and available activation material from previous experiments at the institute of atomic and subatomic physics (Atominstitut), TU Wien. The materials selected were titanium, indium and nickel. The threshold reactions, together with the effective threshold energies and effective cross-sections, are summarized in Table 2.7.

The cross-sections for the relevant reactions are depicted in Figure 2.6 and are based on recent evaluated nuclear data libraries from the United States of America

Element	Reaction	$T_{1/2}$	E_{eff} (MeV)	σ_{eff} (mb)
Titanium	$^{46}\text{Ti}(\text{n,p})^{46}\text{Sc}$	83.79(4) d	5.5	520
Titanium	$^{47}\text{Ti}(\text{n,p})^{47}\text{Sc}$	3.3492(6) d	3.7	230
Titanium	$^{48}\text{Ti}(\text{n,p})^{48}\text{Sc}$	43.71(9) h	7.2	58
Indium	$^{115}\text{In}(\text{n,n}')^{115\text{m}}\text{In}$	4.486(4) h	1.65	350
Nickel	$^{58}\text{Ni}(\text{n,p})^{58}\text{Co}$	70.86(6) d	3.45	600

Table 2.7.: Materials selected for the neutron activation analysis for the fast neutron flux. Uncertainties in $T_{1/2}$ refer to the last digit in parentheses. [19], [24], [20, pp. 198-199]

(ENDF/B–VIII), Japan (JENDL–4.0u) and the Russian Federation (BROND–3.1). For the calculation of the fast flux, the effective cross-sections and energies were adopted from [20, pp. 198-199], as the calculation based on newer datasets would have gone beyond the scope of this work.

2.5.3. Effective Irradiation Time Calculation

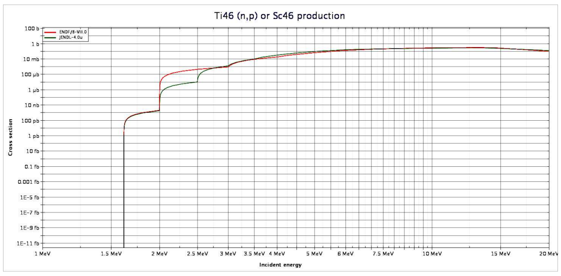
The original goal of the experiment is to determine the neutron flux at a target power of 250 kW, meaning that the samples have to be irradiated only at target power. This ideal condition is generally not achievable in experimental conditions because of different circumstances:

- The reactor power is not always stable at target power (250 kW), especially during startup.
- **CIP**: The samples had to be inserted into the CIP before reactor startup.
- **CIP**: The samples experience neutron activation even after reactor shutdown as the power decreases exponentially.
- **Weber tube**: The samples can be inserted during operation but cannot be removed instantly at the end of irradiation because of the high dose rate.

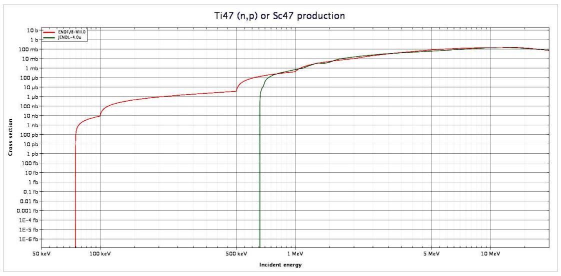
In order to correct the deviation of the neutron flux from the target power (250 kW) during startup, shutdown or power dips, the power measured by reactor instrumentation is integrated over the time of the experiment, which can be approximated using the trapezoidal rule:

$$\int_0^{t_{\text{total}}} P(t)\,dt \approx \sum_{i=1}^{n-1} \frac{P(t_i) + P(t_{i+1})}{2} \cdot (t_{i+1} - t_i) \tag{2.51}$$

where $P(t)$ is the reactor power at time t while t_{total} is the total irradiation duration. The effective irradiation time t_{eff} at the target power P_{ref} (250 kW) can be expressed by normalizing to the target power level:

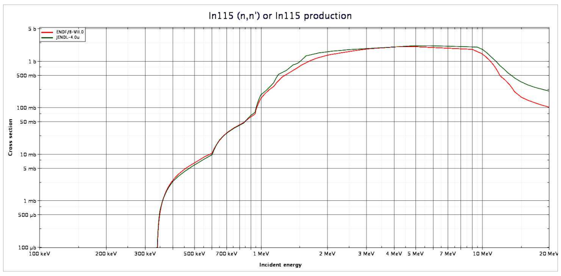


(a) Cross-section of ^{46}Ti(n, p)^{46}Sc reaction [37]

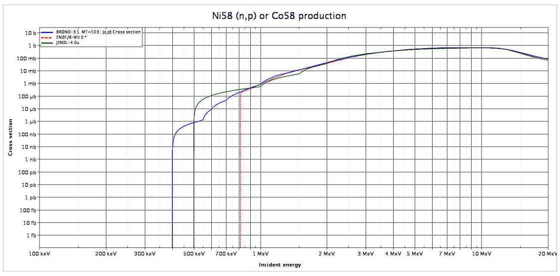


(b) Cross-section of ^{47}Ti(n, p)^{47}Sc reaction [37]

(c) Cross-section of ^{48}Ti(n, p)^{48}Sc reaction [37]



(d) Cross-section of ^{115}In(n, n$'$)^{115m}In reaction [37]



(e) Cross-section of ^{58}Ni(n, p)^{58}Co reaction [38]

Figure 2.6.: Fast neutron reaction cross-sections for neutron activation analysis. Comparison of cross-sections for different isotopes in the fast neutron energy region, showing threshold effects and energy dependence for (n,p) and (n,n′) reactions. The color of the original plots was adjusted for increased contrast and better visibility.

$$t_{\text{eff}} = \frac{1}{P_{\text{target}}} \int_0^{t_{\text{total}}} P(t)\, dt \tag{2.52}$$

The final calculation is done by using the approximation of Equation 2.51:

$$t_{\text{eff}} \approx \frac{1}{P_{\text{target}}} \sum_{i=1}^{n-1} \frac{P(t_i) + P(t_{i+1})}{2} \cdot (t_{i+1} - t_i) \tag{2.53}$$

The computed data from the experiments are depicted in subsection 3.1.5 while the complete set of reactor power plots can be found in section B.2.

2.5.4. Error Propagation in Neutron Flux Calculations

For a general function $f(x_1, x_2, \ldots, x_n)$ with uncertainties $\sigma_{x_1}, \ldots, \sigma_{x_n}$ in all variables, the combined standard deviation is calculated using linear approximation of error propagation theory. If the uncertainties $\sigma_{x_1}, \ldots, \sigma_{x_n}$ are not correlated and small, the error can be calculated with the Gaussian error propagation formula [39, p. 7]:

$$\sigma_f = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2} \tag{2.54}$$

The thermal neutron flux Φ_{th} using the cadmium difference method depends on:

$$\Phi_{\text{th}} = \Phi_{\text{th}}(A_{\text{bare}}, A_{\text{Cd}}, t_{\text{cool,bare}}, t_{\text{cool,Cd}}, N_{\text{bare}}, N_{\text{Cd}}, \bar{\sigma}, S_{\text{bare}}, S_{\text{Cd}}, F_{\text{Cd}}, \lambda) \tag{2.55}$$

Where the number of target nuclei further depends on:

$$N = N(m, M, \theta) \tag{2.56}$$

The saturation factor depends on:

$$S = S(\lambda, t_{\text{irr}}) \tag{2.57}$$

And the decay constant depends on:

$$\lambda = \lambda(T_{1/2}) \tag{2.58}$$

The fast neutron flux Φ_{fast} depends on:

$$\Phi_{\text{fast}} = \Phi_{\text{fast}}(A, N, \sigma_{\text{eff}}, S, t_{\text{cool}}, \lambda) \tag{2.59}$$

With the same additional dependencies for N, S, and λ as listed above.

2.5.5. Gammaspectroscopy

In order to measure the activity through gamma spectroscopy, an high purity germanium (HPGe) detector was employed. This type of detector consists of germanium used as a semiconductor, configured as a reverse-biased diode. By applying a reverse voltage, an electric field is established inside the germanium crystal. When incident photons interact with the semiconductor's electrons, the electrons are promoted to the conduction band. This process produces a charge current, which a preamplifier converts into voltage pulses. The height of each pulse corresponds directly to the amount of energy deposited in the detector. Compared to scintillation detectors such as NaI, HPGe detectors achieve much higher energy resolution. The disadvantages of HPGe detectors are their lower counting efficiency and the need for cryogenic cooling to approximately 77 K to minimize thermally induced leakage currents. [40]

Because of their high energy resolution, HPGe spectrometers are particularly well suited for radionuclide identification and for detecting low activity levels. The detector measures γ-quanta counts per second (CPS) at specific energies. The efficiency depends on geometry and energy, often predetermined by reference measurements. Formation of the full-energy peak occurs only when the entire photon energy is deposited in the detector, which happens primarily through the following processes:

- **Photoelectric effect** below 1022 keV
- **Pair production** above 1022 keV, where positron annihilation produces two 511 keV photons

Photon interactions occur through three primary processes:

1. **Compton scattering**: Partial energy transfer to outer-shell electrons creates a continuum and Compton edge in spectra [40]
2. **Photoelectric effect**: Full energy absorption through inner-shell electron ejection, producing characteristic full-energy peaks [40]
3. **Pair production**: High-energy photons (>1022 keV) create electron-positron pairs, with subsequent annihilation generating 511 keV photons [40]

Escape peaks occur when annihilation photons leave the detector volume. The energy deposition mechanisms directly influence spectral features shown in Figure 2.7.

To determine the activity of radioactive isotopes based on their gamma-ray emissions, the gamma spectrometer software carries out the following calculations [22, p. 1]:

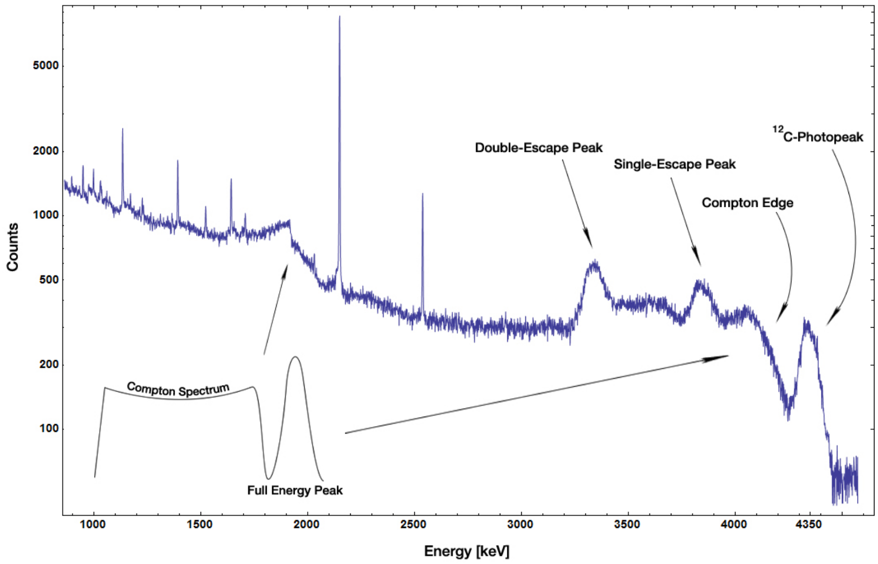


Figure 2.7.: Typical HPGe detector gamma spectrum showing features from Compton scattering, photoelectric effect, and pair production [41]

$$A_0(E_\gamma) = I_0(E_\gamma) \cdot \frac{\lambda}{1 - e^{-\lambda t_\mathrm{m}}} \tag{2.60}$$

If the duration of measurement (t_m) is small compared to the mean lifetime (τ) of the radioactive nuclei ($\lambda t_\mathrm{m} \ll 1$), Equation 2.60 can be simplified to [22, p. 1]:

$$A_0(E_\gamma) \approx \frac{I_0(E_\gamma)}{t_\mathrm{m}} \tag{2.61}$$

To obtain the corrected activity, it is necessary to account for the energy-dependent detector efficiency, the detector dead time, and the emission probability of the measured gamma ray [22, p. 1]:

$$A_\mathrm{corr} = \frac{A_0(E_\gamma)}{\eta(E_\gamma) \cdot R_\mathrm{t} \cdot I_\gamma(E_\gamma)} \tag{2.62}$$

$$\begin{aligned} \text{For thermal neutron flux:} \quad & A_\mathrm{corr} \equiv A_\mathrm{bare},\ A_\mathrm{Cd} \\ \text{For fast neutron flux:} \quad & A_\mathrm{corr} \equiv A \end{aligned}$$

where $A_0(E_\gamma)$: initial activity at start of measurement at energy E_γ (Bq)
$I_0(E_\gamma)$: net peak count rate at energy E_γ (counts/s)
λ : decay constant of the radionuclide (s^{-1})
t_m : measurement time (s)
τ : mean lifetime of the radionuclide (s)
$\eta(E_\gamma)$: detector efficiency at energy E_γ (dimensionless)
$R_\mathrm{t} = \frac{t_\mathrm{live}}{t_\mathrm{real}}$: dead-time correction factor (dimensionless)
$I_\gamma(E_\gamma)$: emission probability for gamma rays of energy E_γ (dimensionless)
A_corr : corrected activity of the sample (Bq)

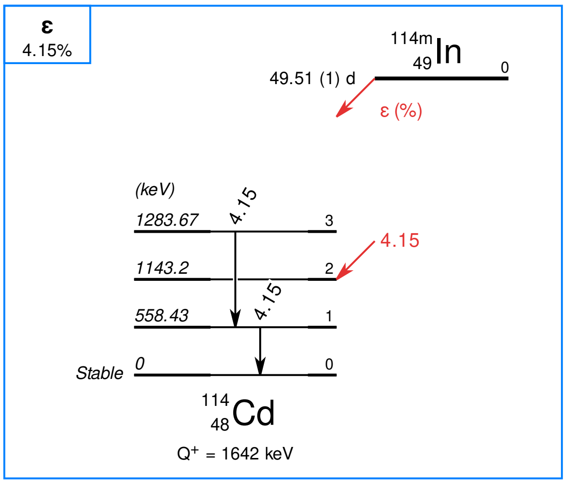
The decay schemes and gamma-ray emission probabilities of the relevant radionuclides are shown in Figure 2.8.

2.6. Nuclear Fusion

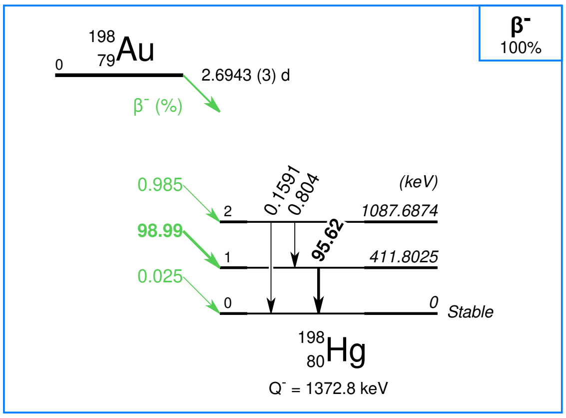
Nuclear fusion describes the process of two or more nuclei combining into larger nuclei and other particles. Fusing two or more nuclei requires vast amounts of energy, as the Coulomb force of the repelling protons must be overcome. If the repelling particles are close enough, the strong nuclear force leads to the attraction of the nuclei and subsequent fusion. [43] The necessary energy is the lowest for isotopes of hydrogen, as only one proton is present. Deuterium-tritium fusion is therefore the most relevant for future fusion reactors, releasing a total of 17.6 MeV per fusion reaction [44]:

$$ {}^2_1\mathrm{D} + {}^3_1\mathrm{T} \rightarrow {}^4_2\mathrm{He}\ (3.5\ \mathrm{MeV}) + {}^1_0\mathrm{n}\ (14.1\ \mathrm{MeV}) \tag{2.63} $$

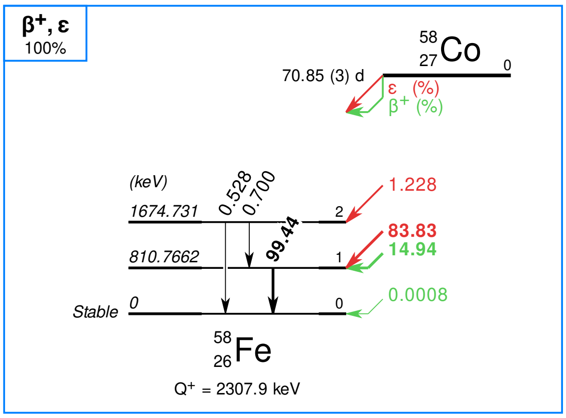
One of the central fusion reactor principles foresees the confinement of the plasma particles in a magnetic field, e.g., the tokamak design, which confines plasma with powerful magnetic coils in an axially symmetrical torus. This requires superconducting magnets, as the necessary magnetic field cannot be provided by conventional magnets if a positive power output is desired. The need for extremely strong and therefore superconducting magnets becomes clear when the relationship between the power density (P_F) of a tokamak reactor and the magnetic field



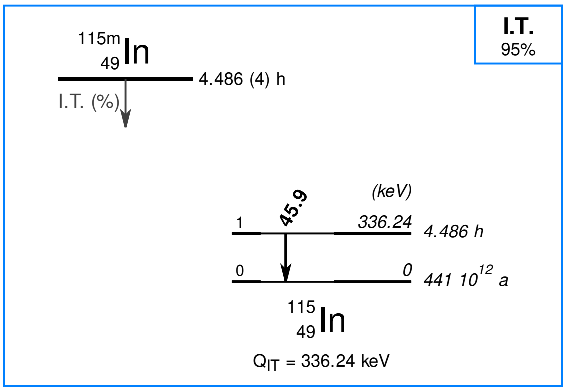
(a) Decay scheme of ^{114m1}In



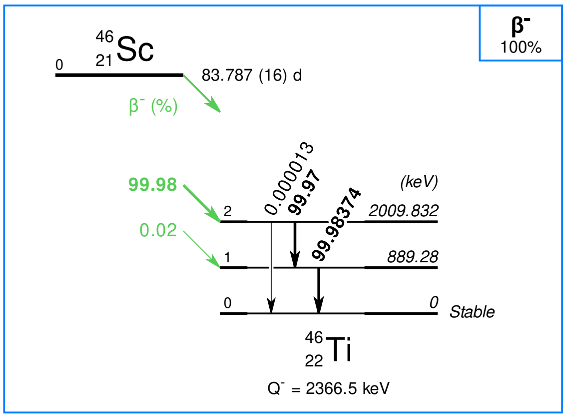
(b) Decay scheme of ^{198}Au



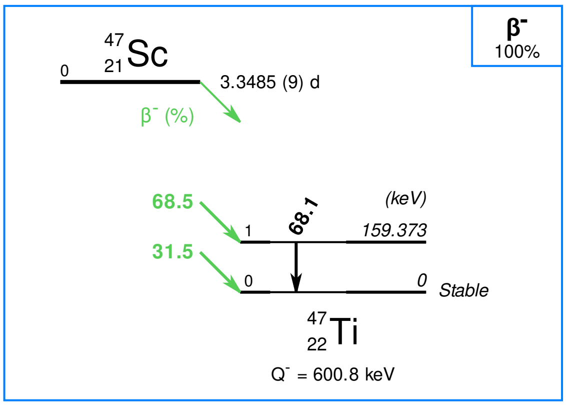
(c) Decay scheme of ^{58}Co



(d) Decay scheme of ^{115m}In



(e) Decay scheme of ^{46}Sc



(f) Decay scheme of ^{47}Sc

Figure 2.8.: Decay schemes of the radioactive isotopes used for neutron flux measurements: ^{114m1}In, ^{198}Au, ^{58}Co, ^{115m1}In, ^{46}Sc, and ^{47}Sc. All decay schemes were sourced from [42].

($B_{T,0}$) is analyzed. According to [45, eq. (1)], P_F is proportional to the fourth power of $B_{T,0}$:

$$P_F \propto \beta_t^4 \, B_{T,0}^4 \tag{2.64}$$

P_F : fusion power density in the tokamak core (arbitrary scaling units)
β_t : ratio of kinetic to magnetic pressure of the plasma (dimensionless)
$B_{T,0}$: toroidal magnetic field at the plasma center (T)

β_t is limited by plasma stability while $B_{T,0}$ is restricted by the achievable peak field ($B_{T,\mathrm{m}}$). $B_{T,\mathrm{m}}$ depends on the superconducting material capabilities. Both low-temperature superconductor (LTS) and HTS are proposed and envisioned for different tokamak reactor designs currently under development and construction. [45] HTS are foreseen for compact fusion reactors, as they provide higher fields necessary for smaller devices. Rare-earth barium copper oxide (REBCO) superconductors are currently the most promising materials, as long conductor lengths can be manufactured and their change of properties under irradiation conditions is well researched. [46]

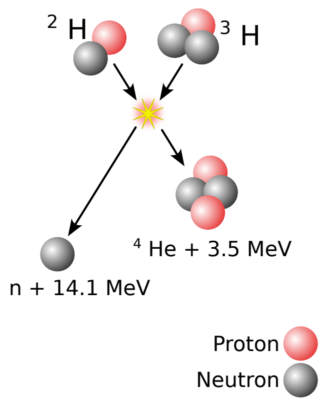


Figure 2.9.: Schematic of deuterium–tritium fusion [44]

2.7. Superconductivity

Certain materials transition into a new state, known as the superconducting state, when their temperature falls below a critical value termed critical temperature (T_c). Superconductivity of mercury was first discovered in 1911 by Heike Kamerlingh Onnes. It is defined by two fundamental properties at low external magnetic fields and low internal electrical currents [47]:

- **Zero electrical resistivity**: As soon as the temperature falls below the critical temperature, the electrical resistance drops to zero. In Figure 2.10a the electrical resistivity of normal conductors and superconductors is compared for decreasing temperature.
- **Magnetic flux expulsion (Meissner effect)**: An external magnetic field is expelled from the inside of the superconductor by the superconductive state (see part below B_{c_1} in Figure 2.10b). The superconductor acts as an ideal diamagnet.

In a superconducting state, a certain order in the system occurs. If the temperature falls below T_c, two electrons form a Cooper pair, which behave like bosons. By condensing into Cooper pairs, the total energy of the system is lowered. These Cooper pairs can be described by a macroscopic wave function (ψ). This microscopic process of the formation of Cooper pairs is described by the Bardeen-Cooper-Schrieffer (BCS) theory. Essentially, Cooper pairs are formed via electron-phonon interaction. [48] The BCS theory is valid for low T_c superconductors, while HTS, e.g., cuprates, cannot be explained by the conventional BCS theory. [43]

The three thermodynamic limitations of a superconductive state comprise the [47, p. 13]:

1. Temperature (T_c)
2. Magnetic field (H_{c_i})
3. Current density (j_d)

The general relation between the magnetic flux density, magnetic field strength and magnetization is defined by [47, p. 3]:

$$B = \mu_0(H + M) \tag{2.65}$$

$$\begin{aligned}
B &: \text{ Magnetic flux density (magnetic induction) (T)} \\
H &: \text{ Magnetic field strength (A/m)} \\
M &: \text{ Magnetization (magnetic moment per unit volume (A/m)} \\
\mu_0 &: \text{ Vacuum magnetic permeability } (\mathrm{T \cdot m \cdot A^{-1}})
\end{aligned}$$

The Meissner effect was first described by the London theory. Essentially, an external magnetic field induces shielding currents at the surface of a superconductor when cooled below T_c. This current expels the outer magnetic field from the inside of a superconductor except for a finite depth. This depth is termed London penetration depth (λ_L) and describes the depth at which the magnetic field drops to $1/e$ times its value. The Ginzburg-Landau (GL) theory was a further development of the London theory and introduced an order parameter (ψ), which describes the system during a phase transition. Within the GL theory, two characteristic length parameters are introduced [48]:

1. **GL penetration depth** (λ): characterizes the depth to which a magnetic field penetrates a superconductor within GL theory.
2. **GL coherence length** (ξ): describes the size of the region over which the superconducting order parameter varies within GL theory.

A new parameter is defined as the ratio of these two characteristic lengths [48, p. 13]:

$$\kappa = \frac{\lambda}{\xi} \tag{2.66}$$

$$\begin{aligned}
\kappa &: \text{ Ginzburg-Landau parameter (dimensionless)} \\
\lambda &: \text{ Ginzburg-Landau penetration depth (m)} \\
\xi &: \text{ Ginzburg-Landau coherence length (m)}
\end{aligned}$$

Based on the GL parameter, there are two types of superconductors.

2.7.1. Type-I Superconductor

A type-I superconductor is defined by the condition [48, p. 14]:

$$\kappa < \frac{1}{\sqrt{2}} \tag{2.67}$$

Below T_c, an external magnetic field is expelled from the superconductor (Meissner state) until a critical field (H_c) is reached. Then, the superconductor transitions into the normal conducting state. [48]

$$\begin{aligned}
H < H_c &: \text{ Meissner state} \\
H > H_c &: \text{ Normal conducting state}
\end{aligned}$$

2.7.2. Type-II Superconductor

The second type of superconductor is defined by [48, p. 15]:

$$\kappa > \frac{1}{\sqrt{2}} \tag{2.68}$$

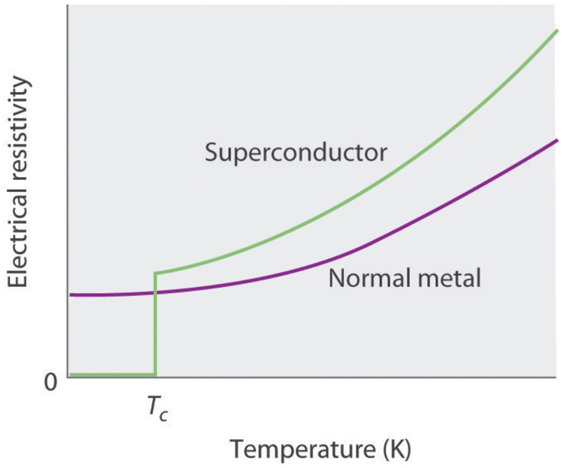
Here, there are two critical fields: The lower (H_{c_1}) and the upper critical field (H_{c_2}). If the external magnetic field is lower than H_{c_1}, the magnetic field is expelled completely from the superconductor except for a small layer characterized by λ. This is depicted in Figure 2.10b as (a). If the external field is above H_{c_1} but below H_{c_2}, the superconductor transitions into a mixed phase, also called Shubnikov state. The superconductor is partially penetrated by flux lines in the form of vortices. The magnetic flux is quantized and the number of flux lines is proportional to the average magnetic field. In Figure 2.10b this is marked as (b). Above H_{c_2}, the superconductor transitions into the normal conducting state. [47]

With the discovery of type II superconductors, their application became widely relevant for technical purposes, as they also exhibit higher critical current densities and can therefore generate strong magnetic fields. One example is magnetic resonance imaging, which uses high magnetic fields to image body parts.

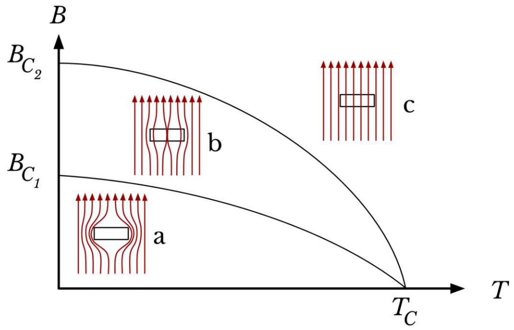
$$\begin{aligned} H < H_{c_1} &: \text{ Meissner state} \\ H > H_{c_2} &: \text{ Mixed state} \\ H > H_{c_2} &: \text{ Normal conducting state} \end{aligned}$$

2.7.3. High Temperature Superconductors

Up until 1986, superconductivity was only achievable with expensive and complex liquid helium cooling, as the only known superconductors were LTS (e.g., Nb-Ti, Nb_3Sn). LTS are defined by $T_c < 30\,K$. One of the few liquids which has a boiling point below 30 K at atmospheric pressure is helium at 4.2 K. In 1986, the forerunner of HTS was discovered during experiments with new compounds of copper oxides, termed cuprates. The compound $La_{2-x}Ba_xCuO_4$ was the first cuprate discovered, exhibiting potential for high T_c properties. With $YBa_2Cu_3O_{7-\delta}$ or yttrium barium copper oxide (YBCO), the first HTS was discovered with a critical temperature of approximately 93 K. HTS are defined by the boiling point of nitrogen at atmospheric



(a) Electrical resistivity of a superconductor and a normal metal [49, Fig. 12.28]



(b) Superconductor interactions with magnetic field [50]

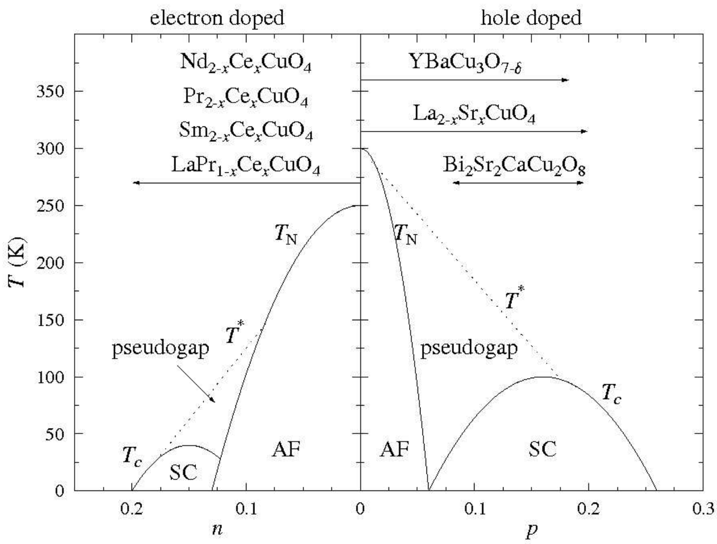
Figure 2.10.: (a) Temperature dependence of the electrical resistivity of a normal metal and a superconductor; (b) Type-II superconductor magnetic field interactions.

pressure: $T_c > 77\,K$. The possibility of cooling by liquid nitrogen enables a cheaper and easier achievement of the superconducting state. [48]

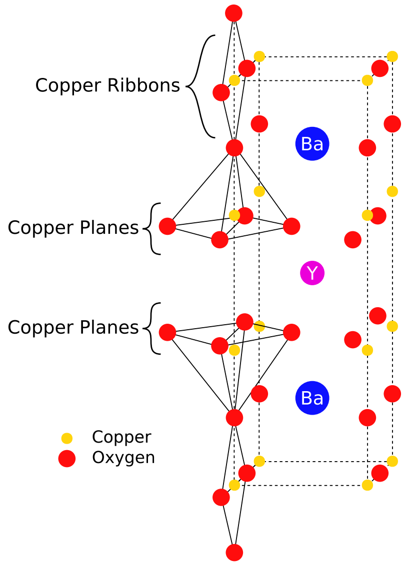
The family of barium copper oxides in combination with rare earth elements such as yttrium, lanthanum, or gadolinium is referred to as REBCO. These compounds are classed as HTS and exhibit both high T_c and H_{c_2}. For experiments conducted as part of this thesis, the SuperPower (SP) SCS4050 2009 superconductor, referred to as SP SCS09, was used. This superconductor is a layered thin-film second-generation (2G) HTS, which features a large number of intrinsic pinning centers, exhibiting high pristine T_c and j_c.

The exact layer structure is described in subsection 3.2.3 and depicted in Figure 3.12. This superconductor features $GdBa_2Cu_3O_{7-\delta}$ or GdBCO as the superconducting layer. GdBCO has a $T_c \approx 92 - 95\,K$, which is slightly higher than YBCO's.

As described in [51], GdBCO coated conductors exhibit superior magnetic field dependence of critical current density compared to YBCO, leading to a higher magnetic field caused by coils made from GdBCO. In REBCO, layers of rare earth elements are sandwiched between two pyramid-shaped CuO_2 in the c-axis direction (see Figure 2.11b). In the same direction, rare earth atoms are periodically alternating with Ba atoms. In the b-axis direction, there are one-dimensional Cu-O-Cu chains. Undoped cuprates are Mott-insulators. By doping them with electrons or holes, expressed by x in the index of chemical formulas, additional charge carriers are introduced into the CuO_2 layers. This is believed to cause their superconducting properties below T_c. Oxygen deficiency is expressed by δ. In Figure 2.11a, a phase diagram of different cuprates is depicted, showing anti-ferromagnetic and superconducting properties depending on the electron or hole doping level, as well as the temperature. [48]



(a) Cuprate phase diagram [52]



(b) Crystal structure of YBCO [53]

Figure 2.11.: (a) Phase diagram of cuprate superconductors; (b) Crystal structure of $YBa_2Cu_3O_{7-\delta}$

2.7.4. Radiation Damage in REBCO Superconductors

In a fusion reactor, irradiation damage caused by neutrons is the primary source of superconductor degradation. The damage pattern resulting from neutrons depends on their energy:

1. **Fast neutrons**: High energy collisions, inducing collision cascades. Both small and big defects are introduced into the REBCO layer [46].
2. **Thermal neutrons**: $n - \gamma$ capture reactions, causing point-like defects, clusters consisting of Oxygen Frenkel pairs and more complex defects [54].

As described in Equation 2.63, the neutrons emitted from deuterium-tritium fusion have a kinetic energy of 14.1 MeV. In a fusion reactor, several layers of components are situated between the fusion plasma and the magnets, which are composed of superconductors. These layers include the blanket, vacuum vessel, insulator, and stabilizer. Elastic collisions with atoms in these layers lead to a decrease in neutron flux density with increasing distance from the plasma and result in moderation of the neutrons. Consequently, the superconductor is irradiated with neutrons featuring a continuous energy distribution. [55]

This continuous neutron flux distribution is similar to the flux spectrum of a TRIGA reactor, except for the presence of thermal neutrons. The thermal flux at the magnets in a fusion device is negligible. To achieve similar irradiation conditions with a TRIGA reactor, the superconductor samples must be shielded from thermal neutrons by covering them with cadmium.

Fast Neutron Damage

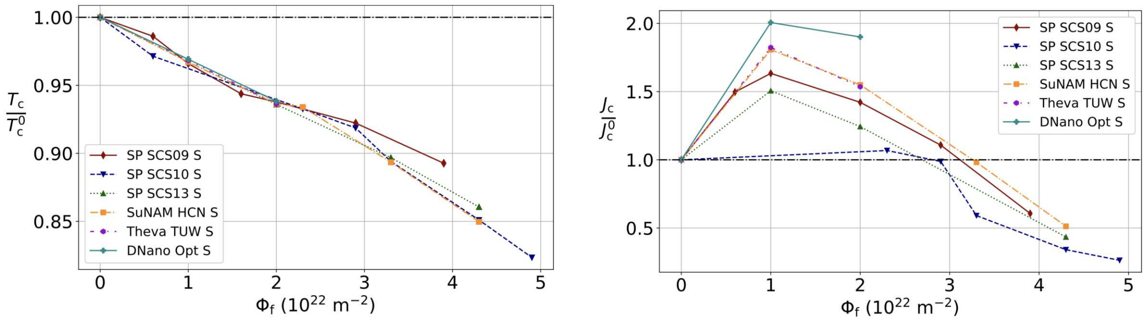
For REBCO coated conductors exposed to fast neutrons, T_c decreases monotonically with increasing fast neutron fluence (Φ_f, see Figure 2.12a). In contrast to this, the critical current exhibits non-monotonic behavior as a function of neutron fluence (see Figure 2.12b). At first, j_c improves with fluence, reaching its maximum at around $\Phi_f \approx 1 \cdot 10^{22}\ \mathrm{m}^{-2}$. This can be explained by the presence of large defects introduced in the REBCO layer, which are of the same order of magnitude as the coherence length. The mean diameter of these defects were determined by previous studies at around 2.5 nm, fitting the coherence length of YBCO: $\xi \approx 1.4$ nm, improving flux pinning. [54] Then, j_c deteriorates with increasing fluence.

The maximum lifetime fluence, also termed crossover fluence, is reached at $\Phi_f \approx 3 - 3.3 \cdot 10^{22}\ \mathrm{m}^{-2}$. Linden, Iliffe, He, *et al.* estimate that large cascade defects only occupy 0.01 % of the superconducting layer [56, p. 8]. It is therefore suspected, that the degradation of j_c is caused by smaller defects, which are not directly visible in conventional scanning transmission electron microscopy images [54]. The improving pinning competes with the loss of superfluid density, which is the universal cause of degradation. [46]

The suspected reason lies in the d-wave symmetry of the order parameter of REBCO HTS. The impurities in form of defects cause pair-breaking and finally the loss of superfluid density. Generally, the loss of superconducting properties depends not only on the number of displaced atoms, but is highly dependent on the type and size of the introduced defects. In contrast to the d-wave symmetry of the order parameter in HTS, LTS feature s-wave symmetry. For s-wave symmetry, introduced defects do not lead to pair-breaking. LTS are therefore more resistant to neutron irradiation. [54]

Thermal Neutron Damage

When unshielded GdBCO is irradiated in a thermal reactor, it is exposed to thermal neutrons. As gadolinium has an extremely high (n,γ)-absorption cross-section, two main reactions, depicted in Table 2.8, are induced. The mother nucleus absorbs a neutron and transitions into an excited state. As the daughter nucleus relaxes by emitting γ-rays, it is knocked out of the lattice with a recoil energy of



(a) Normalized critical temperature

(b) Normalized critical current at 15 T and 30 K

Figure 2.12.: Dependence of normalized T_c and j_c on fast neutron fluence. Source: [54, Fig. 1]

29 to 34 eV, causing a high density of point-like defects and small clusters of few displaced atoms [2], [54]. It was found that T_c and j_c of irradiated, unshielded GdBCO samples degraded 14 to 15 times faster than shielded samples. This effect can be seen in Figure 2.14, which compares the normalized T_c degradation of different shielded (denoted with "S") and unshielded ("U") HTS. On the right, the normalized j_c degradation of the shielded and unshielded SP SCS2009 at 15 T and 30 K is compared.

It should be noted that the faster degradation of GdBCO by thermal neutrons is not directly relevant for the application of HTS in a fusion reactor. This is because the neutron spectrum differs fundamentally between a thermal reactor and a fusion device. In the latter, there are few to no thermal neutrons present. If the samples are irradiated in a thermal reactor, they should be shielded with cadmium foil if a neutron spectrum similar to that of a fusion reactor is desired.

Reaction	$\sigma_{a,th}$ (barn)	Daughter Nucleus	Recoil Energy
$^{155}Gd + n \rightarrow {}^{156}Gd + \gamma$	60 797.69	^{156}Gd	29 eV
$^{157}Gd + n \rightarrow {}^{158}Gd + \gamma$	253 913.3	^{158}Gd	34 eV

Table 2.8.: Neutron capture (n, γ) reactions of ^{155}Gd and ^{157}Gd, showing their thermal neutron absorption cross section [25], the daughter nucleus, and the calculated recoil energy [57, p. 46]

Recovery of Superconducting Properties

REBCO have lower irradiation robustness compared to, e.g., Nb_3Sn, due to the pair-breaking effect of defects on the d-wave symmetry of the order parameter. As REBCO are the most promising materials for high-field magnets necessary in compact fusion reactor designs, the degradation by neutron irradiation drastically

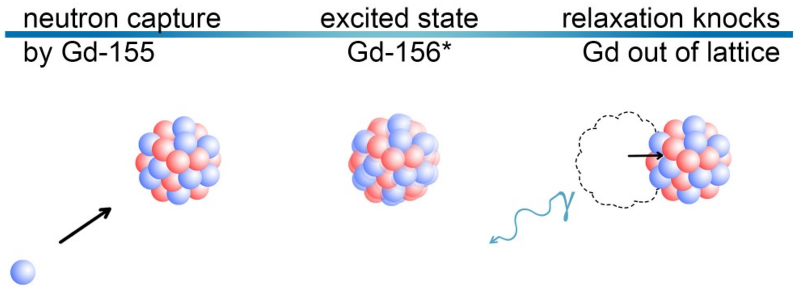
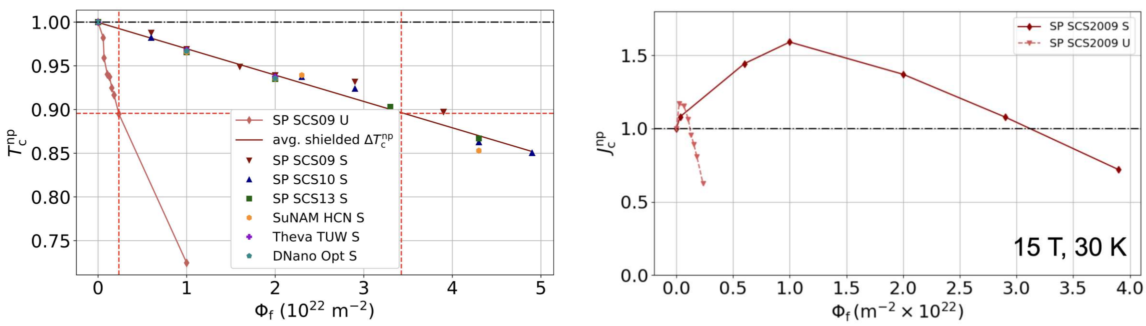


Figure 2.13.: Illustration of the neutron capture reaction by ^{155}Gd, forming an excited ^{156}Gd* nucleus. The relaxation process can knock the Gd atom out of the lattice, causing point-like defects in the superconductor material. [46, p. 20]



(a) Normalized critical temperature [54, Fig. 4]

(b) Normalized critical current at 15 T and 30 K [46, p. 30]

Figure 2.14.: Dependence of normalized T_c and j_c on neutron fluence. Sources: [54, Fig. 4], [46, p. 30]. Note: "np" denotes values normalized to the pristine (unirradiated) state

limits the lifetime of the magnets and jeopardizes the economic viability of these reactor designs. Previous work conducted by Unterrainer, Fischer, Lorenz, *et al.* demonstrated that by thermally treating irradiated REBCO HTS, T_c recovers linearly with annealing temperature, while j_c recovers non-monotonically [1]. By annealing the samples at 150 °C in ambient argon atmosphere, a T_c recovery of 25 % was achieved. Starting at around 160 °C, the oxygen in the REBCO layer becomes increasingly more mobile. Annealing at temperatures above this value requires therefore annealing in pure O_2 environment. A recovery of 60 % was reported at 400 °C in pure ambient O_2 atmosphere. [1]

As deduced by Unterrainer, the degradation of superconducting properties is universal. For different irradiation conditions, including protons, the full neutron spectrum (unshielded) as well as only epithermal and fast neutrons (shielded), there is a strong indication that small defects are the underlying cause for the loss of superfluid density and the resulting degradation. Moreover, Unterrainer notes that defect annealing happens at the same ratio across different samples, leading to the same linear recovery of T_c [2].

As described by Homes' law, the superfluid density is directly proportional to both the normal state direct-current resistivity and T_c [58]:

$$\rho_s \propto \sigma_{dc} \cdot T_c = \frac{1}{\rho_n} \cdot T_c \tag{2.69}$$

where $\rho_s \propto \frac{1}{\lambda^2}$: superfluid density of Cooper pairs (m^{-3})

σ_{dc} : normal state direct-current (dc) resistivity close to T_c ($S \cdot m^{-1}$)

ρ_n : normal-state resistivity close to T_c ($\Omega \cdot m$)

T_c : critical temperature (K)

This demonstrates that T_c is an excellent parameter quantifying the defect density introduced into the superconducting layer, influencing ρ_s. It also establishes a connection between the superfluid density and the normal-state resistivity. The defects that lead to a loss of ρ_s also influence the normal-state resistivity. This can be described by the Matthiessen's rule, which states that the total resistivity (ρ) of a normal state conductor is the sum of the individual resistivities (ρ_i) caused by the independent mechanisms [59, p. 400]:

$$\rho = \sum_i \rho_i = \rho_{\text{Ph-Ph}}(T) + \rho_{\text{e-e}}(T) + \rho_{\text{defects}} + \dots \tag{2.70}$$

$$\begin{aligned}
\text{where} \quad \rho &: \text{total resistivity } (\Omega\cdot\text{m}) \\
\rho_{\text{Ph-Ph}}(T) &: \text{resistivity due to phonon-phonon scattering } (\Omega\cdot\text{m}) \\
\rho_{\text{e-e}}(T) &: \text{resistivity due to electron-electron scattering } (\Omega\cdot\text{m}) \\
\rho_{\text{defects}} &: \text{resistivity due to defects and impurities } (\Omega\cdot\text{m}) \\
\ldots &: \text{other possible contributions}
\end{aligned}$$

It is therefore expected that changes in the defect density can be detected by measuring the resistivity or resistance of REBCO samples during thermal annealing. By comparing the temperatures at which these changes occur to literature data on activation energies, the underlying physical processes responsible for the recovery of T_c can be identified.

2.7.5. Critical Temperature Measurement

The Python script determines two characteristic critical temperatures (T_c and T_c^{high}) using the tangent intersection method (see Figure 2.15):

1. **Data Smoothing and Derivative**: First, the voltage data V(T) is smoothed and the first derivative is computed:

$$S(T) = \text{sgn}(V(T)) \cdot |V(T)| \cdot 10^6 \tag{2.71}$$

$$\begin{aligned}
S(T) &: \text{ scaled and sign-preserved voltage signal at temperature } T \ (\mu\text{V}) \\
V(T) &: \text{ measured voltage at temperature } T \ (\text{V}) \\
\text{sgn}(V) &: \text{ sign function, returns } +1 \text{ for } V > 0, \ -1 \text{ for } V < 0
\end{aligned}$$

$$\frac{dS}{dT} = \text{Savitzky-Golay}(S(T)) \tag{2.72}$$

$$\begin{aligned}
\frac{dS}{dT} &: \text{ smoothed numerical derivative of } S(T) \text{ with respect to } T \ (\mu\text{V/K}) \\
\text{Savitzky-Golay} &: \text{ digital filter used for smoothing and derivative estimation}
\end{aligned}$$

2. **Region Selection**: As described in [2], fits are applied to two regions of the measurement. In order to separate these regions, the maximum of the derivation is located: $k_{\text{max}} = \max\left|\frac{dS}{dT}\right|$. Furthermore, $T_{\text{max. grad}}$ is the temperature at which this maximum occurs. The two regions are defined by:

- **Transition region**: Points where $\left|\frac{dS}{dT}\right| > 0.6 \cdot k_{\text{max}}$ and $T \leq T_{\text{max. grad}}$
- **Ohmic (resistive) region**: This is the region above the transition points where $\left|\frac{dS}{dT}\right| < 0.3 \cdot k_{\text{max}}$ and $T > T_{\text{max. grad}}$.

3. **Linear Fits**: Fit straight lines to both regions:

$$S_{\text{trans}}(T) = a_{\text{trans}}T + b_{\text{trans}} \tag{2.73}$$

$$S_{\text{ohmic}}(T) = a_{\text{ohmic}}T + b_{\text{ohmic}} \tag{2.74}$$

$$\begin{aligned} a_{\text{trans}}, a_{\text{ohmic}} &: \text{ slopes of the fits } (\mu\text{V}/\text{K}) \\ b_{\text{trans}}, b_{\text{ohmic}} &: \text{ intercepts of the fits } (\mu\text{V}) \end{aligned}$$

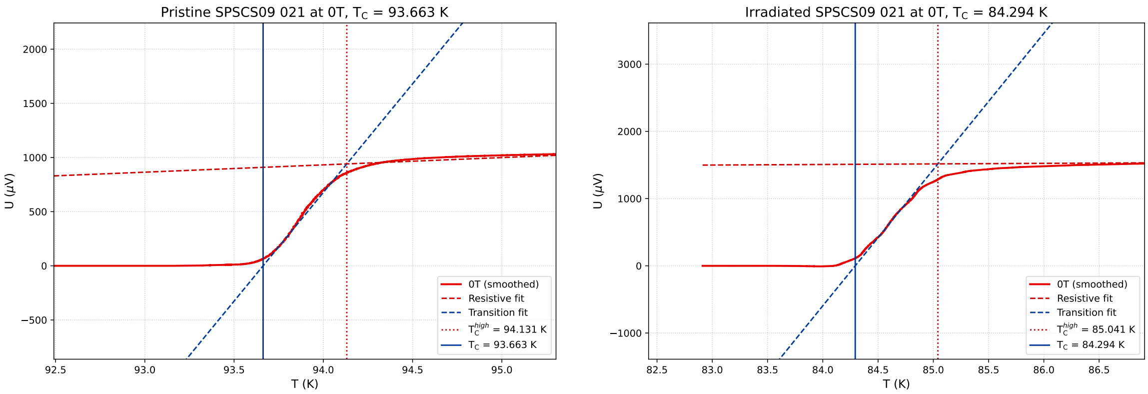
4. **Intersection and x-Axis Intercept**:
   - **High critical temperature** (T_c^{high}): Intersection of the transition and resistive linear fits:
   $$T_c^{\text{high}} = \frac{b_{\text{ohmic}} - b_{\text{trans}}}{a_{\text{trans}} - a_{\text{ohmic}}} \tag{2.75}$$
   - **Low critical temperature** (T_c): x-Axis intercept of the transition fit:
   $$T_c = -\frac{b_{\text{trans}}}{a_{\text{trans}}} \tag{2.76}$$

$$\begin{aligned} T_c^{\text{high}} &: \text{ temperature at intersection of resistive and transition fits (K)} \\ T_c &: \text{ x-intercept of the transition fit (K)} \end{aligned}$$

2.8. Resistivity Measurement

2.8.1. Four-Wire Method

For resistance measurements, the two-wire measurement is a possible measurement technique. Here, only a single test lead is used for applying both the current and measuring the voltage drop across the determining resistance. The main issue with this configuration is that the test current induces a voltage drop in the lead connections. This total lead resistance is added to the determining resistance. Therefore, the measured voltage drop (V_M) differs from the voltage drop across the sample (V_R). For low-resistance measurements, the four-wire (Kelvin) method



(a) SP SCS09 021 (Pristine)

(b) SP SCS09 021 (Irradiated)

Figure 2.15.: T_c of sample 021 (pristine and irradiated)

is the preferred measurement configuration. It uses two sets of test leads. With the first set of test leads, the current is forced through the resistance, while the other set of leads, also called sense leads, measures the voltage drop. This configuration is depicted in Figure 2.16a. As the voltage drop across the sense leads is insignificant, V_M is essentially equal to V_R. In this thesis, the four-wire (Kelvin) method is realized by introducing four probes. It is therefore also termed four-point probe measurement.

2.8.2. Thermoelectric Voltages

For low-level voltage measurements, thermoelectric voltages present one of the biggest error sources. These voltages, also termed thermoelectric electromotive force (EMF), occur when different sections of a circuit are at different temperatures and different materials form a junction. [60] The underlying process is described by the Seebeck effect. There are different methods to avoid thermoelectric voltages impeding the low resistance measurement accuracy. As described in [60], these three approaches are used frequently:

- Current-reversal method
- Delta Method
- Offset-compensated ohms method

Current-reversal Method

The current-reversal method was used for our experiments. Therefore, it will be explained in more detail in the following section. By applying opposite current polarities, we measure two different voltage values [60, p. 3-20]:

$$V_{\text{M+}} = V_{\text{EMF}} + I \cdot R \tag{2.77}$$

$$V_{\text{M-}} = V_{\text{EMF}} - I \cdot R \tag{2.78}$$

By combining the two measured values, the thermoelectric voltage cancels out and we receive the correct voltage value [60, p. 3-22]:

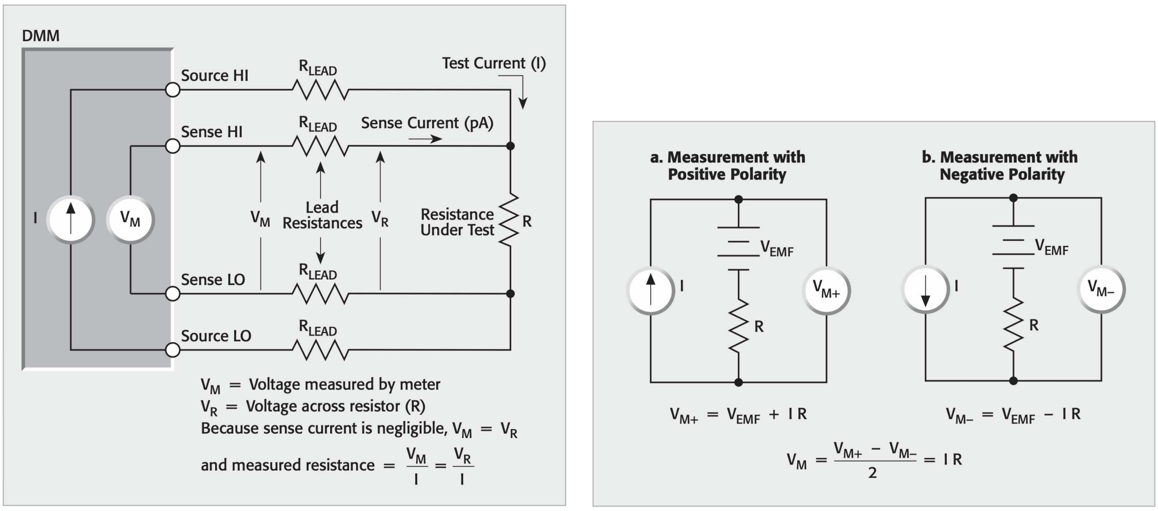
$$V_{\text{M}} = \frac{V_{\text{M+}} - V_{\text{M-}}}{2} = \frac{(V_{\text{EMF}} + I \cdot R) - (V_{\text{EMF}} - I \cdot R)}{2} = I \cdot R \tag{2.79}$$

The resistance can easily be calculated with Ohm's law [60, p. 3-22]:

$$R = \frac{V_{\text{M}}}{I} \tag{2.80}$$

where V_{EMF} : thermoelectric voltage (V)
V_{M} : voltage measured by meter (V)
I : current (A)
R : resistance (Ω)

The schematics for the positive and negative polarity of the current reversal method is shown in Figure 2.16b.



(a) Four-Wire Resistance Measurement [60, Fig. 3-17]

(b) Current reversal method, canceling EMFs. [60, Fig. 3-18]

Figure 2.16.: Illustrations of the four-wire resistance measurement and current reversal method [60]

2.8.3. Calculation of Resistivity

The GdBCO layer of SP 2G HTS has an approximate thickness of 1 µm [61, p. 13]. The bridge has a top area of approximately 4 mm × 1 mm. The resistivity of the superconductor can be calculated with:

$$\rho = R \cdot \frac{A}{l} = R \cdot C \tag{2.81}$$

$$\begin{aligned} \text{where} \quad R &: \text{electrical resistance of a uniform specimen } (\Omega) \\ l &: \text{length of the specimen (m)} \\ A &: \text{cross-sectional area of the specimen } (\mathrm{m}^2) \\ C = \frac{A}{l} &: \text{geometric factor (m)} \end{aligned}$$

The geometric factor is defined as the ratio of the cross-sectional area to the length of our REBCO layer and can be approximated for our SP samples with:

$$C = \frac{A}{l} \approx 4 \times 10^{-6}\ \mathrm{m} \tag{2.82}$$

For the experimental part, it was decided to compute and plot the resistance rather than the resistivity, as the focus was on the qualitative shape of the graphs rather than the exact resistivity values. However, by measuring the dimensions of the individual etched bridges under the microscope and applying Equation 2.81, the resistivity can be accurately calculated and plotted for each REBCO sample.

3. Experimental Methods and Measurements

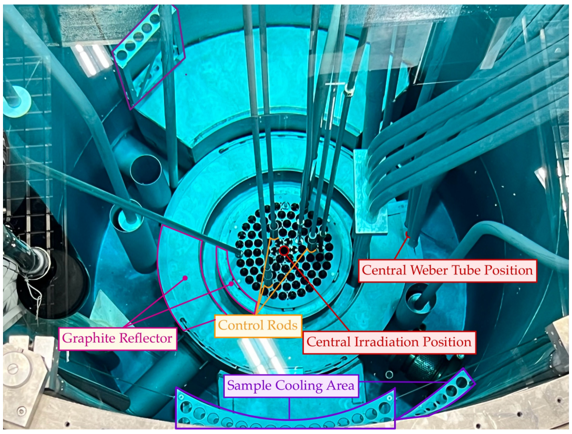
3.1. Neutron Flux Measurement

The TRIGA Mark II reactor at the Atominstitut, TU Wien, is utilized for the irradiation and subsequent analysis of superconductors, with a particular focus on applications in fusion devices. As the neutron field of the TRIGA Mark II reactor differs from a fusion reactor, an accurate determination of the thermal and fast neutron flux is essential to evaluate the significance of the observations made for irradiated superconductors and assess the limitations of these observations.

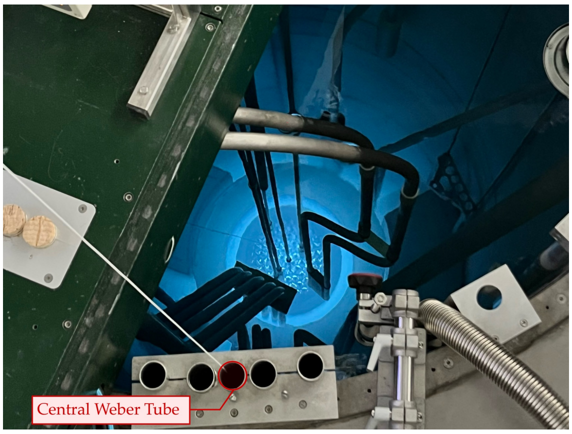
The fast and thermal neutron flux was determined at the CIP, which is located at the central position inside the core, as well as at the central Weber tube position, which is located outside both the core and graphite reflector. These two positions are depicted in Figure 3.1a and Figure 3.1b, while reactor cross-sections for the thermal and fast flux measurements are shown in Figure 3.1c and Figure 3.1d.

Based on the extensive theoretical knowledge compiled and described in section 2.1, the experimental part of the flux measurement can be divided into several steps:

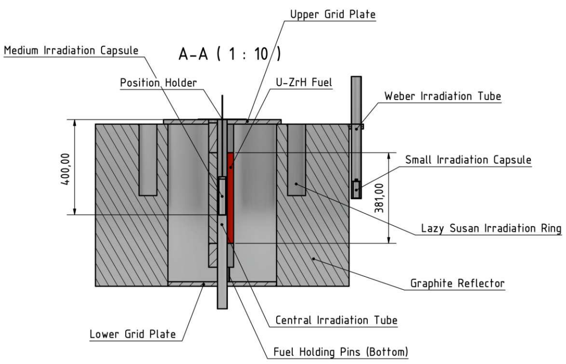
1. Activation and dose rate calculation
2. Development of radiation protection concept
3. Sample selection, preparation and weighing
4. Sample encapsulation in quartz glass tubes
5. Irradiation of the samples at the appropriate irradiation position
6. Unpacking of encapsulated and irradiated samples after adequate t_{cool}
7. Re-weighing of the samples
8. Sample measurement with γ-spectroscopy
9. Calculation of the thermal and fast neutron flux based on the gathered data
10. Clearance measurement of irradiated samples after decay and return of samples



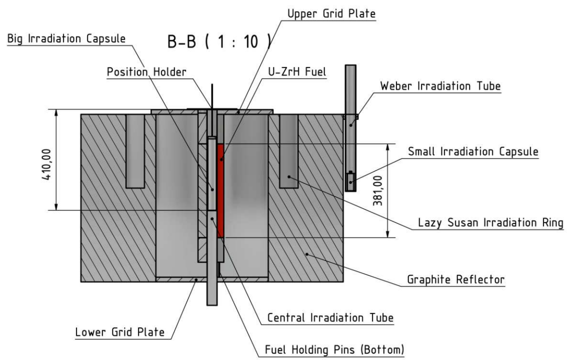
(a) Photograph of the reactor core overview



(b) Photograph of the reactor in operation



(c) Reactor cross-section showing medium irradiation capsule configuration for the thermal flux measurement



(d) Reactor cross-section showing large irradiation capsule configuration for the fast flux measurement

Figure 3.1.: Top: Photographs showing the reactor core overview and the reactor in operation. Bottom: Cross-sectional diagrams of the reactor core showing irradiation capsule configurations for thermal and fast neutron flux measurements. The diagrams indicate the central irradiation and Weber tube positions.

3.1.1. Radiation Protection

In the first step, the mass of the activation probes had to be determined. The activation calculations were done with the online tool *Activation Calculator V2*[1] by the Technical University of Munich [62]. This calculator utilizes the Python script periodictable[2] and enables the activation calculation of different isotopes by the fast and thermal fluence of neutrons.

As the flux monitors were to be irradiated at a reactor power of 250 kW and their activities measured using low-level γ-spectroscopy, with sensitivities below one becquerel, their masses were chosen to be in the milligram range. This decision was initially justified from a radiation protection standpoint. However, it later proved impractical, as the small size of the flux monitors made them difficult to locate during unpacking after irradiation, increasing the risk of loss and contamination of the workspace.

In the subsequent step, reliable calculations of the radiation dose rates for all activated isotopes were required. A Python code was developed to compute the γ-dose rate. This script calculates the remaining activity after an arbitrary time, which can be defined by the user, according to Equation 2.8. Additionally, the γ-dose rate at a user-specified distance is calculated according to Equation 2.11.

The beta dose rates of all the relevant nuclides were estimated by taking the nuclide activity computed with the online tool [62] and then calculating the beta dose rate based on the values presented in [63]. According to the data sheet for ^{198}Au [63, p. 136] (see Figure 3.2), the external exposure values for a 1 MBq point source at 30 cm are $0.124\,\mathrm{mSv\,h^{-1}}$ for beta-radiation and $0.764\,\mathrm{\mu Sv\,h^{-1}}$ for γ- and X-ray radiation. The beta dose corresponds to the superficial or skin dose equivalent, while the gamma dose represents deep or whole-body dose equivalent due to X- and γ-ray components [63, p. 11].

3.1.2. Sample Preparation and Encapsulation

In the next step, the samples were selected. To optimize the use of existing resources, available material was utilized for this experiment. This material existed from previous flux measurements. Indium and gold were available in foil form and were cut into small pieces measuring a few millimeters in length using a clean scalpel. Nickel was available as wire and was cut to size to achieve the desired mass. The wire was then bent into a concentric spiral to achieve a sample geometry as point-like as possible. Titanium was already in disc form with a diameter of

[1] `https://webapps.frm2.tum.de/activation/`
[2] `https://periodictable.readthedocs.io/en/latest/`

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Gold - 198

$^{198}Au_{79}$

Half life: 2.7 days
Specific activity: 9.03E+15 Bq.g^{-1}

Risk group: 3
Risk colour: Yellow

Main emissions (keV)	Gamma or X		Beta (Emax)		Electrons		Alpha	
	E	%	E	%	E	%	E	%
E1	412	96	285	1	8	2		
E2	676	<1	961	99	329	3		
E3	1088	<1			397	1		
% omitted		<1		0		<1		

Exemption levels	
Quantity (Bq)	1E+06
Concentration (Bq.g^{-1})	1E+02

Transport (TBq)	
IAEA ST1 A_1 value	1E+0
IAEA ST1 A_2 value	6E-1

EXTERNAL EXPOSURE (mSv.h^{-1}) for an activity of 1 MBq or 1 MBq.m^{-2} (as appropriate)

Point source (30 cm)

Betas, electrons (skin dose): 1.24E-1

Gammas, X rays (deep tissue dose): 7.64E-4

Infinite plane source

Betas, electrons (skin)	
10 cm	1.1E-01
1 m	8.2E-03
Photons (skin)	
10 cm	2.8E-03
1 m	1.8E-03
Photons (deep dose)	
10 cm	2.7E-03
1 m	1.7E-03

10 ml glass vial (100 cm): 6.69E-5

Contact with 50 ml glass beaker: 2.36E-1

Contact with 5 ml plastic syringe: 3.63E+0

The values above do not include Bremsstrahlung radiation.

CONTAMINATION

Contamination skin dose (mSv.h^{-1})	
Uniform deposit (1kBq.cm^{-2})	1.68E+0
0.05 ml droplet (1 kBq)	9.53E-1

Uniform deposit

Droplet

Detection — Recommended probes*	
Alpha	
Beta	+ +
Gamma	+ +
X rays	+

Derived limits (Bq.cm^{-2})
Removable contamination
4E+1
Fixed contamination
7E+1

* If no probes are indicated the recommended technique is to use a wipe test in association with a probe or liquid scintillation technique

SHIELDING (mm)

Betas and electrons (Total absorption)	
Glass	1.6
Plastic	3.0

Gamma and X rays (half and tenth value thickness)	½	1/10
Lead	4	12
Steel	24	58

INTERNAL EXPOSURE FOR WORKERS

COMMITTED EFFECTIVE DOSE PER UNIT INTAKE (Sv.Bq^{-1})

Ingestion	f_1	
All compounds	0.100	1.0E-09

Inhalation		1 µm	5 µm
All unspec. compounds	F	2.3E-10	3.9E-10
Halides & nitrat.	M	7.6E-10	9.8E-10
Oxid.& hydrox.	S	8.4E-10	1.1E-09

Highest dose organ: Lower large intestine — **20 mSv ALI**$_{ingestion}$: 2.0E+07 (Bq) — **20 mSv ALI**$_{inhalation}$: 1.8E+07 (Bq)

MAXIMUM RECOMMENDED ACTIVITIES IN LOW LEVEL OR INTERMEDIATE LEVEL LABORATORIES (Bq)

Subject to external exposure requirements which may be more restrictive

PHYSICOCHEMICAL STATE	Volatility factor (k)	Supervised area		Controlled area		
		Bench	Fume hood	Bench	Fume hood	Glove box
All compounds	0.01	5E+05	5E+06	2E+06	2E+07	2E+09

Figure 3.2.: Overview of different dose rate scenarios for ^{198}Au. The beta dose rate for our ^{198}Au point source was estimated using the data highlighted with two red boxes. [63, p. 136].

approximately 5 mm and the desired mass. The activation probes were then cleaned with isopropyl alcohol and acetone.

Then, the neutron flux monitors were welded into the quartz glass tubes under normal atmosphere, using both gas and acetylene burners. The latter proved to be challenging because of the extremely hot flame of acetylene. The tubes generally had to be completely sealed in order to prevent leakage of the reactor coolant water into the samples. The sealing was checked by submerging the tubes in a bucket of water. The gas welding process is depicted in Figure 3.3a, while the finished tubes with the activation probes can be seen in Figure 3.3b.

(a) Gas welding process of quartz glass tubes

(b) Finished welded quartz glass tubes

Figure 3.3.: The welding process of quartz glass tubes and the resulting welded quartz glass tubes

3.1.3. Irradiation of Samples

For irradiation in the central Weber tube position, the different elements as well as the shielded and unshielded thermal flux monitors were irradiated subsequently in the small irradiation capsules, as this irradiation position is generally unused and there was no time constraint. For the CIP, the available irradiation time was extremely limited. This lead to the requirement of irradiating both the shielded and unshielded samples together. For this, a quartz glass rod with a length of 4.5 to 5 cm was placed in the quartz glass tubes as a spacer, separating the shielded and unshielded activation probes and therefore preventing any influence of the cadmium on the unshielded flux monitor.

Two capsules were used for the irradiation at the CIP: The medium capsule (see Figure B.13) was used for the irradiation of unshielded and shielded gold, while two quartz glass tubes were placed in the big capsule (see Figure B.9), irradiating both unshielded and shielded indium, as well as titanium. Generally, to every independent irradiation experiment, a nickel monitor was added to ensure comparability and the possibility of normalization of the different flux values.

In the top row of Figure 3.1, an overview of the reactor with its most important components as well as the reactor in operation can be seen. The distinctive blue glow emitted during operation is caused by Cherenkov radiation. In the bottom row of Figure 3.1c, cross-sectional diagrams of the reactor core are depicted for the irradiation configuration of both the medium and big irradiation capsules. Schematics of the TRIGA reactor for both the thermal and fast irradiation configurations, as well as the technical drawings of the irradiation and quartz glass capsules, are presented in section B.2.

3.1.4. Sample Unpacking Procedure

After adequate cooling time in the sample cooling area (see Figure 3.1a), the activation probes were subsequently taken out of the reactor pool and transported to the radiochemistry department. There, the samples were unpacked in a fume hood, which also features lead shielding. The unpacking procedure consisted of multiple steps (see Figure 3.4): First, the irradiation capsules had to be opened. This was either done with distance tools for the small capsules, as depicted in Figure 3.4a, or by hand for the medium and large capsules. A finger dosimeter, which is a passive thermoluminescence dosimeter, was used to more accurately measure the radiation dose received when handling the irradiated capsules by hand.

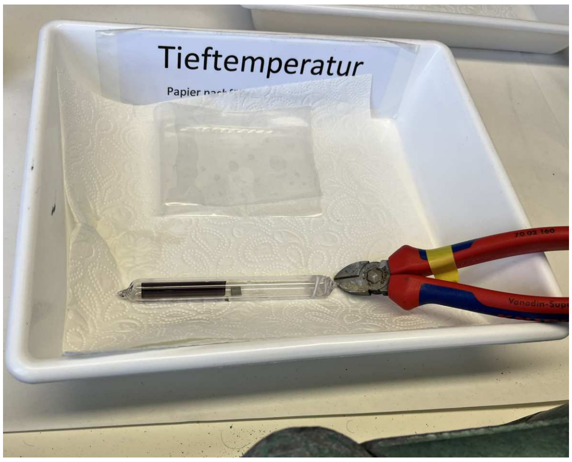
After opening the capsules, the quartz glass tubes containing the activation probes were placed in a plastic bag and sealed. The glass tubes were broken with pliers (see Figure 3.4b). Afterwards, the activation probes had to be visually located. This proved to be difficult, as the activation probes were extremely small and difficult to differentiate from the shards. This also led to the loss of one nickel monitor.

The waste was discarded in the appropriate waste category, which differentiated burnable and non-burnable radioactive waste. After this, the activation probes were weighed. The masses of the activation probes for the Weber tube and CIP are presented in Table 3.2 and Table 3.1.

Element	Mass (mg)	Ni-Mass (mg)
Gold	0.91 ± 0.01	2.13 ± 0.01
Gold (shielded)	0.73 ± 0.01	"
Indium	1.37 ± 0.01	2.33 ± 0.01
Indium (shielded)	1.80 ± 0.01	"
Titanium	2.06 ± 0.01	"

Table 3.1.: Mass of the activation probes irradiated at the central irradiation position.

(a) Unpacking of the samples using distance tools



(b) Preparation for opening the quartz glass tube

Figure 3.4.: Unpacking procedure of the activation probes: (a) opening the irradiation capsules with distance tools, (b) preparation of the tools for breaking the quartz glass tube.

Element	Mass (mg)	Ni-Mass (mg)
Gold	1.94 ± 0.01	2.08 ± 0.01
Gold (shielded)	1.83 ± 0.01	3.81 ± 0.01
Indium	15.09 ± 0.01	4.53 ± 0.01
Indium (shielded)	14.97 ± 0.01	lost
Titanium	2.04 ± 0.01	1.37 ± 0.01

Table 3.2.: Mass of the activation probes irradiated at the Weber tube

3.1.5. Effective Irradiation Time

The effective irradiation times was calculated based on the presented relations in subsection 2.5.3. The computed values are presented in Table 3.3.

Position	Element	$t_{nom.}$ (min)	t_{eff} (min)	$t_{plateau}$ (min)
CIP	Gold	15	15.5	14.3
	Ti & In	15	15.4	14.1
Weber	Gold	15	15.3	15.2
	Gold (shielded)	15	15.3	15.2
	Indium	30	30.4	30.2
	Indium (shielded)	30	30.2	30.0
	Titanium	420	415.0	413.9

Table 3.3.: Comparison of the nominal irradiation times ($t_{nom.}$), the effective irradiation times (t_{eff}) at 250 kW and plateau times ($t_{plateau}$) for the various activation probes in the CIP and Weber tube. Values are rounded to one decimal place.

The power profile of gold irradiation at the CIP is shown in Figure 3.5. Three operational phases are marked in the plot: Startup (blue), plateau (green) and shutdown (red). It can be seen that during startup, the power was raised too fast and dropped because of the reactor protection system reducing power.

For the irradiation at the CIP, the control rods were dropped into the core for shutdown. For the Weber tube irradiation, the control rods were lowered into the core slowly using the driveshaft, leading to a slower power drop compared to the irradiation at the CIP. This was done to reduce strain on the control rod dampers at the bottom. It was also clear that the irradiation time would be corrected using the Python script, eliminating the need for a fast shutdown. The complete set of reactor profiles is depicted in section B.2.

3.1.6. Correction of the Neutron Flux at the CIP

The samples were irradiated using position holders with historical markings indicating the distance between the upper grid plate and middle of the fuel, translating to the maximum flux. These markings turned out to be incorrect, leading to irradiation below the flux maximum (see Figure 3.6). In order to correct the flux data at the CIP, axial flux distribution data from [64] was analyzed. This data was obtained by irradiating a thin gold wire at a power level of $P_{th} = 1$ kW.

In order to correct the flux, we introduce the position correction factor $P(x)$, which is defined as the fraction of the maximal thermal flux value divided by the thermal

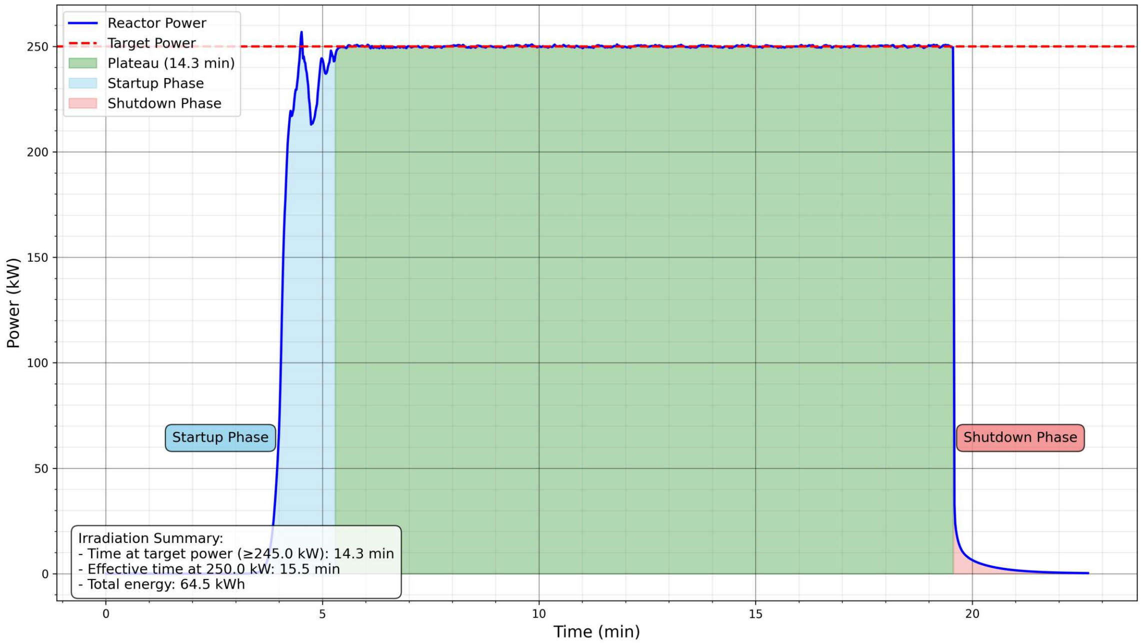


Figure 3.5.: Reactor power profile during gold irradiation in the CIP. Three operational phases are marked: Startup (blue), plateau (green) and shutdown (red).

flux value at position x. As the thermal flux is directly proportional to the activation per unit mass, we can therefore conclude that the following relation applies:

$$P(x) = \frac{\Phi_{\text{th,max}}}{\Phi_{\text{th}}(x)} \propto \frac{A_{\text{max}}/m_{\text{max}}}{A(x)/m(x)} \tag{3.1}$$

$$\begin{aligned} \text{where} \quad P(x) &: \text{position correction factor at position } x \text{ (dimensionless)} \\ \Phi_{\text{th,max}} &: \text{maximum thermal neutron flux } (\text{cm}^{-2}\,\text{s}^{-1}) \\ \Phi_{\text{th}}(x) &: \text{neutron flux at position } x \; (\text{cm}^{-2}\,\text{s}^{-1}) \\ A_{\text{max}} &: \text{activation at the maximum flux position (Bq)} \\ m_{\text{max}} &: \text{mass of the sample at the maximum flux position (g)} \\ A(x) &: \text{activation at position } x \text{ (Bq)} \\ m(x) &: \text{mass of the sample at position } x \text{ (g)} \end{aligned}$$

As the ideal axial neutron flux distribution can be described by a cosine function, we can model the activation per unit mass with a general function:

$$\frac{A(x)}{m(x)} := I(x) = a \cdot \cos(b \cdot (x - c)) + d \tag{3.2}$$

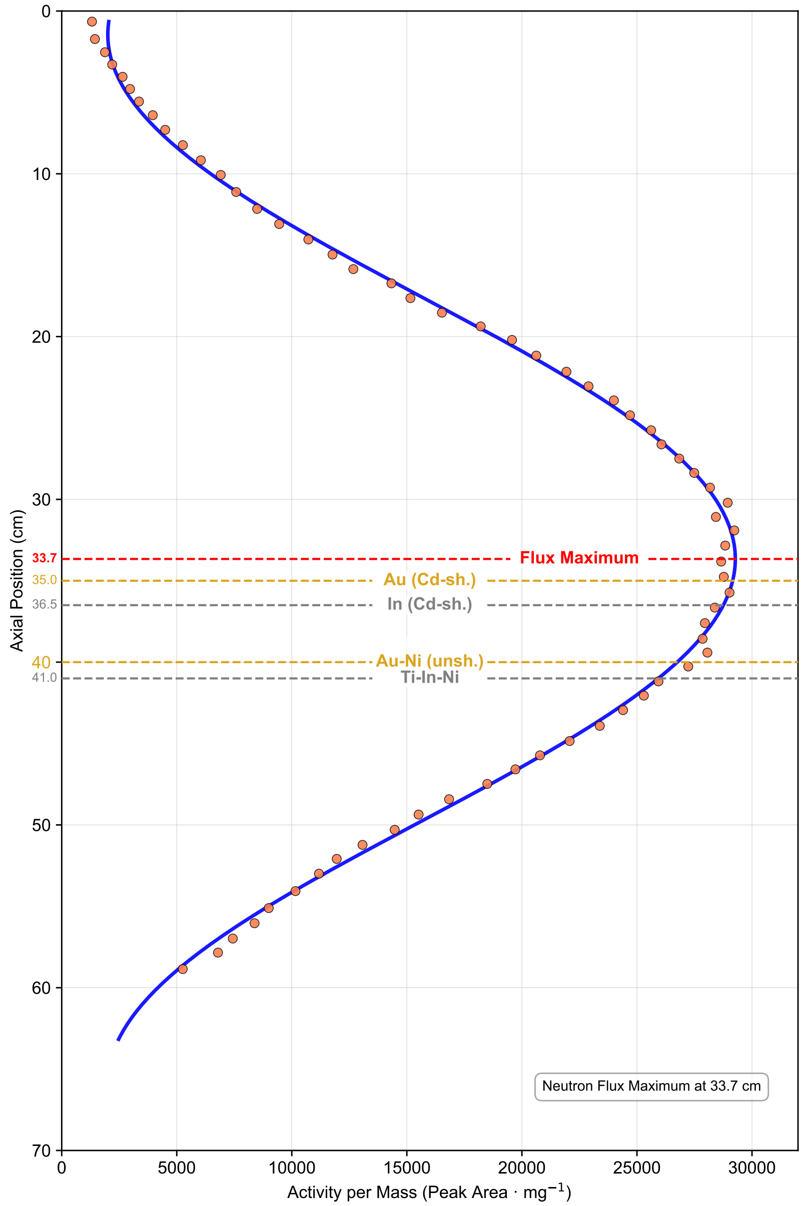


Figure 3.6.: Irradiation Position of different activation probes in the CIP. The axial position start of 0 cm marks the upper grid plate of the reactor. Data sourced from [64]

Fitting our data with the Python script, we get:

$$I(x) = 13636.1518 \cdot \cos(0.0975 \cdot (x - 33.6619)) + 15639.4294 \tag{3.3}$$

Yielding the maximum flux at the position:

$$I(x_{\text{max}}) = c \approx 33.7 \text{ cm} \tag{3.4}$$

A similar position was determined in 2005 for the thermal flux maximum at around 33 cm by measuring the axial flux with a cobalt wire [23, p. 11]. The fast axial flux was measured at the TRIGA Reactor in Vienna in 1981 with nickel, published in [16]. Interestingly, the center of the active fuel element zone was specified as 31.5 cm [16, Fig. 64]. The fast flux maximum was read from [16, Fig. 65] and is located at about 35 cm. The difference between the center of the active zone and the flux maximum is explained by the flux depression of the control rods. They are inserted from above and move the flux downwards [16, p. 100].

From this, we can conclude that the axial position of the flux maximum depends on the reactor power, as irradiation at lower power means that the control rods are inserted further into the core, while at full power they are further out, leading to a shift in the position of the flux maximum. It is not clear at which power the axial fast flux distribution was measured in [16], as nickel was irradiated at reactor powers 51 kW, 100 kW and 250 kW [16, p. 96].

Finally, the position correction factor can be expressed as:

$$P(x) = \frac{a \cdot \cos(b \cdot (c - c)) + d}{a \cdot \cos(b \cdot (x - c)) + d} = \frac{a + d}{a \cdot \cos(b \cdot (x - c)) + d} \tag{3.5}$$

Using the fitted parameters from our model:

$$P(x) \approx \frac{15182 + 14094}{15182 \cdot \cos(0.053 \cdot (x - 33.662)) + 14094} \tag{3.6}$$

The calculated position correction factor ($P(x)$) for the different positions are shown in Table 3.4.

Position	Position (cm)	Activity (Peak Area $\cdot$ mg^{-1})	P
Maximum	33.7	29 276	1.0000
Au (Cd-sh.)	35.0	29 160	1.0040
Au-Ni (unsh.)	40.0	26 752	1.0943

Table 3.4.: Neutron flux correction factors for different irradiation positions. The correction factors convert flux measurements at specific positions to the maximum flux value. Position and activity data sourced from [64]

The corrected thermal neutron flux regarding both the thermal absorption cross-section and position for the CIP can therefore be calculated by introducing the position correction factor. To be precise, the factor P_{bare} has to be multiplied with the unshielded activity and P_{Cd} with the shielded activity in the second term of Equation 2.5.1:

$$\Phi_{\mathrm{th}} \approx \frac{P_{\mathrm{bare}} \cdot A_{\mathrm{bare}} \cdot e^{\lambda \cdot t_{\mathrm{cool,bare}}}}{N_{\mathrm{bare}} \cdot \bar{\sigma}_A(T_{\mathrm{n}}) \cdot S_{\mathrm{bare}}} - F_{\mathrm{Cd}} \cdot \frac{P_{\mathrm{Cd}} \cdot A_{\mathrm{Cd}} \cdot e^{\lambda \cdot t_{\mathrm{cool,Cd}}}}{N_{\mathrm{Cd}} \cdot \bar{\sigma}_A(T_{\mathrm{n}}) \cdot S_{\mathrm{Cd}}} \tag{3.7}$$

The final thermal and fast neutron flux values, computed with the Python script, are presented and discussed in section 4.1, while the underlying data used for the calculations is summarized in section B.1.

3.2. Resistivity Measurement during Annealing

The in-situ resistivity measurement of irradiated REBCO samples required multiple experimental subprojects, which are detailed in the following sections. The custom sample holder, combined with dedicated software, enables continuous monitoring of resistivity changes during thermal annealing, providing insight into the evolution of defect structures and material properties. The following subsections describe the sample holder design, the implementation of the four-point probe measurement technique, chemical etching and patterning of the samples, as well as other procedures like encapsulation and T_c measurement.

3.2.1. Sample Holder and Four-Point Measurement

Two sample holder generations were designed and manufactured during the course of this work. The first, termed V1, is depicted in Figure 3.7. It was designed for square-shaped superconductor samples. With progressing time, SP SCS09 was selected as the main superconductor for irradiation and annealing experiments. As the size of this superconductor was cut to 4 mm × 27 mm, the sample holder had to be further developed to accommodate the new sample dimensions, resulting in the sample holder V2, shown in Figure 3.8. This version will be explained in detail, as all the relevant resistance measurements were conducted with it.

The contacting of the superconductor samples proved to be a challenging task, as the junctions had to fulfill several requirements. They had to be:

- Ohmic contacts with low resistance
- Resistant to corrosion in high-temperature O_2 atmosphere

- Able to allow expansion and contraction while maintaining a secure contact

Different approaches were tested, of which two will be explained subsequently. In the first configuration (see Figure 3.8c), the sample was placed on top of a sample carrier made of glass. Then, per side, two metal contact beams were pressed down on the sample (see Figure A.8 and Figure A.10). Between the two contact beams, a cable lug was placed. Inside this lug, the positive or negative voltage and current lines of the four-wire measurement were crimped together. The exposed surfaces of the contact beams, as well as the cable lugs, were sputtered with a silver target to minimize oxidation during the annealing procedure.

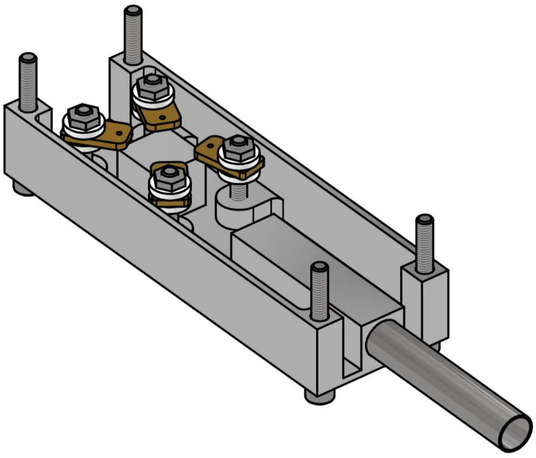
This configuration presented a number of challenges: The sample was pressed down firmly, allowing no movement of the sample. As the superconductor expands and contracts during heating and cooling, it was not clear if the lack of possible movement lead to sudden changes in the resistance observed during testing. Two additional issues were identified in the setup: The current and voltage lines were joined together at the cable lug. Additionally, the cable lug itself did not make direct contact with the sample but was instead connected via a contact beam.

This lead to improved contact arms (see Figure A.5), machined from MACOR® Machinable Glass Ceramic while featuring four separate contact points. These contacts were realized by bending gold-plated needles to a loop. This loop would then be pressed onto the silver layer of the sample. The needles were connected to the cables by crimping them together using a butt splice connector. Afterwards, conductive silver was dripped into the splice connector. The solvent was evaporated with a heat gun. The exact used components were:

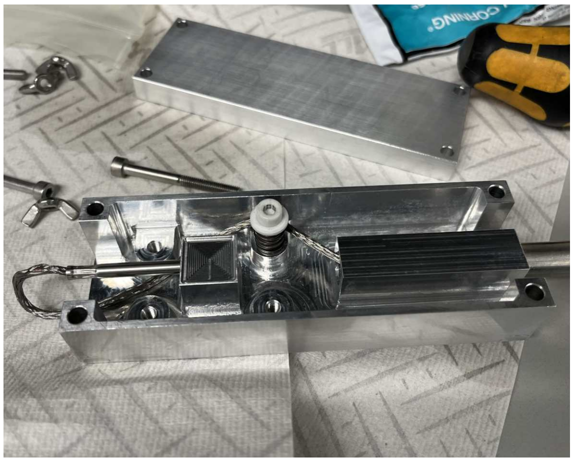
- **HELUTHERM 400 cable**: Nickel conductor with a diameter of $0.75\,\mathrm{mm}^2$ and an outer silk glass layer for temperature insulation of up to $400\,^{\circ}\mathrm{C}$
- **RS PRO silver conductive lacquer** for PCB solder masking for high temperature applications
- **PT100 temperature sensor** measuring the temperature inside the sample holder at the sample position. This sensor was coupled with an Adafruit PT100 RTD Temperature Sensor Amplifier (MAX31865), which was connected to an ESP32 microcontroller. The microcontroller was read out via the serial interface and the data was subsequently merged with the resistance data from the four-wire measurement.

This configuration is depicted in Figure 3.9a and Figure 3.8d. An ohmic test was carried out by applying negative (see Figure 3.9b) and positive current polarity (see Figure 3.9c) to a $4\,\mathrm{mm} \times 27\,\mathrm{mm}$ cut-out printed circuit board (PCB) with a $35\,\mu\mathrm{m}$ copper layer while measuring the voltage drop. This test was conducted at at room temperature while continuously increasing the current. As described in [65, pp. 28-30], the more linear the I–V curves are, the more ohmic the contacts are. When comparing the data to an ideal linear curve using a regression fit method, a correlation coefficient greater than 0.9999 is desirable [65, p. 28]. The ohmic test of the contact arrangement depicted in Figure 3.9a confirmed that the junctions were ohmic contacts, as the least squares regression yielded a correlation coefficient greater than 0.999999. The tested contact configuration was used for the final resistance measurements of the superconductor samples.

An assembly drawing of the sample holder V1, as well as a complete set of assembly and technical drawings for all components of the sample holder V2, is presented in section A.1.



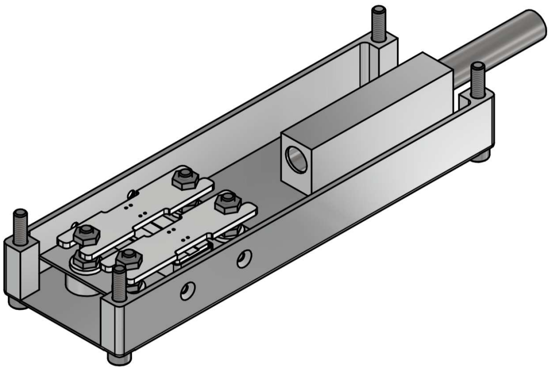
(a) Sample holder V1 (technical drawing)



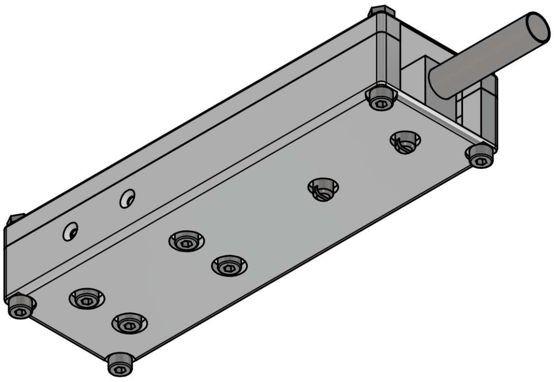
(b) Sample holder V1 (photo)

Figure 3.7.: Sample holder V1: (a) technical drawing and (b) photo

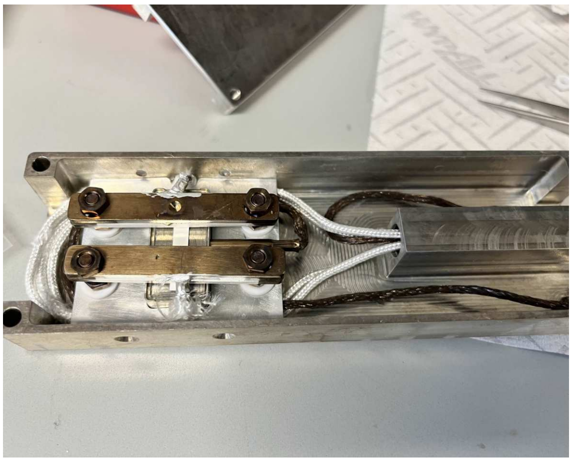
The schematic circuit diagram of the single and triple bridge superconductor measurements is depicted in Figure 3.10a. For the triple bridge measurement, the current injection and voltage measurement contacts were switched to investigate whether this arrangement, in contrast to the conventional four-wire Kelvin configuration (outside current injection, inside voltage measurement), would reveal any differences. This was motivated by strange resistance behavior of the SP SCS09 samples, prompting concerns that the measured voltage drop might not originate solely from the REBCO layer bridge. No significant difference was detected by alternating the measurement layout. It was later determined that this anomalous resistance behavior originated from a different source of error, as presented in section 4.2. All voltage and resistance measurements were carried out with the Keithley 2440 SMU (see Figure 3.10b).



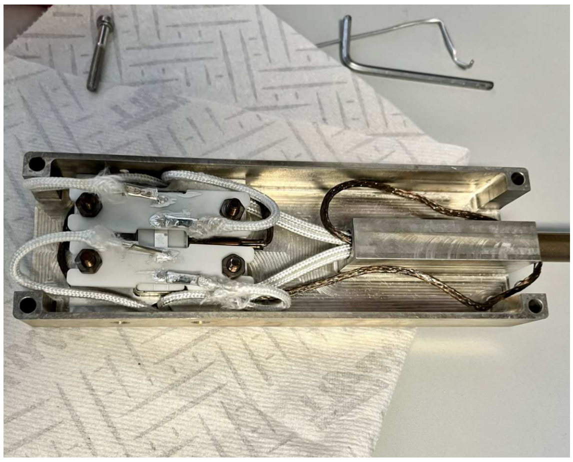
(a) V2 drawing (top view)



(b) V2 drawing (bottom view)



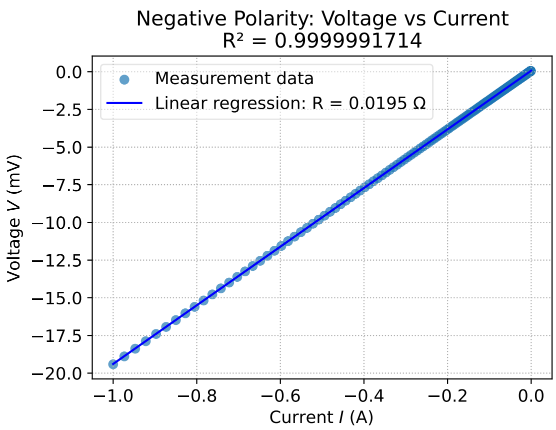
(c) Sample holder V2 (metal contact arms)



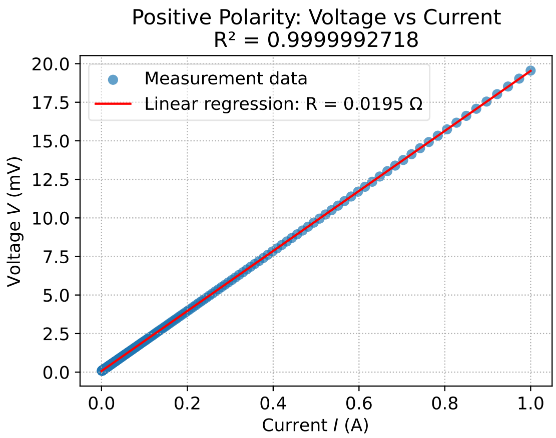
(d) Sample holder V2 (MACOR® contact arms)

Figure 3.8.: Overview of sample holder designs and versions: (a) and (b) show technical drawings of V2 (top and bottom views), (c) and (d) show photographs of V2 (old and new versions)

(a) Photograph of sample contacts

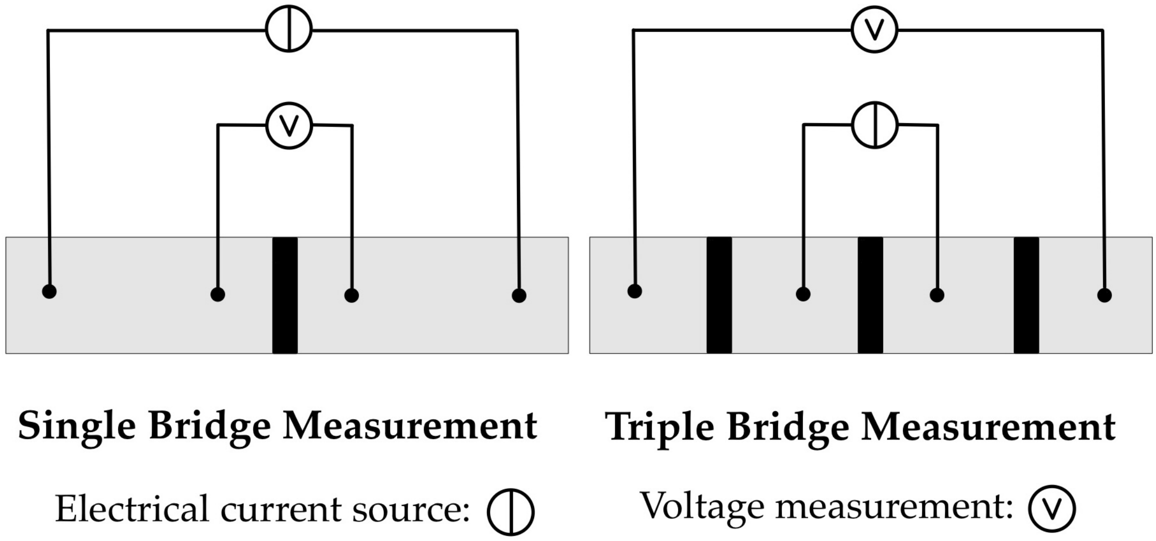


(b) Negative current polarity



(c) Positive current polarity

Figure 3.9.: Top: Photograph of electrical contacts. Bottom: Ohmic contact test for negative and positive current polarity



(a) Schematic diagram of the four-point probe resistance measurement setup (not to scale). The black stripes (bridges) represent the exposed REBCO layer, across which the voltage drop is measured.

(b) Keithley 2440 Source Measure Unit (SMU) used in the measurements

Figure 3.10.: Experimental setup for resistance measurement. Top: measurement circuit for the single and triple bridge measurements; Bottom: Keithley 2440 SMU

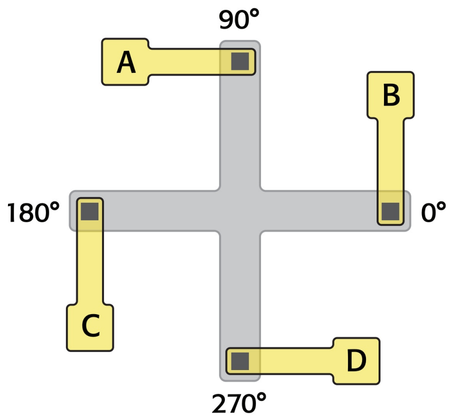
3.2.2. Van der Pauw Structure and Photolithography

The initial approach for contacting the HTS samples and measuring their resistivity was to develop a technique which enables precise chemical etching of patterns. Van der Pauw measurements can be executed by etching different structures, e.g., a Greek cross into HTS (see Figure 3.11a). The key advantage of van der Pauw measurements lies in the ability to measure both the average resistivity while also measuring the Hall constant for an arbitrary sample shape. [66] In contrast, linear four-point probe measurements only measure the resistivity along the sensing direction.

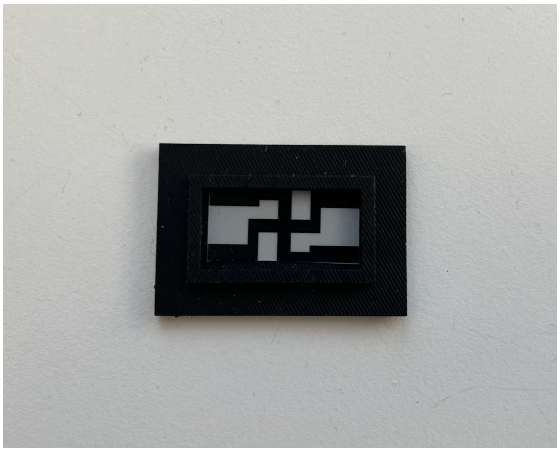
In order to etch different structures and layers, photolithography used in PCB production was selected. To protect certain sections and layers of the superconductor, POSITIV 20, a photo-sensitive lacquer [67], was used. This lacquer was sprayed on the surface of the HTS. In order to apply a homogeneous and thin layer, a spincoater was constructed by my supervisor, Alexander Bodenseher, using a computer cooling fan, placed in a cardboard box and connected to a variable power supply (see Figure 3.11c). By adjusting the voltage and fan speed, and consequently the centrifugal force, the thickness of the sprayed-on lacquer could be varied. The superconductor samples were positioned in the middle of the fan with double-sided tape. The blades of the fan were removed prior to the experiment.

After the application of POSITIV 20, the sample was placed in a 3D printed mask holder (see Figure 3.11b). Then, a translucent film with a printed pattern, acting as a mask, was stacked on top of the superconductor. This film is used for the production of high resolution PCB artworks. Any arbitrary pattern can be printed on this film with a laser printer. In the next step, the 3D printed mask holder with the sample and the mask is placed inside a LPKF UV exposure unit for three to five minutes, exposing the coated surface.

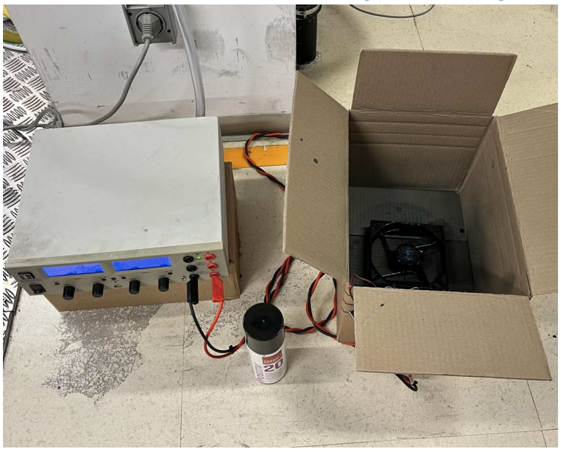
By subsequently immersing the sample into a sodium hydroxide solution (10 g/l $NaOH$ in H_2O) for 60 seconds at room temperature, the lacquer of the unmasked (exposed) sections is dissolved. [67] A pattern structure applied to a Bruker HTS sample is depicted in Figure 3.11d. Now, the uncovered parts can be etched, while the protected sections should remain unaffected by the etching solution. Immense efforts were made to apply this process to the samples. Due to time constraints, which were dominated by our irradiation slot at the TRIGA reactor, this etching approach had to be abandoned, as the etching results were not satisfactory. The samples had to be etched and characterized before irradiation. The most promising attempt of applying a photoresist layer to a superconductor sample is shown in Figure 3.11d. Even for this attempt, the protective layer was not homogeneous enough to protect the covered layers during etching.



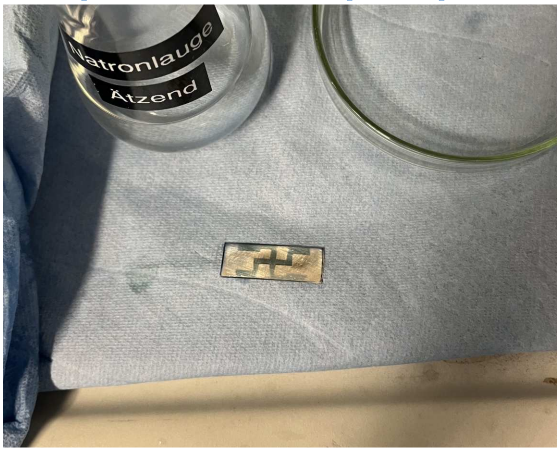
(a) Greek cross Van der Pauw arrangement [66, Fig. 2]



(b) 3D printed mask for UV development of photoresist



(c) Self-made spincoater



(d) Developed photoresist layer on Bruker HTS tape

Figure 3.11.: Top row: Greek cross Van der Pauw arrangement [66, Fig. 2] (left), 3D printed mask for UV development of photoresist (right). Bottom row: Self-made spincoater (left), developed photoresist layer on Bruker HTS tape (right)

3.2.3. Preparation and Chemical Pattern Etching of Samples

SuperPower HTS consist of many layers (see Figure 3.12): on the outside, there is a stabilizing copper layer, which is followed by a silver layer. The REBCO layer is sputtered on top of a buffer stack and a Hastelloy substrate layer. The buffer stack is, inter alia, used to provide lattice matching between the substrate and the REBCO layer.

For our experiments, we first completely removed the copper layer by exposing it to a mixture of sodium persulfate and water (see Figure 3.13a). Then, a metal pin was used to mark the bridge structure by carefully drawing a line into the silver layer. Afterwards, the parts where the silver layer should be preserved were covered with Kapton tape, which is a polyimide film (see Figure 3.13b). For this step, it is important to use pristine Kapton tape, as the tape collects dust at the edges. This dust subsequently leads to frayed borders on the etched bridge structure. After applying the Kapton tape, it was cut along the outer edges of the superconductor with a scalpel.

The sample was then immersed in a solution of ammonium hydroxide, hydrogen peroxide and methanol (see Figure 3.13c), exposing the REBCO layer (see Figure 3.13d). For this step it is extremely important to closely monitor the etching process, as too long exposure leads to the damage of the REBCO layer. The ideal etching time with a fresh solution was determined to be approximately two minutes, as no H_2 bubbles were observed beyond this time mark. The backside was left completely exposed, revealing the substrate layer made of Hastelloy.

The compositions of the etching mixtures listed in Table 3.5 were adopted from [68], whereas the reaction temperature and duration were optimized through trial-and-error experiments using various REBCO HTS tapes. Microscope images of the etched single bridge (SP SCS09 021) and triple bridge (SP SCS09 027) samples can be seen in Figure 3.14.

Material	Etching Mixture (Vol. Prop.)	Temp. [°C]	Time [min]
Copper	$H_2O : Na_2S_2O_8$ (4 : 1)	40 °C	10
Silver	$NH_4OH : H_2O_2 : CH_4O$ (1 : 1 : 4)	20 °C	2
REBCO	HCl	unknown	unknown

Table 3.5.: Mixtures used for etching of the different layers of superconductors [68]

3.2.4. Encapsulation and Irradiation of Superconductors

The superconductor were welded into quartz glass tubes with a gas burner. In addition, one nickel monitor was added to each tube as well as sacrificial copper

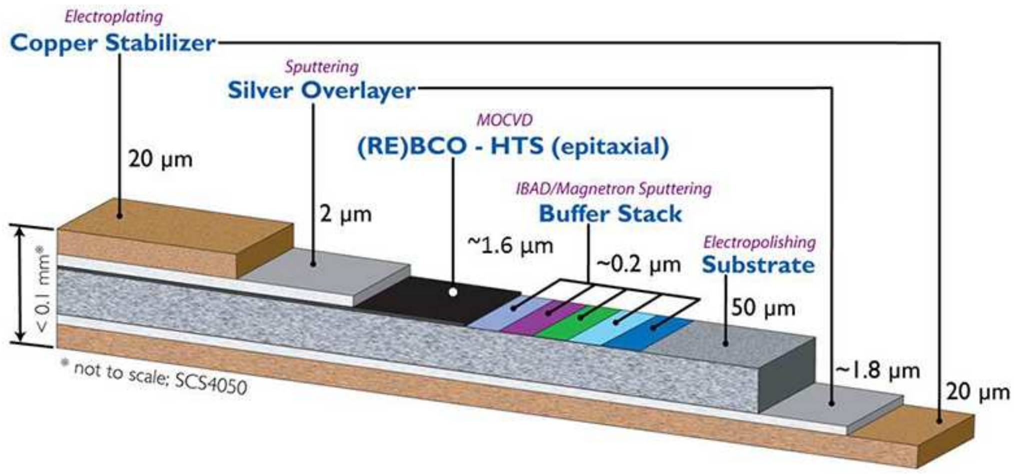
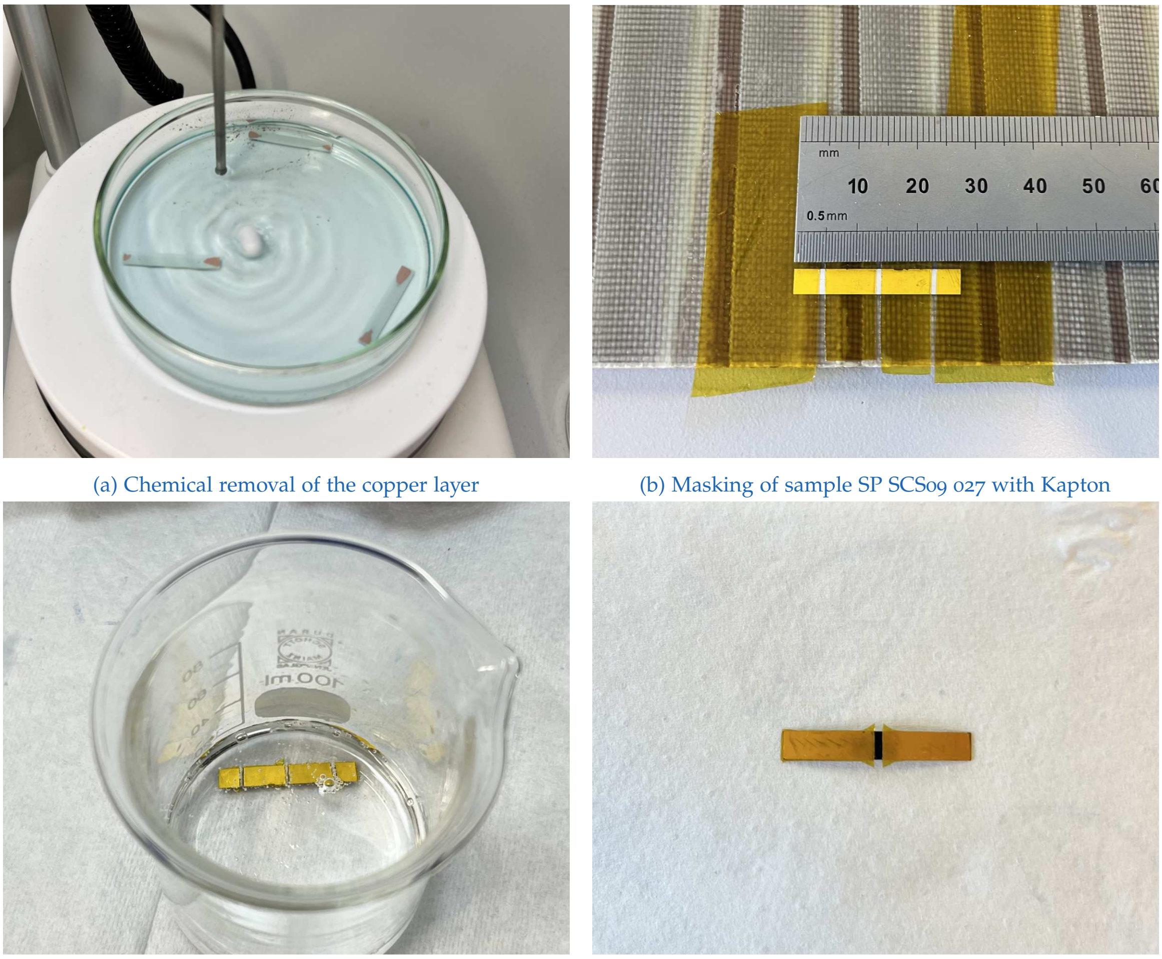


Figure 3.12.: Layer structure of a SuperPower 2G HTS conductor [69, p. 2]



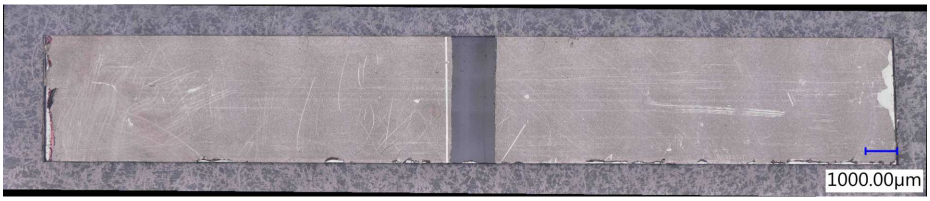
(a) Chemical removal of the copper layer

(b) Masking of sample SP SCS09 027 with Kapton

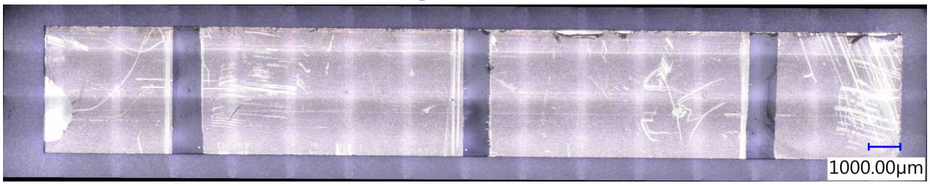
(c) Etching of the silver layer, exposing the REBCO layer

(d) Final etched structure of a single bridge configuration

Figure 3.13.: Process steps overview: (a) Copper removal by etching, (b) Masking of triple bridge, (c) Etching triple bridge, (d) Etched single bridge



(a) Sample SP SCS09 021



(b) Sample SP SCS09 027

Figure 3.14.: Microscope images of the superconductor samples with etched bridge structures: (a) and (b) show the top view of samples SP SCS09 021 and SP SCS09 027, respectively

wool, enveloped in aluminum foil. As described in [2], this material is added to bind radicals which form during irradiation and to absorb oxygen. First, the superconductor samples, along with the other materials, were placed in the quartz glass tubes. Then, the air was pumped out with a vacuum pump and the tube was subsequently flushed with helium. This was done in order to preserve the exposed REBCO layer. After this, the quartz glass tube was separated and sealed using a gas burner. The finished tubes of samples SP SCS09 018–021 are depicted in Figure 3.15.

After this, the samples were irradiated at the CIP (see Figure 3.1a). The total neutron fluence was calculated by evaluating one of the four nickel monitors. Samples SP SCS09 018 – 021 were irradiated together, while SP SCS09 025 was irradiated separately. The fluence values are presented in Table 3.6, although the fluence for sample SP SCS09 025 has yet to be calculated and is therefore indicated as "to be determined".

Sample	Neutron Fluence ($1/m^2$)	Position	Shielding
SP SCS09 018 – 021	$3.0350 \cdot 10^{21}$	CIP	Unshielded
SP SCS09 025	to be determined	CIP	Unshielded

Table 3.6.: Neutron fluence, sample position and shielding condition for different samples

3.2.5. Critical Temperature Measurement

The critical temperatures of the pristine, annealed and irradiated samples were measured with a 17 T cryostat (see Figure 3.16a). It consists of a composite su-

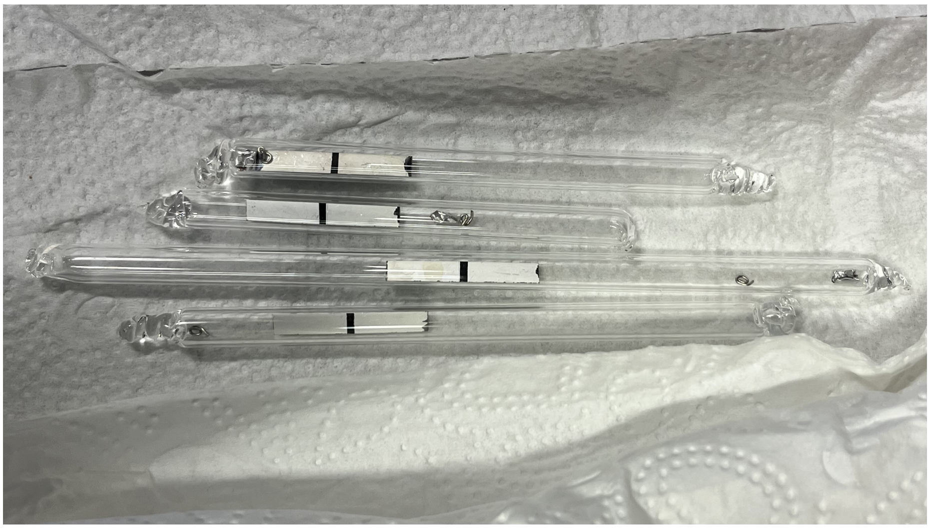


Figure 3.15.: Photograph of the encapsulated superconductor samples (SP SCS09 018–021), Ni-monitors, and sacrificial copper wool wrapped in aluminum foil that were irradiated in the experiments.

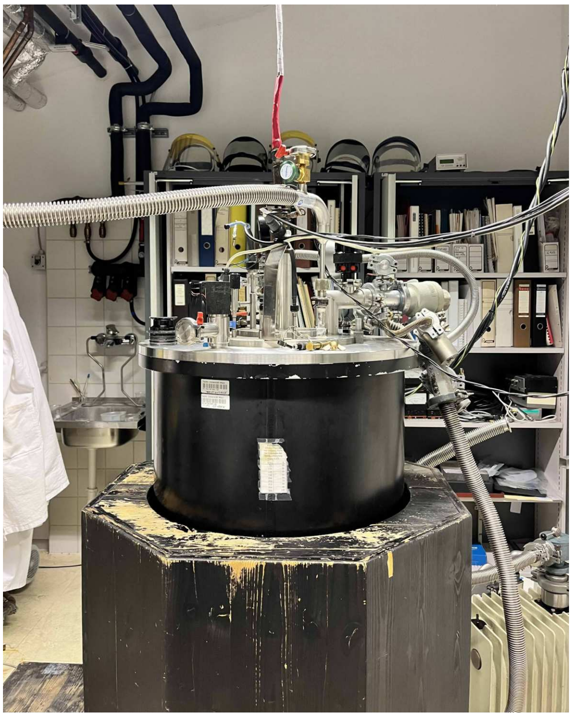
perconducting magnet, providing the necessary magnetic flux density. All the measurements conducted during the course of this thesis were done at a zero-field (0 T). In the center of the device is the variable temperature insert (VTI), which is connected to a helium reservoir. Negative pressure is created with a pump, drawing helium into the VTI. An adjustable resistive heating element enables exact adjustment of the temperature. [70]

The critical temperature was determined by first applying a current and measuring the voltage drop during a continuous decrease of the temperature. The data was then processed as described in subsection 2.7.5, yielding critical temperature values T_c and T_c^{high}.

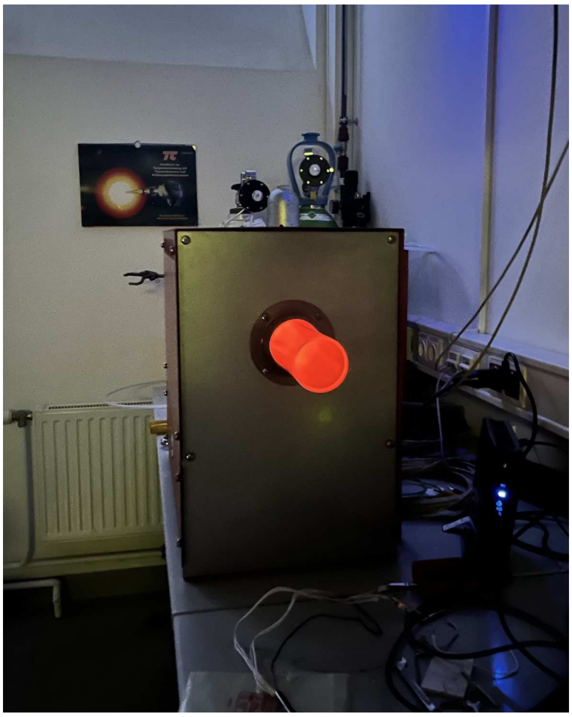
The measurement script of the 17 T cryostat conducts two measurement runs. In the first measurement, a quick temperature ramp is used in order to identify the approximate transition temperature. Based on this, a sensitive temperature sweep is performed, starting above the approximate transition temperature and ramping down slowly. The samples were placed inside the sample holder described in [70, Fig. 3.11], allowing reliable measurements inside the cryostat. The complete set of T_c measurement plots and the corresponding data is presented in section A.3.

3.2.6. Thermal Annealing and Resistivity Measurement

The pristine and irradiated samples used during the experiments are described in Table 3.7. They were annealed in a tube furnace, depicted in Figure 3.16b. The



(a) 17 T cryostat used for critical temperature measurements



(b) Tube furnace used for the annealing procedure

Figure 3.16.: Experimental setups: (a) 17 T cryostat for T_c measurements, (b) tube furnace for annealing

complete setup can be seen in Figure 3.17. It consists of a tank connector on the left side, which directs the gas from a gas cylinder (Ar or O_2) into a quartz tube. The sample holder is connected to a metal rod and pushed into the middle of the quartz tube, where the heating elements of the furnace are located. This metal rod contains the measurement and instrumentation cables of the four-wire measurement and the temperature sensor. The metal rod with the cables is sealed at the end with Teflon tape and heat-shrink tubing to prevent the gas atmosphere from leaking out.

Name	Chemical Formula	Description
Bruker	$YBa_2Cu_3O_{7-\delta}$	Yttrium-based HTS
SuperPower SCS4050 2009	$GdBa_2Cu_3O_{7-\delta}$	Gadolinium-based HTS

Table 3.7.: Overview of the superconductor types used in this thesis

As the experiments are conducted at ambient pressure, the quartz glass tube is also connected on the right side to a copper tube, which is immersed in water to allow adjustment of the gas flow. A target of around one bubble per second was set. Additionally, lead shielding was introduced as the γ-radiation dose rate of the irradiated samples was substantial. All samples except for SP SCS09 027 were thermally annealed in a pure oxygen environment, as otherwise the oxygen escapes the $GdBa_2Cu_3O_{7-\delta}$ crystal lattice. Oxygen out-diffusion changes the oxygen doping levels and leads to a drastic worsening of T_c. For sample SP SCS09 027, insufficient gas flow was provided, leading to an erroneous measurement.

The three different annealing schemes used for our experiments are depicted in Table 3.8. An overview of the different annealing schemes and etching patterns used for the SP SCS09 superconductor samples can be seen in Table 3.9. For all samples, the voltage (V) was measured using a four-point probe measurement in combination with the current-reversal method to compensate for thermoelectric voltages, as described in subsection 2.8.2. An alternating current of 10 mA was applied for all measurements, and the resistance was calculated using Ohm's law. The relevant resistance versus time and temperature plots during annealing are discussed in section 4.2, while the complete set of resistance plots is presented in section A.2.

Scheme	Temp. Profile (°C)	Time Profile (h:min)	Heating Rate (°C/min)
I	50 – 450 – 50	8:00 – 4:00 – 8:00	0.83
II	50 – 400 – 50	7:18 – 0:30 – 7:18	0.8
III	50 – 400 – 50	3:30 – 0:30 – 3:30	1.67

Table 3.8.: Overview of the three different thermal annealing schemes used for SP SCS09 superconductor samples, showing temperature profiles, time durations, and calculated heating/cooling rates

Sample ID	Condition	Scheme	Chemical Etching	Environment
Bruker Reference	Pristine	I	single bridge	O_2
SP SCS09 024	Pristine	I	single bridge	O_2
SP SCS09 025	Pristine & Irr.	I & II	single bridge	O_2
SP SCS09 021	Irradiated	II	single bridge	O_2
SP SCS09 022	Pristine	II	single bridge	O_2
SP SCS09 026	Pristine	II	silver only	O_2
SP SCS09 027	Pristine	II	triple bridge	Atmosphere
SP SCS09 028	Pristine	III	single bridge	O_2

Table 3.9.: Overview of the Bruker reference sample and SP SCS09 superconductor samples with annealing schemes, chemical etching, and atmosphere. SP SCS09 025 used Scheme I for annealing steps 1-2 and Scheme II for steps 3-5.

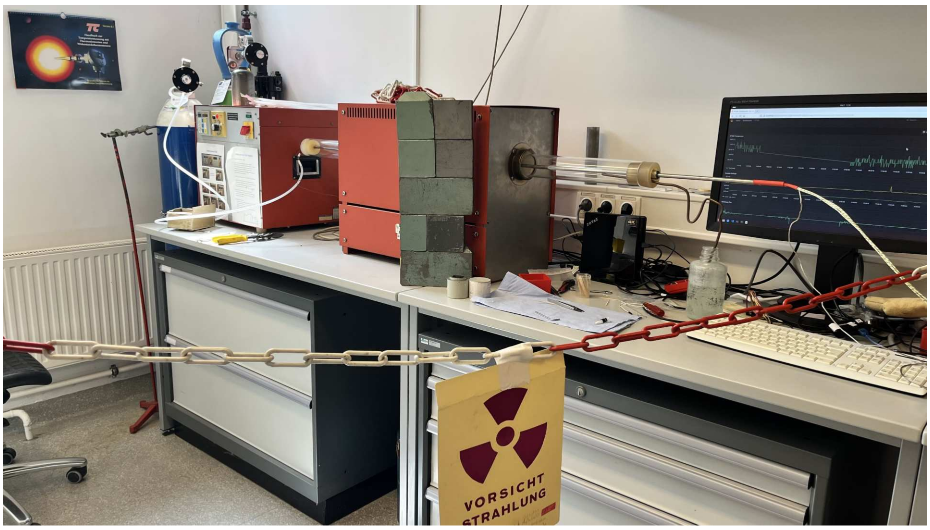


Figure 3.17.: Tube furnace configuration with lead shielding for measurements of radioactive samples.

4. Results and Discussion

4.1. Neutron Flux

The subsequent sections present the results for determining the thermal and fast neutron flux in the TRIGA Mark II reactor at the Atominstitut, Vienna, with a focus on measurements at the central irradiation position (CIP) and the central Weber tube position, which is located outside the graphite reflector (see Figure 3.1).

4.1.1. Thermal Neutron Flux

The initial requirement for the neutron flux measurement was that the flux at the CIP and at the central Weber tube position should be determined at full thermal power of 250 kW. Due to this requirement, several challenges emerged. For the thermal flux, the main challenge was the need for correction of the thermal neutron absorption cross-section. This results from the fact that at a thermal power of 250 kW, neutrons are not in thermal equilibrium with its surroundings, leading to a hardening of the neutron spectrum. This results in a decreased neutron absorption cross-section. By approximating the neutron temperature (see subsubsection 2.5.1) with the moderator temperature, a corrected thermal absorption cross-section was obtained.

During this calculation, another challenge emerged: In a TRIGA reactor, moderation occurs both in the hydrogen of the fuel (U−ZrH) as well as in the light water of the reactor pool. This demanded another estimation of the ratio of moderation at the two irradiation positions. With these corrections, improved thermal flux values were computed. Additionally, a positional correction was necessary for the obtained flux values at the CIP, as the irradiation was not performed at the center of the core, where the flux maximum is, but slightly too low. This was corrected by the introduction of a position correction factor. This factor was calculated based on the thermal flux data from Petz [64], measured axially at the CIP at $P_{\mathrm{th}} = 1$ kW shortly after measurements conducted as part of this thesis.

For both thermal and flux measurements, the irradiation time was corrected using a Python script. This was done as samples are irradiated not only at full power, but also during reactor startup or shutdown. In order to obtain the effective irradiation time at target power of 250 kW, a normalization was performed.

The thermal neutron flux values at the CIP and the central Weber tube position are presented in Table 4.1 and Table 4.2. They represent the same flux values in two different units: The first table uses the SI units $1/m^2/s$, while the second uses $1/cm^2/s$, often used in reactor physics. The original approach to determine the thermal flux with both gold and indium was abandoned and only gold was used. The reason was that, for the selected reaction $^{113}In(n, \gamma)^{114m1}In$, key parameters, such as the cadmium correction factor, were not available.

The three correction factors and the corresponding neutron flux values are listed in the table. As the absorption cross-section is the primary source of error in the neutron flux calculation, an error was computed only for the calculation using the uncorrected $\sigma_{a,th}$. This error calculation was performed using the uncertainties package as part of the final Python script. The final thermal neutron flux value of 9.51×10^{16} $1/m^2/s$, using three correction steps, is similar to the values found in other publications. These reference values are depicted in Table 4.3.

The flux value from Weber, Böck, Unfried, *et al.* [17, Tab. 2] is from 1986. The thermal flux was determined at a reactor power of $P_{th} = 10$ kW and was specified for an energy range of $E < 0.55$ eV. This measurement was conducted with the old core, as a new core was loaded in January of 2013 [18, Tab. 5.8]. The new core features only low-enriched (20 % ^{235}U) fuel elements with stainless steel cladding. In contrast, the old fuel elements were encased in aluminum, which absorbs fewer neutrons. The substantial difference in the flux value measured with the old core, compared to the value reported in the more recent publication [18, Tab. 5.2], could be attributed to the different neutronics of the new core. However, the observed increase in flux with the new core contradicts the expected decrease caused by the stainless steel cladding. The thermal flux measurements in [18] were conducted at $P_{th} = 1$ kW. It should be noted that the energy ranges used in [17] ($E < 0.55$ eV) and [18] ($E < 0.69$ eV) are not correct for thermal neutrons, as the transition energy (E_{ET}) between thermal and epithermal neutrons is approximately 0.1 eV.

It is generally difficult to determine precise neutron flux values because of the nature and way of interaction of neutrons with matter. Due to errors or inconsistencies in neutron flux measurements, it is of limited value to compare the exact values reported in different publications when determining parameters are not specified. The value from Böck and Villa [27, p. 31] most certainly originates from the original calculations made by General Atomic for the TRIGA Mark II reactor. Therefore, it is also of limited value, since the core used for the calculation was different.

Position	Irradiated	Correction	Therm. Flux ($1/m^2/s$)	Rel. Unc. (%)
CIP	^{197}Au	t_{irr}	$(6.02 \pm 0.43) \times 10^{16}$	7.07
		$t_{irr} + \sigma_{a,th}$	8.35×10^{16}	unknown
		$t_{irr} + \sigma_{a,th}$ + Pos.	9.51×10^{16}	unknown
Weber	^{197}Au	t_{irr}	$(3.19 \pm 0.11) \times 10^{15}$	3.51
		$t_{irr} + \sigma_{a,th}$	3.73×10^{15}	unknown

Table 4.1.: Thermal neutron flux for two positions of the TRIGA reactor. The flux in the top row uses the corrected irradiation time and the uncorrected thermal cross-section of gold (98.7 barns), while the corrected flux accounts for the higher neutron temperature at 250 kW using the corrected cross-section (71.2 barns). Pos. stands for positional correction of the flux. All values are expressed in SI units ($1/m^2/s$).

Position	Irradiated	Correction	Therm. Flux ($1/cm^2/s$)	Rel. Unc. (%)
CIP	^{197}Au	t_{irr}	$(6.02 \pm 0.43) \times 10^{12}$	7.07
		$t_{irr} + \sigma_{a,th}$	8.35×10^{12}	unknown
		$t_{irr} + \sigma_{a,th}$ + Pos.	9.51×10^{12}	unknown
Weber	^{197}Au	t_{irr}	$(3.19 \pm 0.11) \times 10^{11}$	3.51
		$t_{irr} + \sigma_{a,th}$	3.73×10^{11}	unknown

Table 4.2.: Thermal neutron flux for two positions of the TRIGA reactor. The flux in the top row uses the corrected irradiation time and the uncorrected thermal cross-section of gold (98.7 barns), while the corrected flux accounts for the higher neutron temperature at 250 kW using the corrected cross-section (71.2 barns). Pos. stands for positional correction of the flux.

Publication	Therm. Flux ($1/m^2/s$)	Simulation Code
Weber, Böck, Unfried, *et al.* (1986) [17]	6.1×10^{16}	STAY'SL
Böck and Villa (2003) [27]	1×10^{17}	–
Cagnazzo (2018) [18]	1.13×10^{17}	Serpent
	1.16×10^{17}	MCNP6

Table 4.3.: Thermal neutron flux values from selected publications for the CIP

4.1.2. Fast Neutron Flux

The fast neutron flux data are presented in Table 4.4 ($1/m^2/s$) and Table 4.5 ($1/cm^2/s$). The only correction applied was for the irradiation time. The position correction was not performed due to the lack of recent axial fast flux data and uncertainty whether thermal and fast flux scale equally. Three different materials were successfully utilized: nickel, indium, and titanium, yielding fast neutron flux values by applying the method of effective threshold energies. The presented data represent values of a distinctive pair (E_{eff}, σ_{eff}), whereas other studies, e.g., [17], [18], often report energy groups ($E > 0.1$ MeV, $E > 1$ MeV). A principal challenge with the method of effective threshold energies is determining the pair of effective cross-section and threshold energy (E_{eff}, σ_{eff}) for the neutron spectrum at the measurement location, which is generally unknown prior to the experiment.

Different pairs of (E_{eff}, σ_{eff}) values may be employed in fast neutron flux calculations, and it is essential to specify both values and their sources, since the neutron spectrum used to derive these parameters should be similar to the spectrum present during the neutron flux determination. These values can deviate drastically for the same reaction depending on the source and reactor spectrum, which can subsequently lead to substantial differences in the computed fast flux. This can also be observed for two relevant publications conducted at the same reactor as this thesis. For the reaction $^{58}\mathrm{Ni}(\mathrm{n,p})^{58}\mathrm{Co}$, [2, p. 54] utilizes $\sigma_{\mathrm{eff}} = 0.106$ b, while [18, Tab. 2.1] uses $\sigma_{\mathrm{eff}} = 0.6$ b. Both publications lack further information, making evaluation of differences in the flux values difficult. For the neutron flux values presented in Table 4.4, existing values from the literature (see Table 2.7) were utilized and should therefore be viewed critically.

A more accurate fast flux calculation would include determining σ_{eff} and E_{eff} analytically, e.g., with a Python script. As computing these values requires the detailed energy-dependent neutron flux spectrum, it has to be determined first if it is not available from previous academic studies. This is done in two steps consisting of:

1. **Neutron activation analysis**: In the first step, foils of different elements are irradiated. This was done as part of this thesis with a limited selection of materials. The elements should be selected so their activation energies overlap and cover the whole spectrum of interest. A comprehensive selection of activation elements is presented in [17, Tab. 1], as well as in [16] for fast flux measurements. After irradiation, the activation of the foils is measured before their decay.
2. **Neutron spectrum unfolding**: Using special software code such as SAND-II, flux de-convolution analysis is performed, yielding the desired energy-dependent flux spectrum. To achieve this, an initial seed spectrum must be provided. The code then iteratively adjusts the spectrum based on the

activation data from the irradiated foils. This approach was most recently applied for the TRIGA reactor in Vienna in [18] at 1 kW for the CIP.

For the flux values at the Weber tube position, an additional aspect must be considered. As σ_{eff} is calculated with a certain flux spectrum, this value is not the same for different irradiation positions if the spectrum changes. For measurements inside the core of a thermal or epithermal reactor, the flux can be described by a fission or moderation spectrum. As the Weber tube position is located outside of the graphite reflector, it can be assumed that the spectrum deviates from that at the CIP. This leads to an even greater uncertainty in our calculated values, although it was not possible to quantify these uncertainties exactly. For a precise fast flux calculation at the Weber tube position, the energy-dependent flux spectrum must first be determined using the method proposed previously, as this was not done before at this specific position. Subsequently, σ_{eff} and E_{eff} should be computed analytically.

Position	Irradiated	E_{eff} (MeV)	Fast Flux (1/m^2/s)
CIP	^{115}In (unsh.)	1.65	2.20×10^{16}
	^{115}In (sh.)	1.65	2.36×10^{16}
	^{58}Ni (Au)	3.45	6.37×10^{15}
	^{58}Ni (Ti-In)	3.45	6.40×10^{15}
	^{47}Ti	3.70	3.20×10^{15}
	^{46}Ti	5.50	8.54×10^{14}
	^{48}Ti	7.20	1.92×10^{14}
Weber	^{115}In (unsh.)	1.65	1.12×10^{14}
	^{115}In (sh.)	1.65	1.26×10^{14}
	^{58}Ni (Au, unsh.)	3.45	4.01×10^{13}
	^{58}Ni (Au, sh.)	3.45	3.45×10^{13}
	^{58}Ni (In, unsh.)	3.45	3.62×10^{13}
	^{58}Ni (Ti)	3.45	3.72×10^{13}
	^{47}Ti	3.70	1.77×10^{13}
	^{46}Ti	5.50	5.45×10^{12}
	^{48}Ti	7.20	1.20×10^{12}

Table 4.4.: Fast neutron flux measured at different effective energy thresholds (E_{eff}), sorted by increasing energy. All values were calculated with the corrected irradiation time. The terms in the parentheses indicate: The activation probe was unshielded or shielded with Cd during irradiation and alongside which other material the Ni-Monitor was irradiated. All values are expressed in SI units (1/m^2/s). Uncertainties are not shown as they are dominated by the effective cross-section uncertainties, which are not well established for these threshold reactions.

Position	Irradiated	E_{eff} (MeV)	Fast Flux (1/cm^2/s)
CIP	^{115}In (unsh.)	1.65	2.20×10^{12}
	^{115}In (sh.)	1.65	2.36×10^{12}
	^{58}Ni (Au)	3.45	6.37×10^{11}
	^{58}Ni (Ti-In)	3.45	6.40×10^{11}
	^{47}Ti	3.70	3.20×10^{11}
	^{46}Ti	5.50	8.54×10^{10}
	^{48}Ti	7.20	1.92×10^{10}
Weber	^{115}In (unsh.)	1.65	1.12×10^{10}
	^{115}In (sh.)	1.65	1.26×10^{10}
	^{58}Ni (Au, unsh.)	3.45	4.01×10^{9}
	^{58}Ni (Au, sh.)	3.45	3.45×10^{9}
	^{58}Ni (In, unsh.)	3.45	3.62×10^{9}
	^{58}Ni (Ti)	3.45	3.72×10^{9}
	^{47}Ti	3.70	1.77×10^{9}
	^{46}Ti	5.50	5.45×10^{8}
	^{48}Ti	7.20	1.20×10^{8}

Table 4.5.: Fast neutron flux measured at different effective energy thresholds (E_{eff}), sorted by increasing energy. All values were calculated with the corrected irradiation time. The terms in the parentheses indicate: The activation probe was unshielded or shielded with Cd during irradiation and alongside which other material the Ni-Monitor was irradiated. Uncertainties are not shown as they are dominated by the effective cross-section uncertainties, which are not well established for these threshold reactions.

4.2. Resistivity Measurement during Annealing

This section presents resistance measurements of pristine and neutron-irradiated high-temperature superconductors (HTS) during thermal annealing. Neutron irradiation introduces defects into the REBCO layer of an HTS, degrading superconducting properties by reducing superfluid density through pair-breaking. For GdBCO, thermal neutron irradiation degrades T_c 14 to 15 times faster than in thermally shielded samples [1].

According to Homes' law, the superfluid density (ρ_s) relates to normal-state resistivity ρ_n and T_c:

$$\rho_s \propto \frac{T_c}{\rho_n}$$

Defects causing pair-breaking and loss of ρ_s also influence normal-state resistivity, described by Matthiessen's rule as a sum of different resistivity contributions:

$$\rho = \sum_i \rho_i = \rho_{\text{Ph-Ph}}(T) + \rho_{\text{e-e}}(T) + \rho_{\text{defects}} + \ldots$$

By in-situ measuring resistance during annealing, defect recovery temperatures can be detected and compared to literature activation energies to identify underlying physical processes.

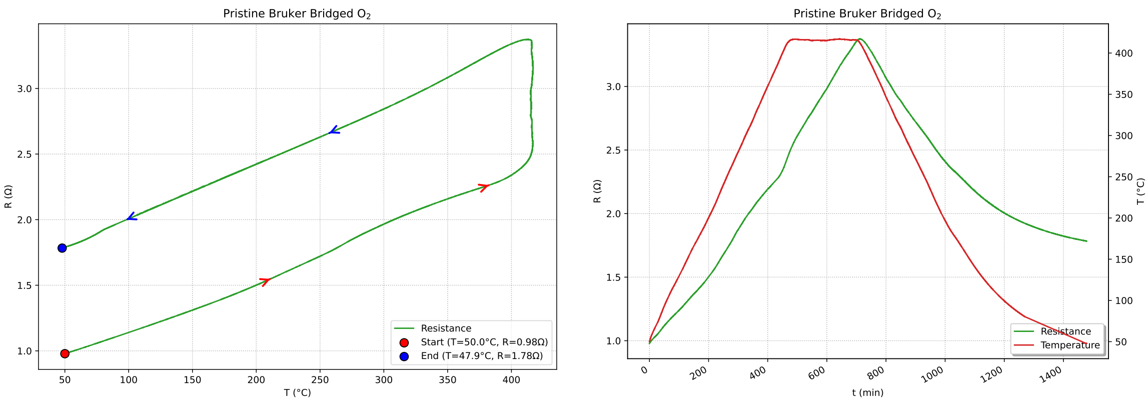
In Figure 4.1, data gathered by in-situ measurement of a pristine Bruker sample during thermal annealing in ambient O_2 atmosphere is depicted. Figure 4.1a displays the resistance plotted against temperature. The red dot indicates the start of the measurement at 50 °C, with the corresponding resistance value presented in the legend. The blue dot represents the end of the measurement at 50 °C. The red arrows show heating, while blue arrows indicate the cooling phase. Figure 4.1b shows the resistance R (in ohms, green, left y-axis) as a function of time t (in minutes, x-axis). The temperature is plotted as an overlay in red, with units displayed in °C on the right y-axis. This sample was annealed using Scheme I, which ramped from 50 °C to 450 °C over eight hours, then maintained this temperature for four hours before ramping down to 50 °C in eight hours.

For the Bruker sample, a linear increase in resistance during heating can be observed. This can be explained by the temperature-dependent contributions $\rho_{\text{Ph-Ph}}(T)$ and $\rho_{\text{e-e}}(T)$. After reaching the oven target temperature of 450 °C, something interesting happens: the resistance continues to rise linearly, though with a different, increased slope. This linear behavior cannot be fully explained and indicates that the underlying process is occurring at a constant rate over time.

As described in [71], the superconducting properties of GdBCO strongly rely on oxygen stoichiometry. Depending on the temperature and oxygen pressure, a resulting oxygen deficiency can cause a phase transition between the superconducting orthorhombic phase and a non-superconducting tetragonal phase. This loss of superconducting properties could also lead to an increase in normal-state resistivity. According to [72], [73], which investigate oxygen nonstoichiometry behavior of REBCO compounds, it is not expected that the experimental environment used for our annealing experiments leads to a substantial change in oxygen stoichiometry, which would cause a change in resistivity. It has to be noted that the data presented in [72], [73] refers to exposed REBCO. The samples annealed in our experiments are covered with silver, except for the exposed bridge structure. This could lead to a change in oxygen kinetics, but the probability of this causing the observed linear increase of resistance is considered to be low. Another possible source of increase in resistance could be the diffusion of Hastelloy components (e.g., nickel) into the REBCO layer, if the buffer layer does not retain the material.

As soon as the temperature starts to decrease, the resistance again falls linearly with temperature. The linear increase during heating and decrease during cooling are consistent with expected behavior. The gain of around 0.8 Ω when comparing the resistance values at the start and end of the measurement corresponds to the approximate increase in resistance during dwelling at the maximum temperature, caused by the unidentified process.

The time-temperature profile was subsequently revised by introducing Scheme II, which reduces the maximum temperature to 400 °C and limits the dwell time at the maximum to 30 minutes. The oven temperature does not correspond exactly to the sample temperature due to heat loss. Therefore, the temperature at the sample is reduced from 415 °C (Scheme I) to 365 °C (Scheme II). This was done to streamline the annealing process and minimize possibly significant damage to the REBCO layer due to extended exposure at high temperatures.



(a) R vs T

(b) R vs Time

Figure 4.1.: Reference sample (Bruker, pristine): Resistance vs temperature and time in O_2

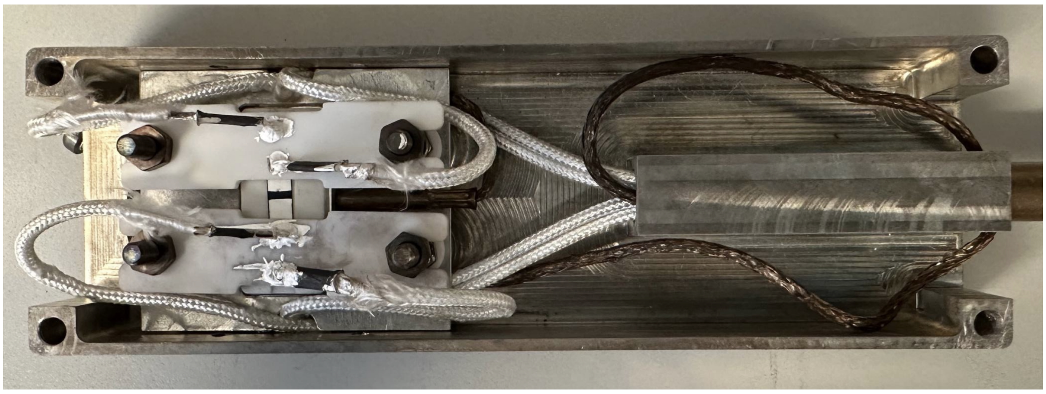


Figure 4.2.: Sample holder V2 with MACOR® contact arms and SP SCS09 sample

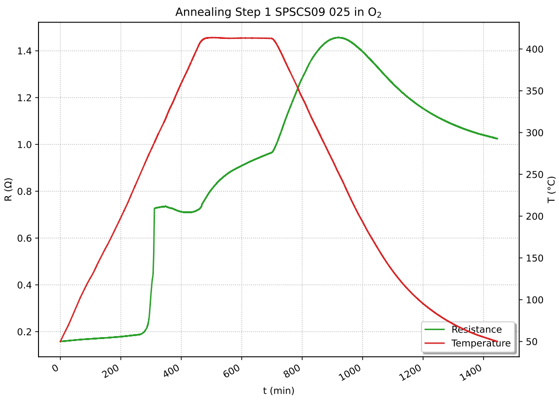
Surprisingly, the measured behavior of the SP SCS09 samples was completely different from that of the Bruker samples. The sample holder V2 with the contact arrangement used for the plots of both the Bruker and SP SCS09 samples is depicted in Figure 4.2. The R vs t curves of three annealing steps for pristine SP SCS09 025 are presented in Figure 4.3a, Figure 4.3b, and Figure 4.3c.

For steps one and two, annealing Scheme I was used, while for the third step, Scheme II was applied. In these plots, the resistance displays a strange behavior which was not explainable by existing physical models or by comparison with available literature. The curves differ fundamentally from the Bruker plots, exhibiting decreasing resistance values as the temperature increases and, conversely, increasing resistance as the temperature decreases.

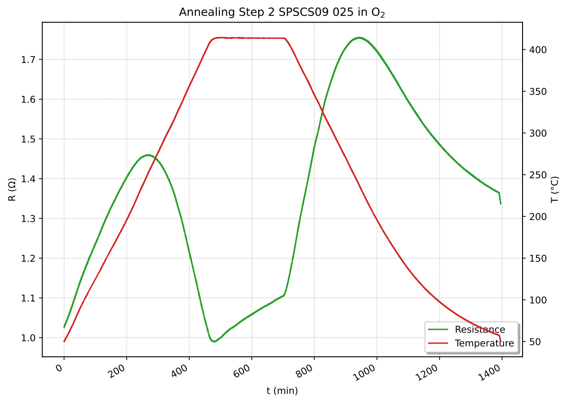
In Figure 4.3d, an overlay plot of SP SCS09 025 showing resistance (R) versus temperature (T) for annealing steps one to six is presented. The annealing of the pristine tape with Scheme II in steps three to five displayed no hysteresis and a completely reversible resistance path. After step five, the sample was irradiated and subsequently annealed following Scheme II. As expected, irradiation caused an initial increase in resistance, observed as a difference between the last pre-irradiation and first post-irradiation measurements. During annealing step six, the irradiated sample exhibited, for the first time, a decrease in resistance in the resistance versus temperature curve.

In Figure 4.4a (R vs T) and Figure 4.4b (R vs t), the first annealing step using Scheme II after irradiation of sample SP SCS09 021 is depicted. As a comparison, the same annealing procedure was conducted with the pristine sample SP SCS09 022, which can be seen in the row below: Figure 4.4c (R vs T) and Figure 4.4d (R vs t).

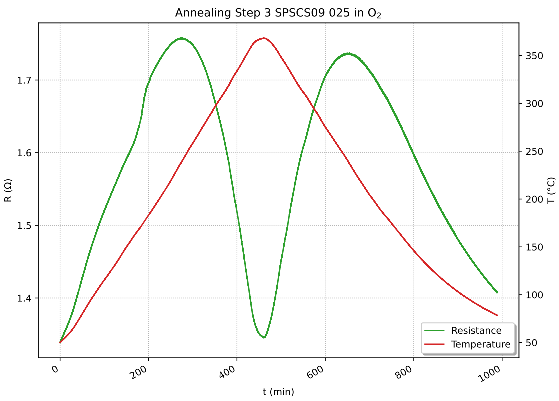
The mystery of the seemingly random behavior of the resistance curves of the SuperPower HTS was solved shortly after the final measurements conducted as part of this thesis. My supervisor, Alexander Bodenseher, hypothesized that during



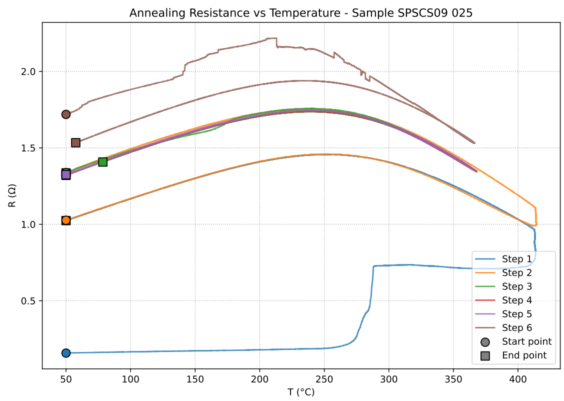
(a) Step 1: Resistance vs Time



(b) Step 2: Resistance vs Time

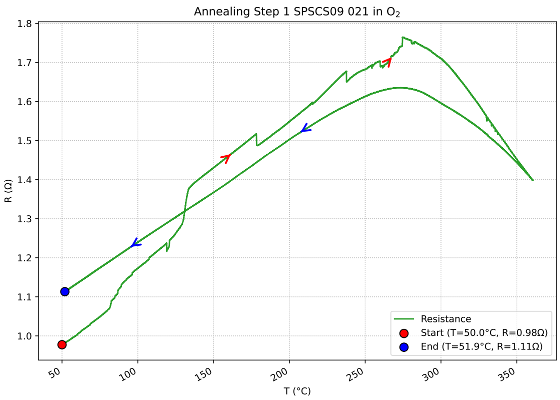


(c) Step 3: Resistance vs Time

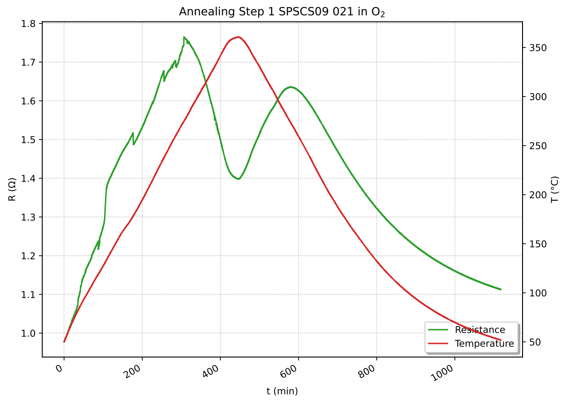


(d) Overlay: Resistance vs Temperature (Steps 1–6)

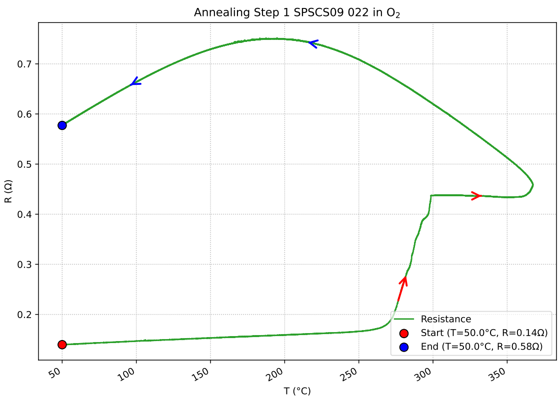
Figure 4.3.: Annealing steps 1–3 for SP SCS09 025: Resistance vs Time, and overall Resistance vs Temperature overlay for steps 1–6. Steps 1–5 were done for pre-irradiation (pristine) while step 6 was performed after irradiation



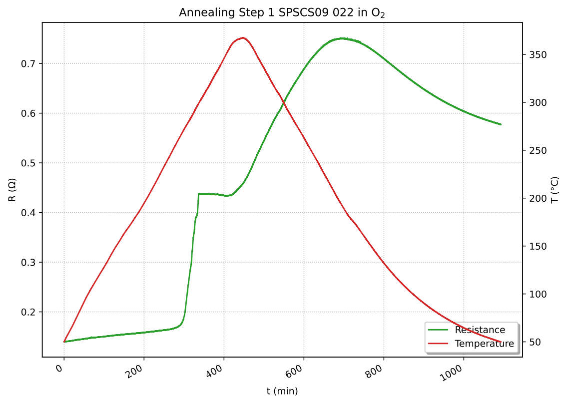
(a) SP SCS09 021 (Irradiated): Resistance vs Temperature



(b) SP SCS09 021 (Irradiated): Resistance vs Time



(c) SP SCS09 022 (Pristine): Resistance vs Temperature

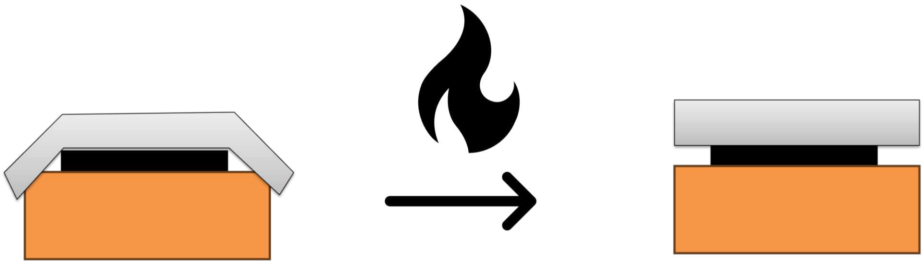


(d) SP SCS09 022 (Pristine): Resistance vs Time

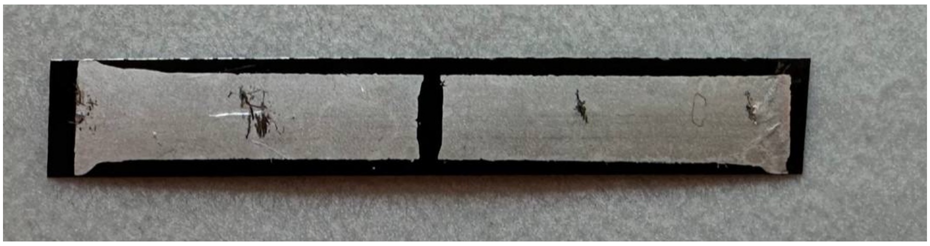
Figure 4.4.: Resistance vs Temperature and Resistance vs Time during annealing of samples SP SCS09 021 (irradiated) and its pristine reference SP SCS09 022 in O_2

the manufacturing or cutting of the superconducting cables, the edges of the silver layer are bent downward, causing a slight short circuit with the substrate. During annealing and subsequent cooling, the silver layer relaxes and contracts, altering the current flow. This is depicted in Figure 4.5a.

To prevent this short circuit and confirm the theory, Bodenseher removed the silver layer from the borders of the sample by chemical etching (see Figure 4.5b). The resulting resistance versus temperature curves of pristine and irradiated SuperPower samples, depicted in Figure 4.6, exhibit a more familiar structure. After annealing, the expected decrease in resistance was measured for the irradiated sample, and a corresponding recovery of $\Delta T_c = 3.73\,\mathrm{K}$ was detected. By plotting $\left|\frac{dR}{dT}\right|$ as a function of temperature, maxima and minima can be detected, indicating the onset of defect annealing. [74]



(a) Short circuit of the silver layer with the substrate and relaxation during annealing.



(b) SuperPower sample with etched borders to prevent short circuits.

Figure 4.5.: Top: Short circuit and relaxation; Bottom: Mitigation by etching the borders [74, p. 25]

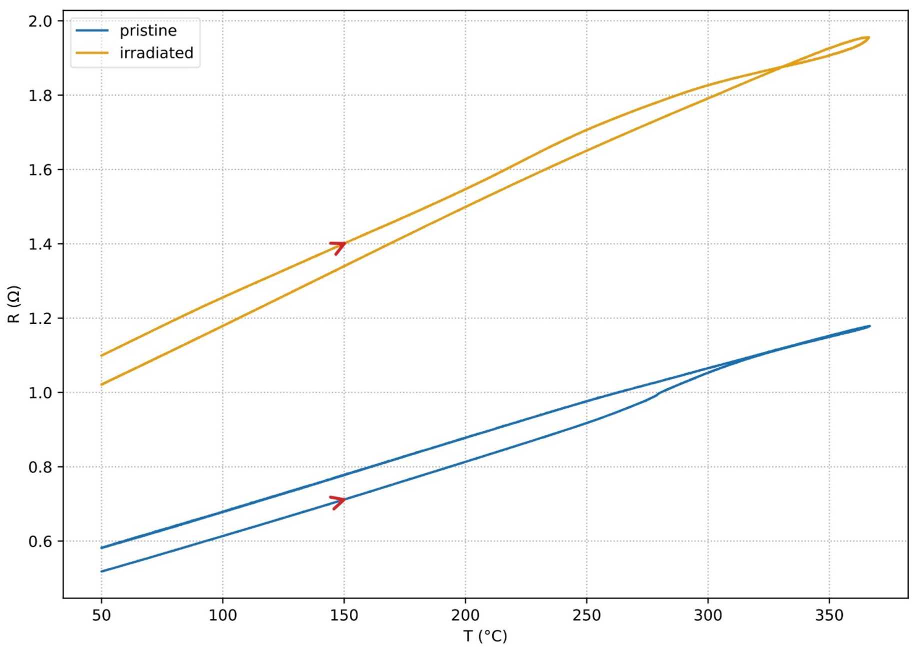


Figure 4.6.: R vs T for a pristine and irradiated SuperPower sample with etched borders [74, p. 26]

5. Conclusion and Outlook

5.1. Neutron Flux

With this thesis, comprehensive work was conducted in the field of neutron flux measurement. Existing literature was analyzed, compared, and discussed, and several key problems in determining the thermal and fast neutron flux were identified and either solved or an approach to their solution was presented. The key neutron flux values are presented in section 4.1.

To further improve these values, several steps are subsequently proposed, mainly concerning the fast flux. The most important improvement concerns the calculation of σ_{eff} and E_{eff} for different fast reactions. This should be performed analytically, for example using a Python code, with the energy-dependent neutron flux spectrum of the specific TRIGA reactor. For the Weber tube position, the neutron spectrum was not measured in previous studies and must therefore be determined. In subsection 4.1.2, a workflow to determine the energy-dependent spectrum was briefly presented.

Furthermore, the position correction must be performed for the fast flux, as the current data certainly does not represent the flux maximum. This can be achieved by measuring the fast flux axially, for example by irradiating a nickel wire, fragmenting it, and measuring the individual parts. Alternatively, the exact relation between the thermal and fast flux could be derived and then confirmed by simulating the flux conditions of the TRIGA reactor using Monte Carlo-based software like Monte Carlo N-Particle (MCNP) or Serpent.

5.2. Resistivity Measurement during Annealing

Two sample holder generations were successfully designed and manufactured. The first generation, termed V1, was designed for square-shaped superconductor samples. As the SP SCS 09 with outer dimensions of 4 mm × 27 mm was selected as the primary superconductor for irradiation and annealing experiments, an advanced second generation was necessary. For this second sample holder generation, named

V2, different four-point resistivity measurement configurations were realized in an iterative approach.

The initial approach for contacting the HTS samples was to use a van der Pauw measurement configuration, which allows measurement of the average resistivity and Hall coefficient across the sample. By using photolithography, it was intended to chemically etch a pattern onto the superconductor. This approach was abandoned due to time constraints and because the results were not yet satisfactory. A simpler masking and chemical etching process was therefore developed, which successfully resulted in $4\,\mathrm{mm} \times 1\,\mathrm{mm}$ etched bridge structures, exposing the REBCO layer. With a sophisticated contact arrangement, the resistance was successfully measured in-situ during annealing of Bruker and SuperPower HTS samples.

The Bruker sample, used as a reference, yielded expected behavior apart from a linear rise in resistance observed during the $415\,^{\circ}\mathrm{C}$, $4\,\mathrm{h}$ dwell of Scheme I. For pristine and irradiated SuperPower samples, the resistance curves displayed strange behavior which could not initially be explained. This was solved later by my supervisor, Alexander Bodenseher, who proved that a short circuit had occurred between the silver and substrate layers, most likely caused by a bent-down silver layer during production. To mitigate this, the silver layer was etched around the borders. By measuring the resistivity of these samples and plotting $\left|\frac{dR}{dT}\right|$, the onset of defect annealing can be detected.

In order to more reliably detect defects introduced only by neutron irradiation, the samples should be annealed and measured prior to irradiation. This eliminates contributions from the intrinsic defect structure. Before the first annealing, the samples must be etched with the new structure, featuring etching along the borders.

For future work, it is recommended to resume the van der Pauw pattern etching process. Successfully implementing this technique would enable measurement of both the average resistivity and the Hall coefficient of the sample. Further advancements in contacting methods will also be necessary. Although sample holder V1 already incorporates a contacting solution suitable for the van der Pauw method, this approach requires revision, as the original implementation did not work as intended. These improvements can be realized by integrating the design features of sample holders V1 and V2 into a subsequent generation, V3.

With the tools and methodologies developed in this thesis, the original objective – detecting defect annealing at specific temperatures introduced by neutron irradiation – can be reliably achieved.

Appendix

Appendix A.

In-Situ Resistivity Measurement

A.1. Technical Drawings of Sample Holders

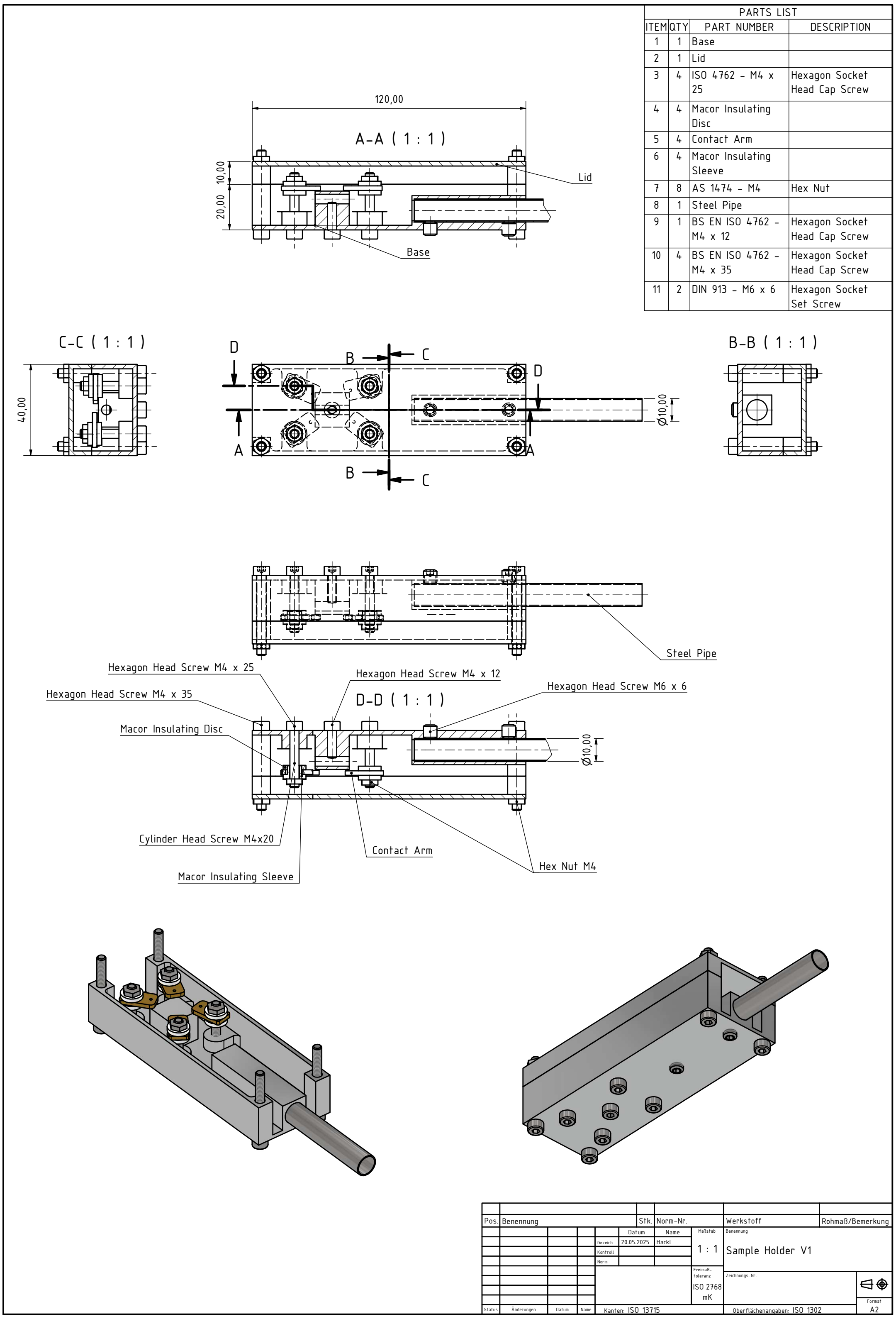
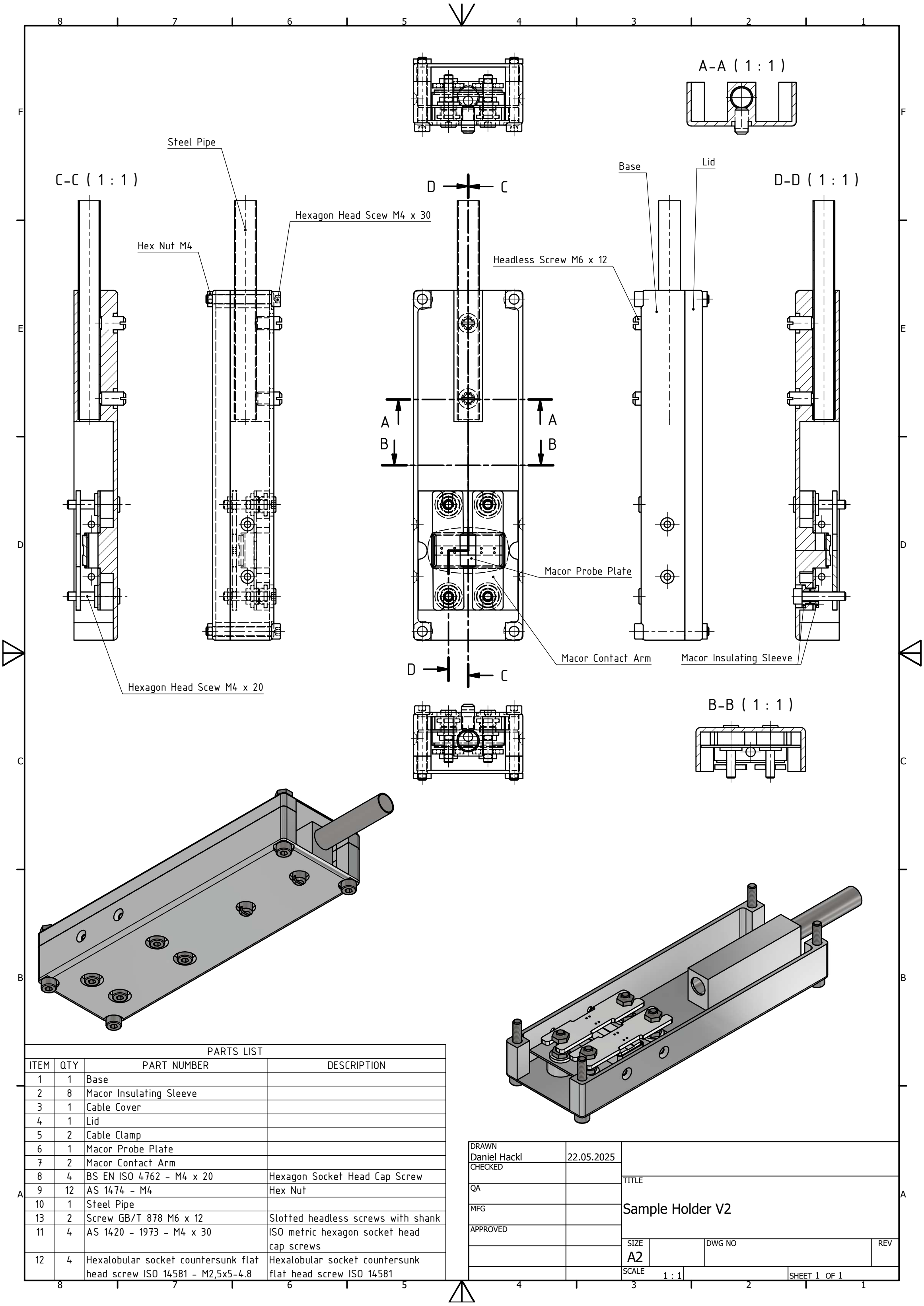


Figure A.1.: Technical drawing of Sample Holder V1



PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Base	
2	8	Macor Insulating Sleeve	
3	1	Cable Cover	
4	1	Lid	
5	2	Cable Clamp	
6	1	Macor Probe Plate	
7	2	Macor Contact Arm	
8	4	BS EN ISO 4762 - M4 x 20	Hexagon Socket Head Cap Screw
9	12	AS 1474 - M4	Hex Nut
10	1	Steel Pipe	
13	2	Screw GB/T 878 M6 x 12	Slotted headless screws with shank
11	4	AS 1420 - 1973 - M4 x 30	ISO metric hexagon socket head cap screws
12	4	Hexalobular socket countersunk flat head screw ISO 14581 - M2,5x5-4.8	Hexalobular socket countersunk flat head screw ISO 14581

Figure A.2.: Technical drawing of Sample Holder V2

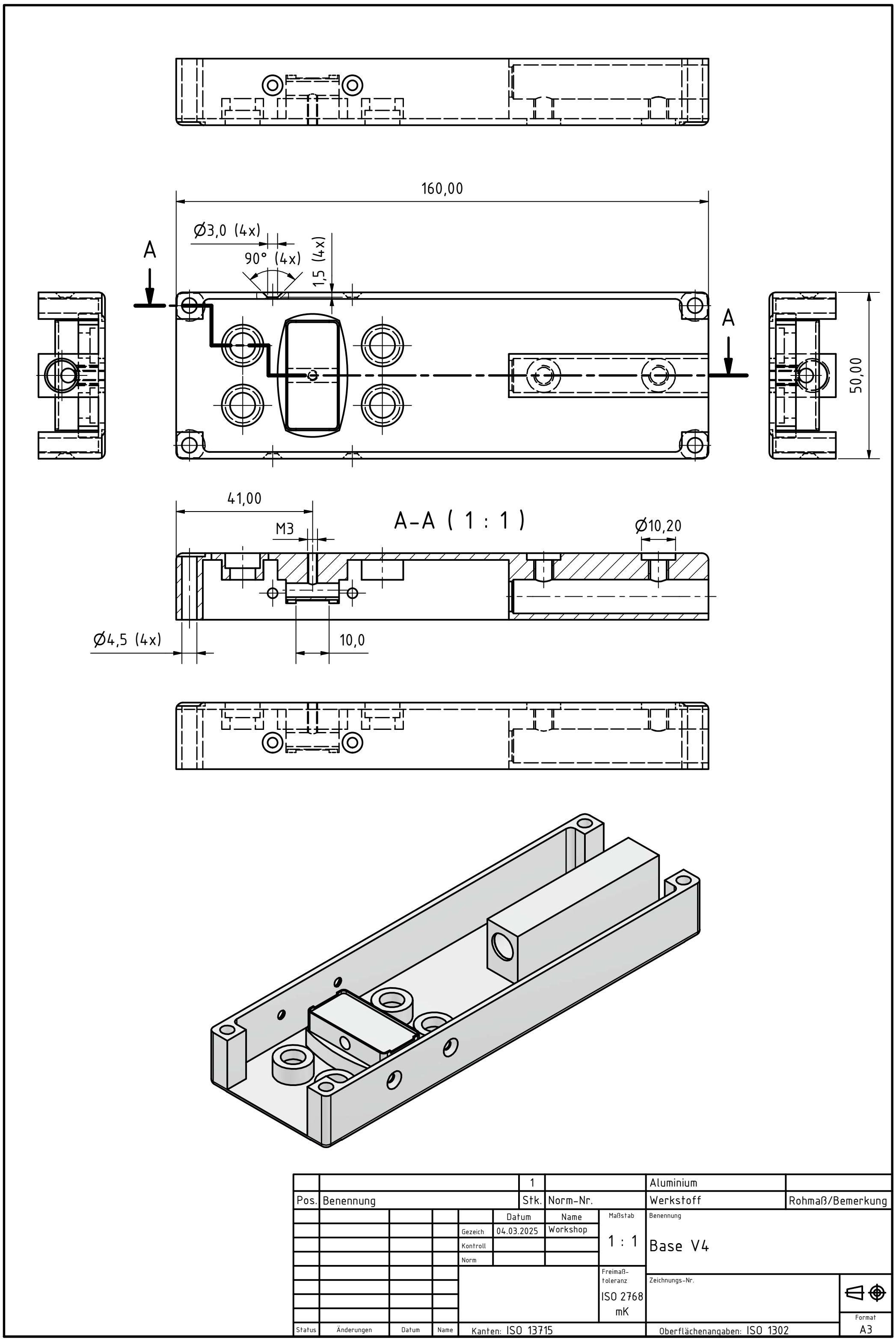


Figure A.3.: Base V4 component of the sample holder V2. Designed by Daniel Hackl, final drawing by Heinz Matusch [75]

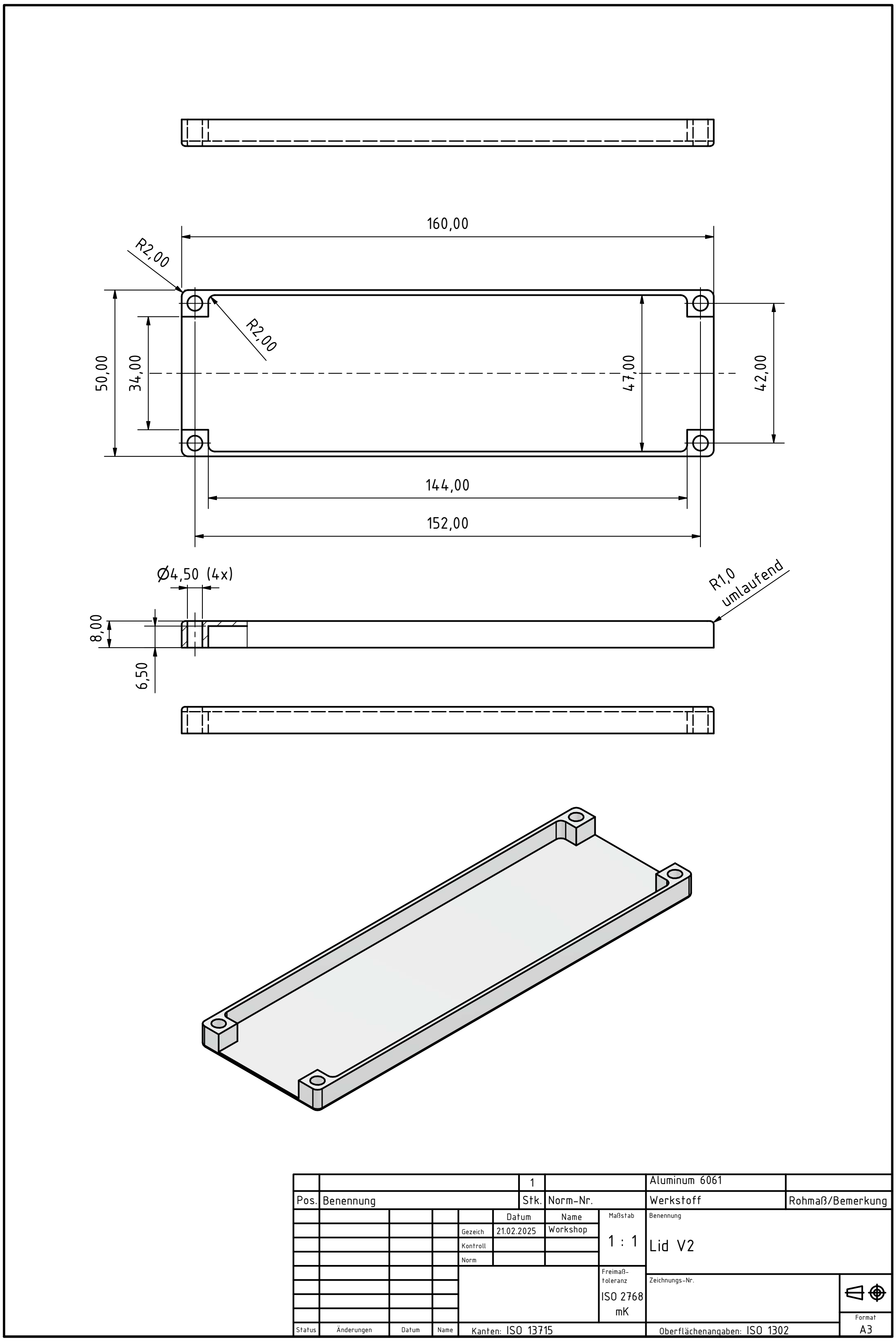


Figure A.4.: Lid V2 component of the sample holder V2. Designed by Daniel Hackl, final drawing by Heinz Matusch [75]

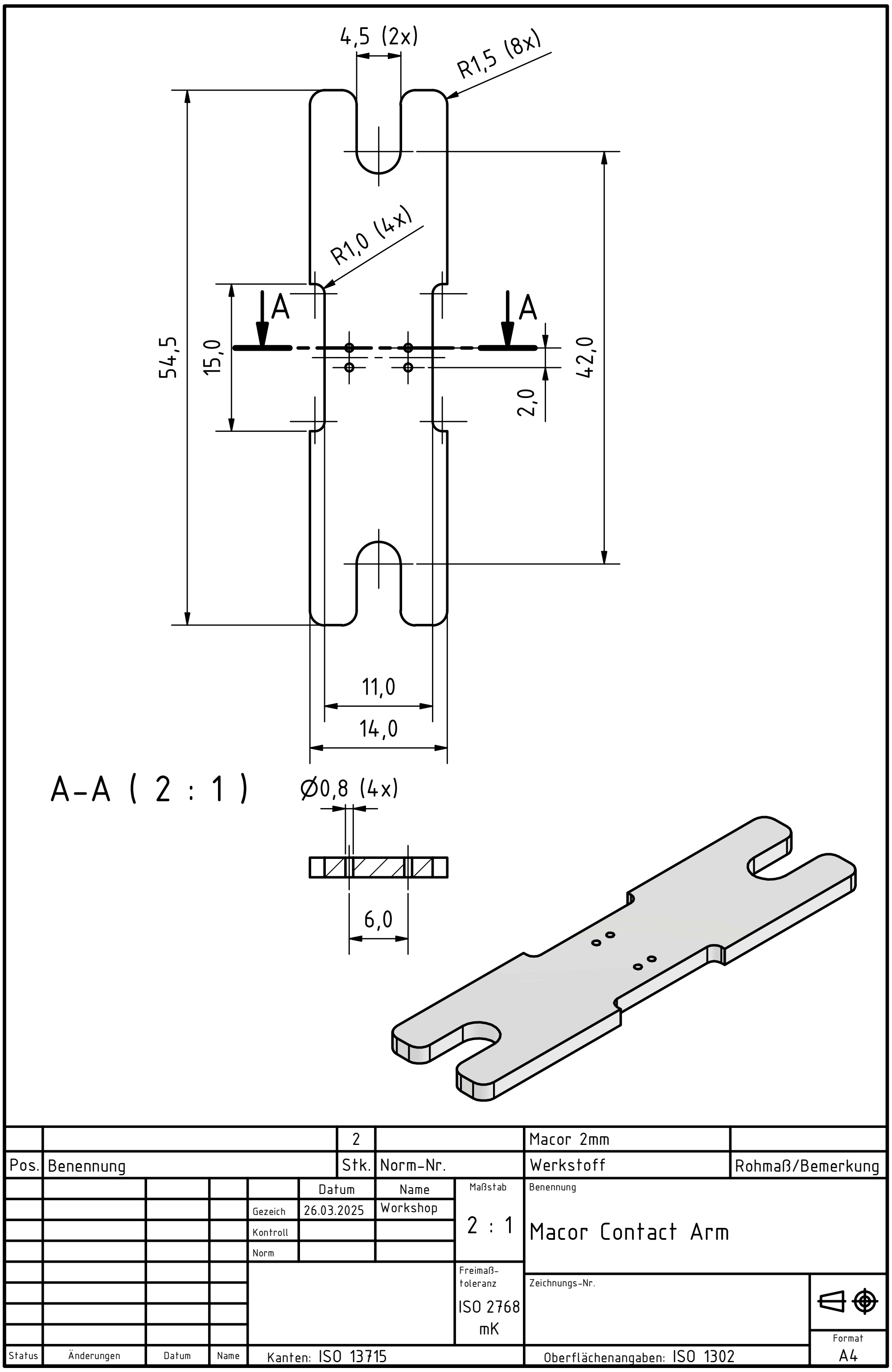


Figure A.5.: Designed in cooperation with Mechanische Werkstatt (Atominstitut, TU Wien); final drawing by Heinz Matusch [75]

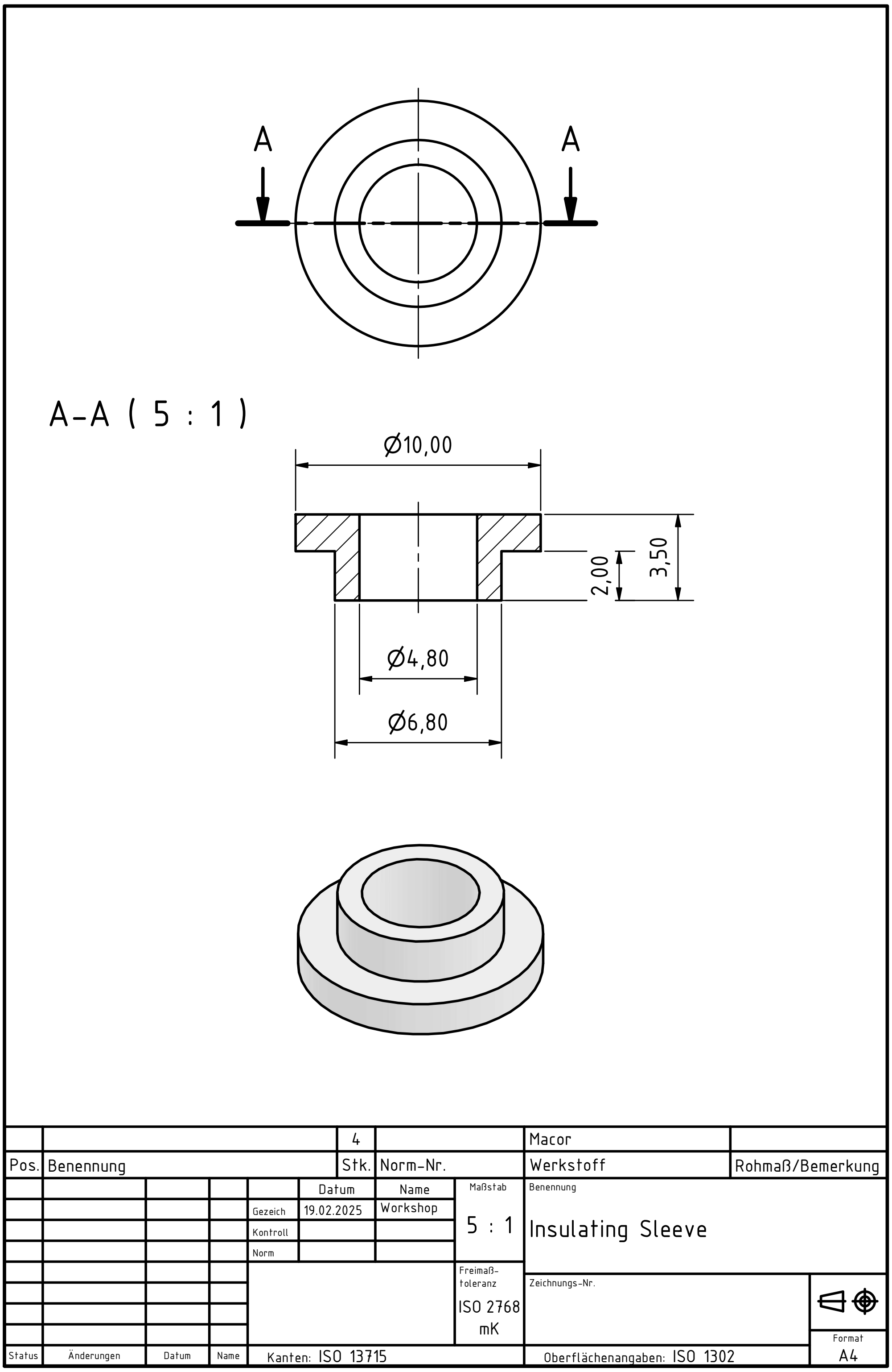


Figure A.6.: MACOR® insulating sleeve component of the sample holder V2. Designed by Daniel Hackl, final drawing by Heinz Matusch [75]

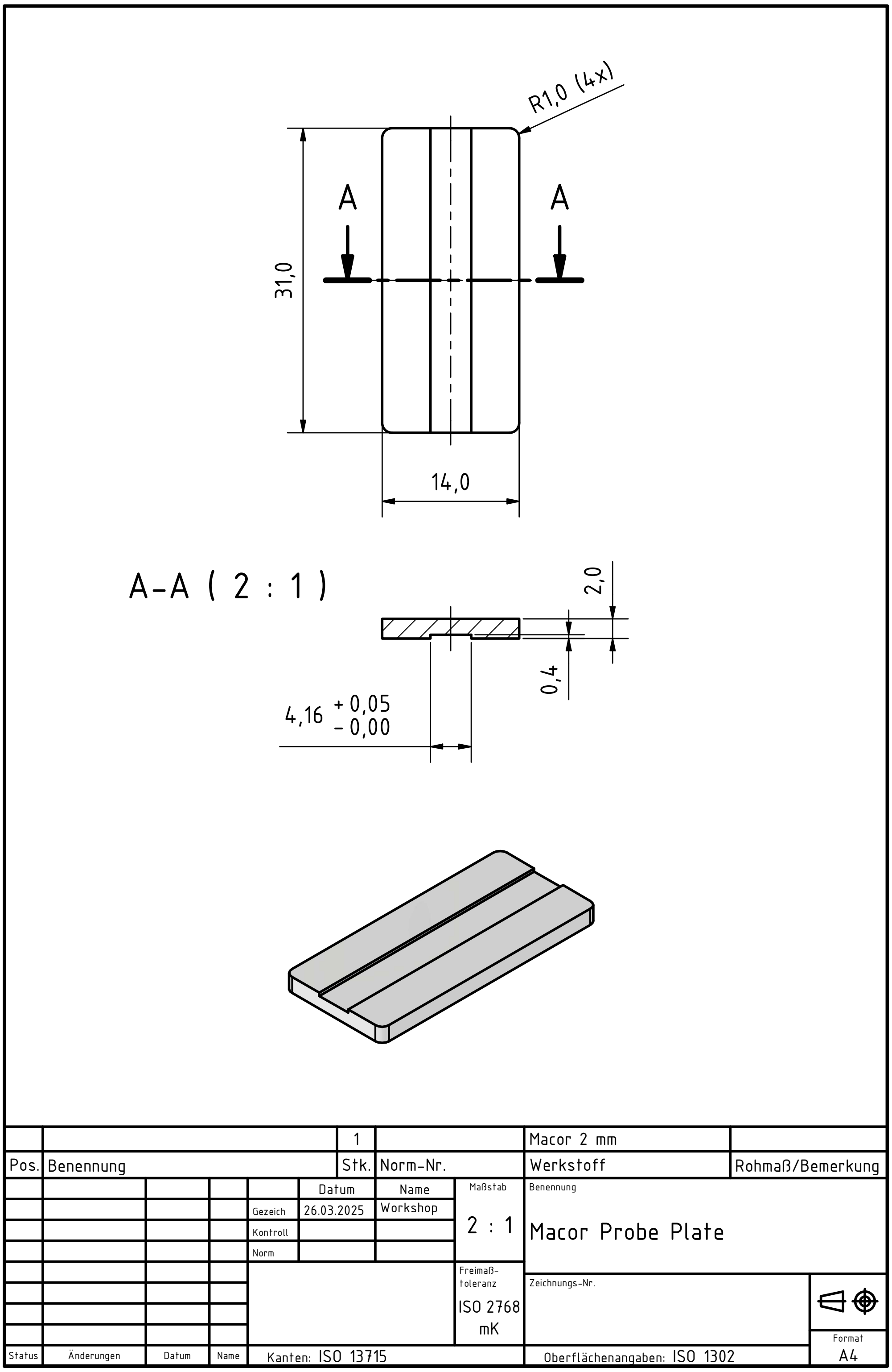


Figure A.7.: MACOR® probe plate component of the sample holder V2. Designed in cooperation with Mechanische Werkstatt (Atominstitut, TU Wien); final drawing by Heinz Matusch [75]

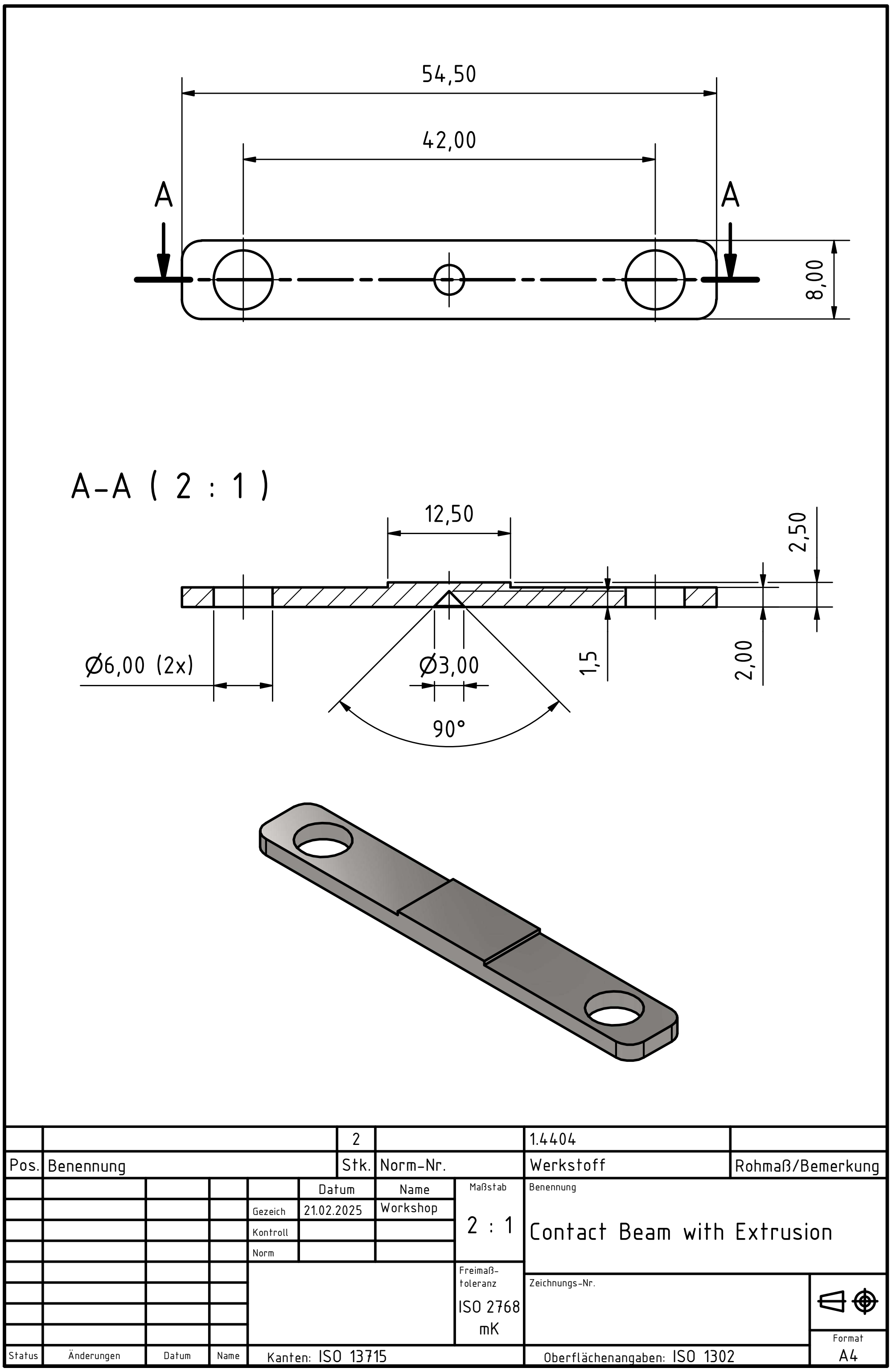


Figure A.8.: Contact beam with extrusion component of the sample holder V2. Designed by Daniel Hackl, final drawing by Heinz Matusch [75]

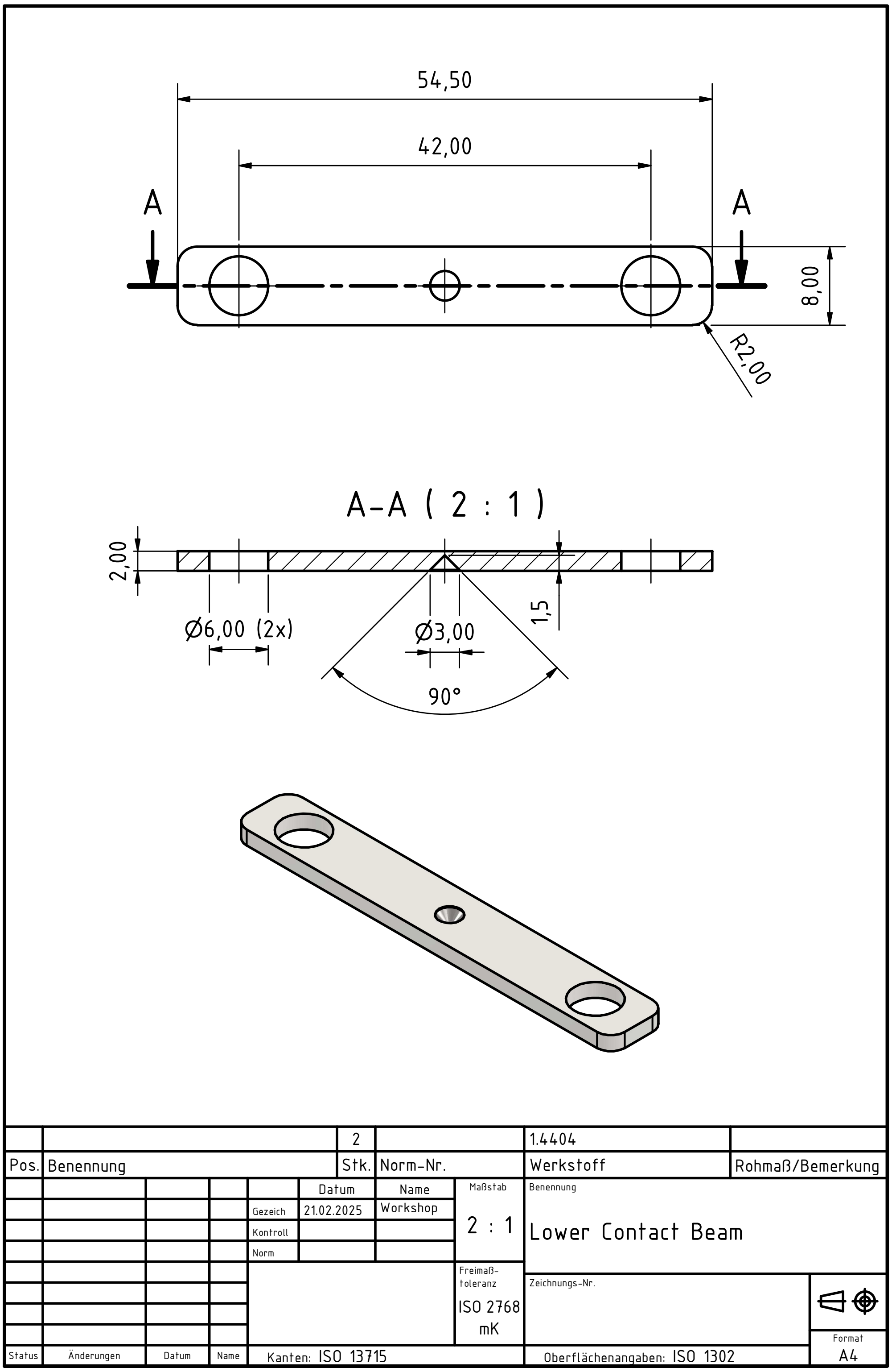


Figure A.9.: Lower contact beam component of the sample holder V2. Designed by Daniel Hackl, final drawing by Heinz Matusch [75]

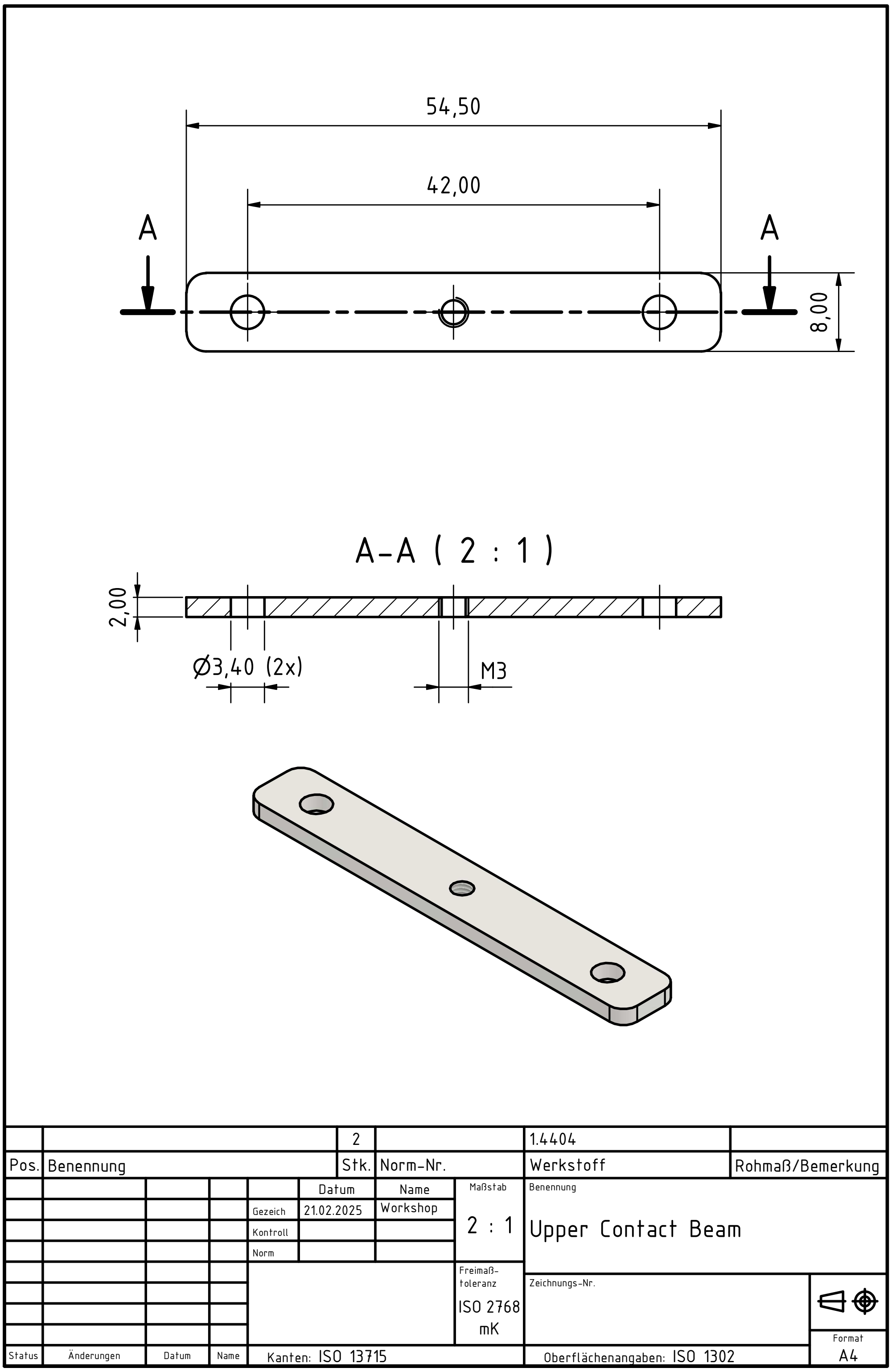


Figure A.10.: Upper contact beam component of the sample holder V2. Designed by Daniel Hackl, final drawing by Heinz Matusch [75]

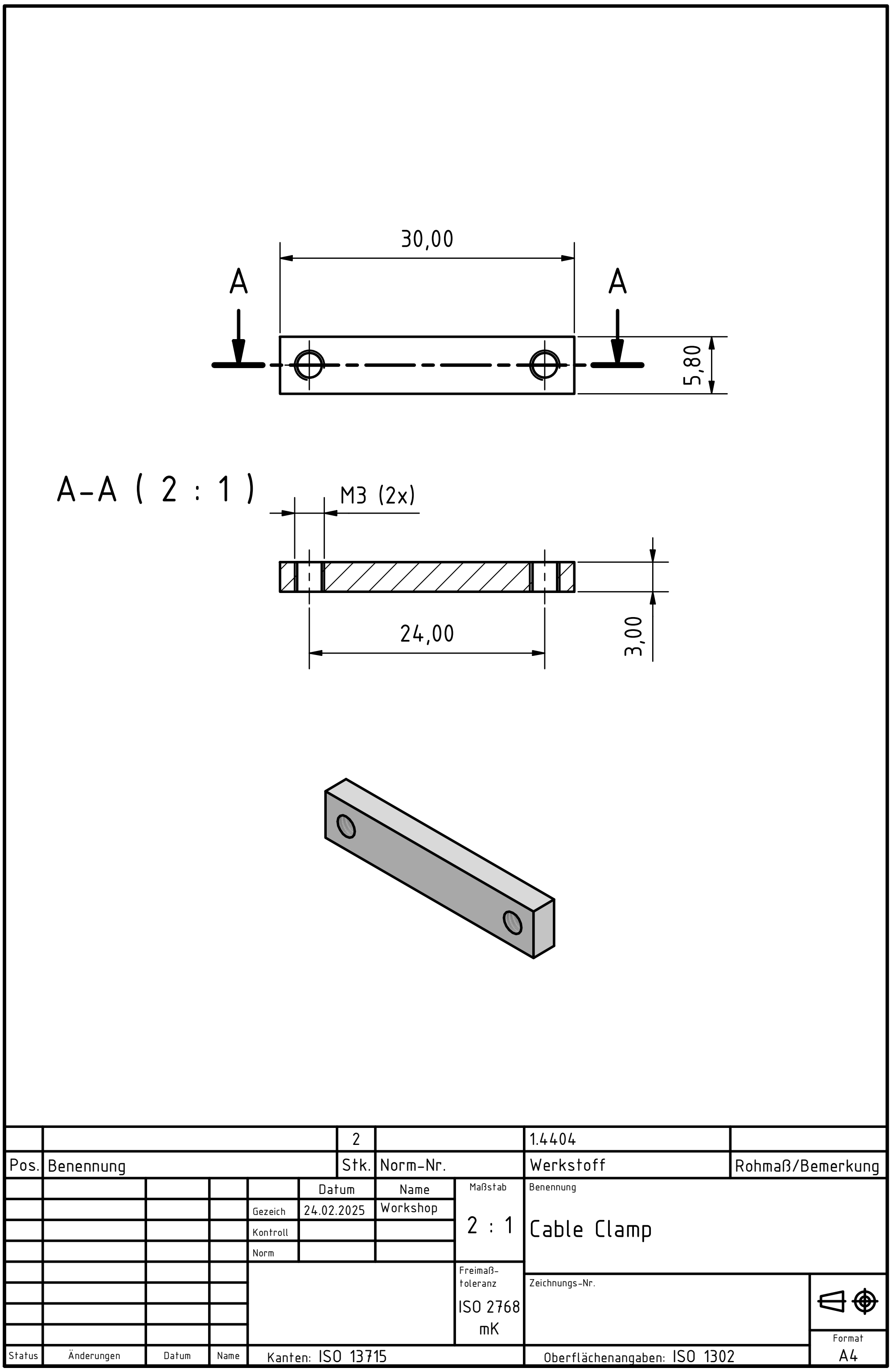
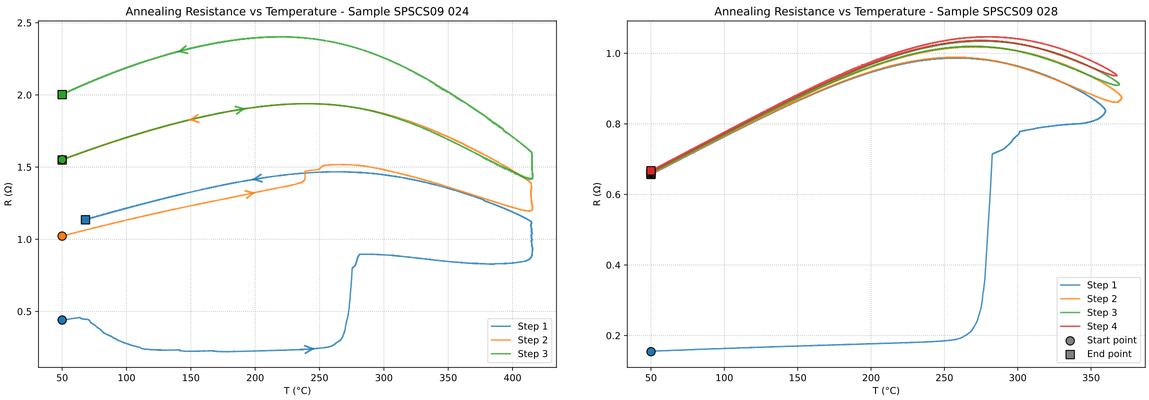


Figure A.11.: Cable clamp component of the sample holder V2. Designed by Daniel Hackl, final drawing by Heinz Matusch [75]

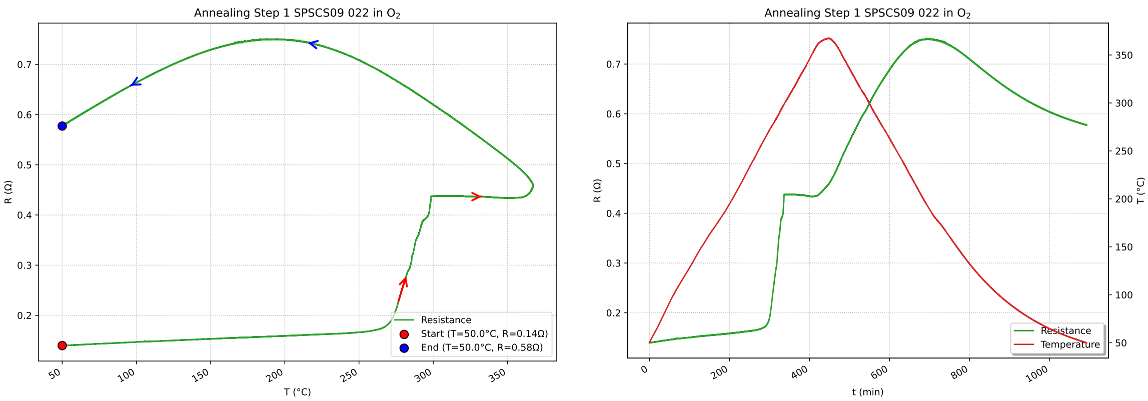
A.2. Resistance Measurements during Annealing



(a) SP SCS09 024: R vs T overlay

(b) SP SCS09 028: R vs T overlay

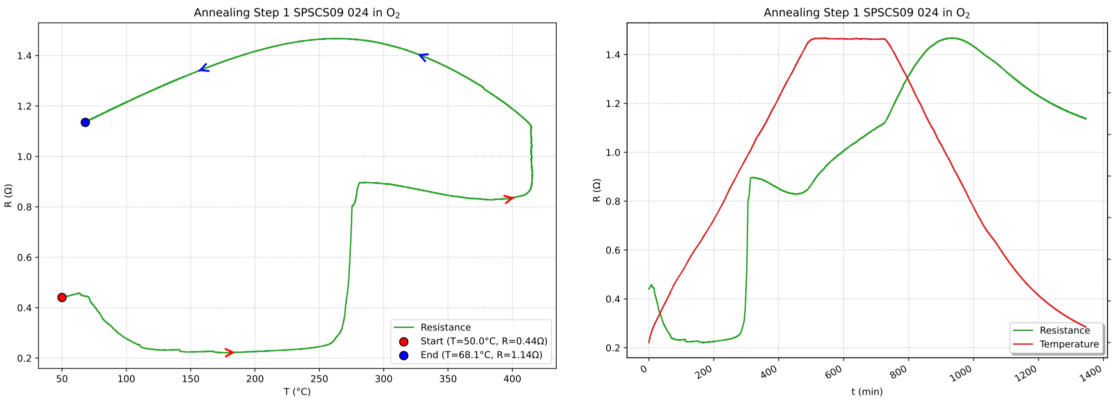
Figure A.12.: Resistance vs temperature overlays for samples SP SCS09 024 and SP SCS09 028



(a) R vs T

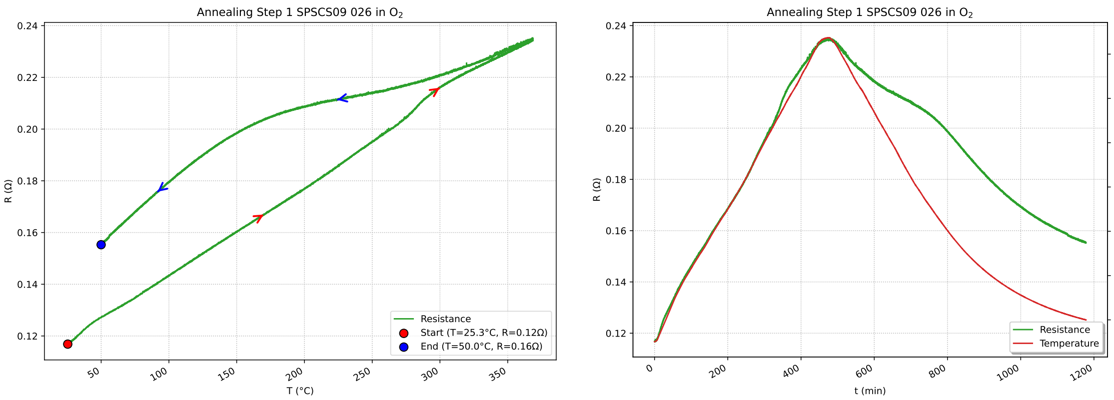
(b) R vs Time

Figure A.13.: SP SCS09 022 (pristine): Resistance vs temperature and time during annealing step 1



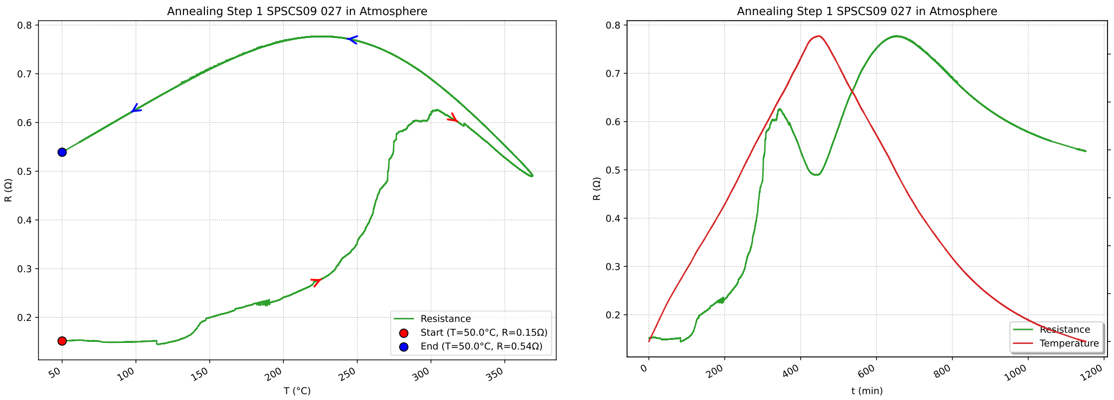
(a) R vs T (b) R vs Time

Figure A.14.: SP SCS09 024 (pristine): Resistance vs temperature and time during annealing step 1



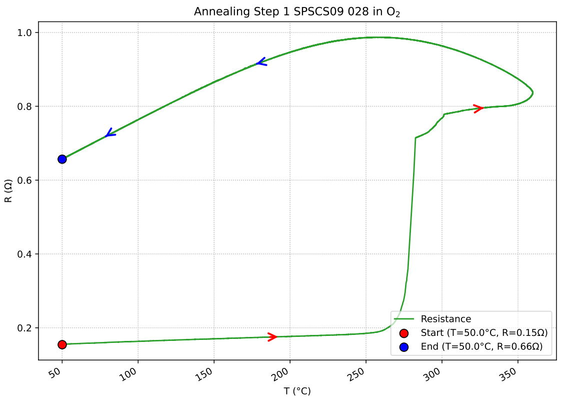
(a) R vs T (b) R vs Time

Figure A.15.: SP SCS09 026 (pristine, silver only): Resistance vs temperature and time during annealing step 1

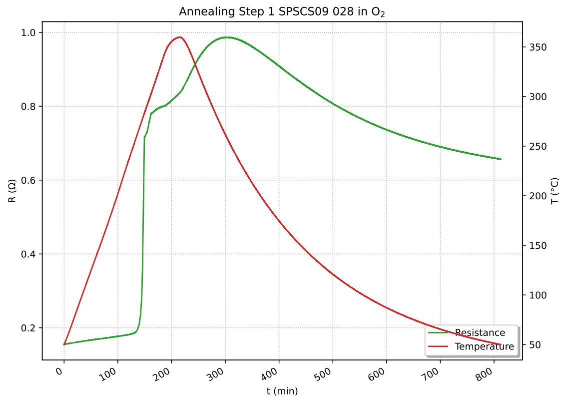


(a) R vs T (b) R vs Time

Figure A.16.: SP SCS09 027 (triple bridge, atmosphere): Resistance vs temperature and time during annealing step 1

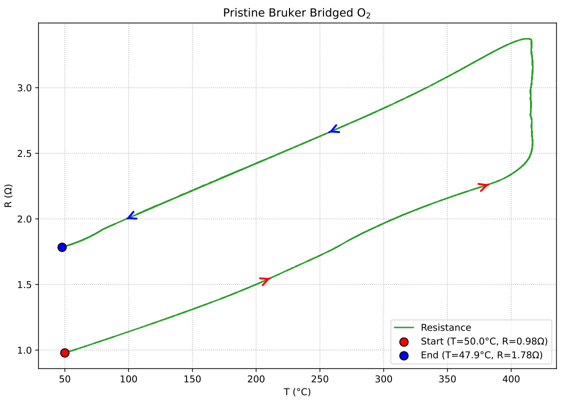


(a) R vs T

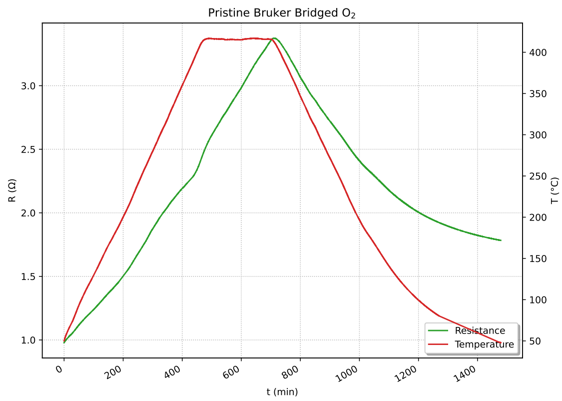


(b) R vs Time

Figure A.17.: SP SCS09 028 (Scheme III): Resistance vs temperature and time during annealing step 1



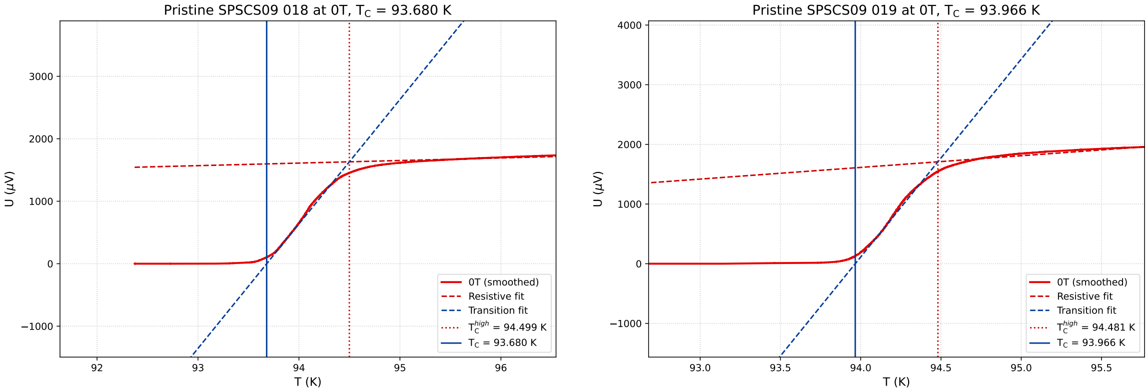
(a) R vs T



(b) R vs Time

Figure A.18.: Reference sample (Bruker, pristine): Resistance vs temperature and time in O_2

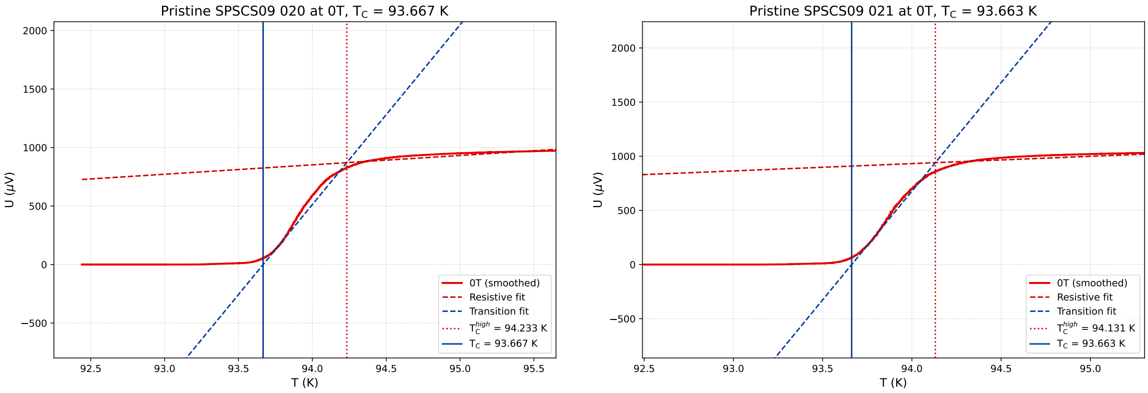
A.3. Critical Temperature Measurements



(a) SPSCS09 018 (Pristine)

(b) SP SCS09 019 (Pristine)

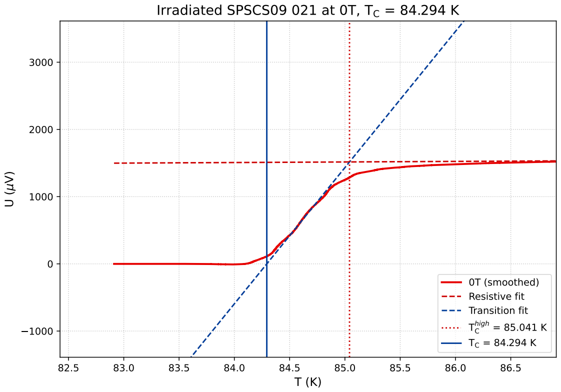
Figure A.19.: Samples 018 – 019 critical temperature measurements



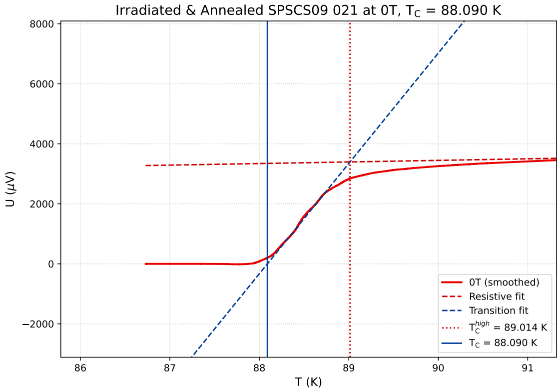
(a) SP SCS09 020 (Pristine)

(b) SP SCS09 021 (Pristine)

Figure A.20.: Samples 020 – 021 pristine critical temperature measurements

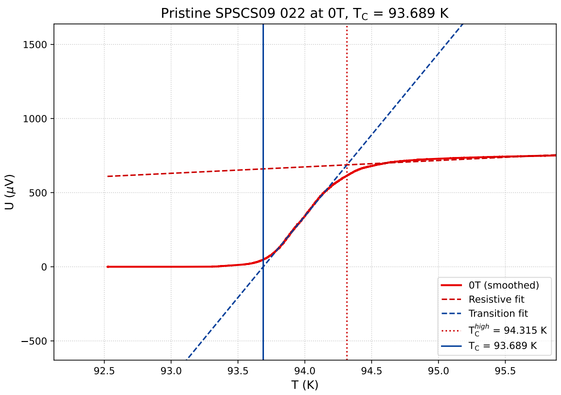


(a) SP SCS09 021 (Irradiated)

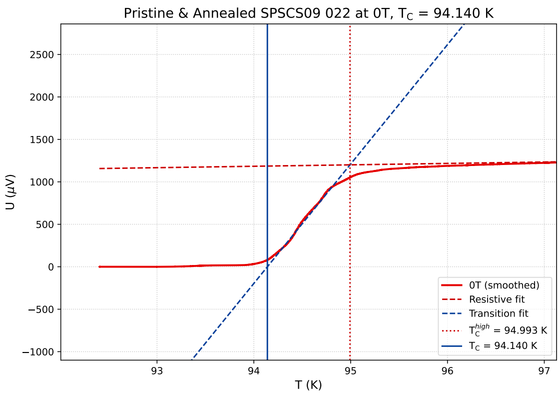


(b) SP SCS09 021 (Irr. & Ann.)

Figure A.21.: Sample 021 under different treatment conditions

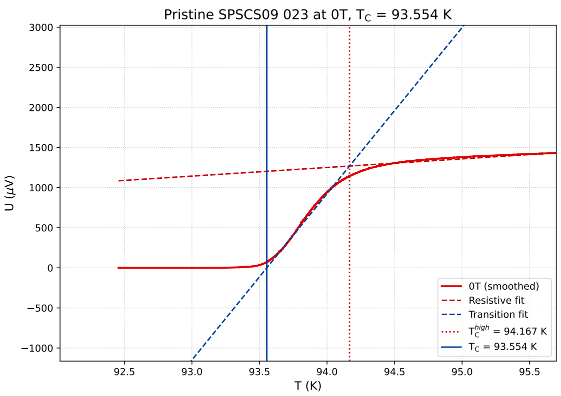


(a) SP SCS09 022 (Pristine)

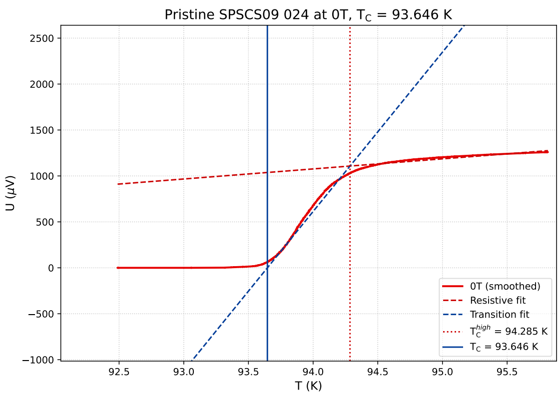


(b) SP SCS09 022 (Prist. & Ann.)

Figure A.22.: Sample 022 under different treatment conditions

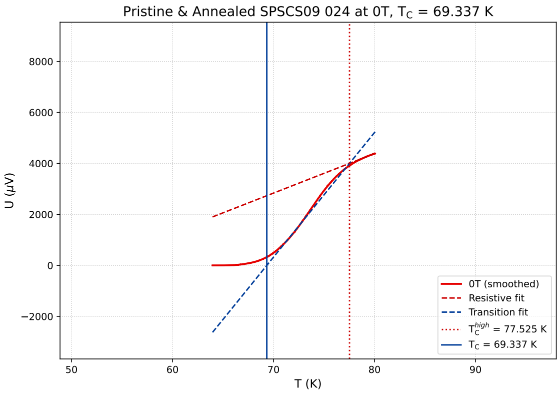


(a) SP SCS09 023 (Pristine)

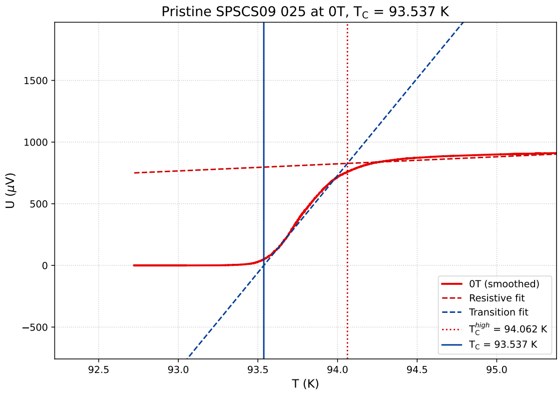


(b) SP SCS09 024 (Pristine)

Figure A.23.: Samples 023 – 024 pristine critical temperature measurements

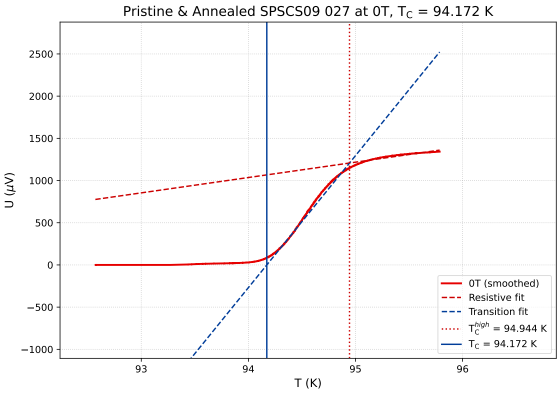


(a) SP SCS09 024 (Prist. & Ann.)

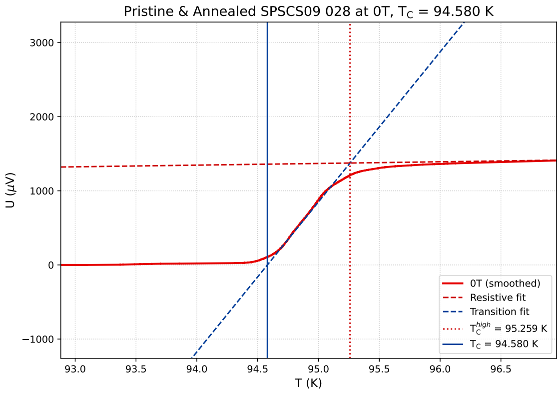


(b) SP SCS09 025 (Pristine)

Figure A.24.: Sample 024 (annealed) and sample 025 critical temperature measurements



(a) SP SCS09 027 (Prist. & Ann.)



(b) SP SCS09 028 (Prist. & Ann.)

Figure A.25.: Samples 027 – 028 critical temperature measurements

Appendix B.

Neutron Flux Determination

B.1. Irradiation Data

Activation Probe	Nuclide	A (Bq)	σ_A(Bq)	γ-geometry	Mass (mg)
Au (unshielded)	^{198}Au	134606.5	3953.88	medium	0.91
Au (shielded)	^{198}Au	31269.38	927.11	medium	0.73
Au-Ni-Monitor	^{58}Co	515.11	17.33	small	2.13
In (unshielded)	^{114m}In	6258.47	220.44	small	1.37
In (unshielded)	^{115m}In	1602.37	53.04	small	1.37
In (shielded)	^{114m}In	4946.39	173.45	small	1.80
In (shielded)	^{115m}In	2478.50	79.66	small	1.80
Ti	^{46}Sc	8.30	0.27	small	2.06
Ti	^{47}Sc	233.91	12.35	small	2.06
Ti	^{48}Sc	50.46	1.36	small	2.06
Ti-In-Ni-Monitor	^{58}Co	644.91	21.93	small	2.33

Table B.1.: Activation parameters for samples irradiated in the CIP position, including specific activities and detector geometry.

Activation Probe	$t_{irr,eff}$ (s)	End of Irradiation	Start of γ-measurement
Au (unshielded)	929.06	2025-02-17 10:43:30	2025-03-04 17:21:13
Au (shielded)	929.06	2025-02-17 10:43:30	2025-03-04 17:36:56
Au-Ni-Monitor	929.06	2025-02-17 10:43:30	2025-03-04 16:51:31
In (unshielded)	923.22	2025-02-17 10:20:00	2025-02-18 17:44:35
In (shielded)	923.22	2025-02-17 10:20:00	2025-02-18 17:09:04
Ti	923.22	2025-02-17 10:20:00	2025-02-18 20:10:42
Ti-In-Ni-Monitor	923.22	2025-02-17 10:20:00	2025-02-18 18:04:49

Table B.2.: Irradiation timing parameters for samples at the CIP position, showing effective irradiation time and measurement schedule.

Activation Probe	Nuclide	A (Bq)	σ_A(Bq)	γ-geometry	Mass (mg)
Au (unshielded)	^{198}Au	195718.7	5611.90	medium	1.94
Au (shielded)	^{198}Au	38083.46	1113.46	medium	1.83
Au-Ni-Monitor (unsh.)	^{58}Co	3.43	0.13	small	2.08
Au-Ni-Monitor (sh.)	^{58}Co	5.61	0.20	small	3.81
In (unshielded)	^{114m}In	3429.79	122.88	small	15.09
In (unshielded)	^{115m}In	590.65	21.29	small	15.09
In (shielded)	^{114m}In	1135.16	42.46	small	14.97
In (shielded)	^{115m}In	575.45	20.30	small	14.97
In-Ni-Monitor (unsh.)	^{58}Co	14.04	0.48	small	4.53
Ti	^{46}Sc	1.42	0.05	small	2.04
Ti	^{47}Sc	36.11	1.91	small	2.04
Ti	^{48}Sc	9.18	0.25	small	2.04
Ti-Ni-Monitor	^{58}Co	59.55	2.57	small	1.37

Table B.3.: Activation parameters for samples irradiated in the Weber tube position, including specific activities and detector geometry.

Activation Probe	$t_{\text{irr,eff}}$ (s)	End of Irradiation	Start of γ-measurement
Au (unshielded)	919.38	2025-02-14 15:43:08	2025-02-18 19:51:28
Au (shielded)	919.49	2025-02-17 15:43:05	2025-02-18 19:13:49
Au-Ni-Monitor (unsh.)	919.38	2025-02-14 15:43:08	2025-02-20 14:58:53
Au-Ni-Monitor (sh.)	919.49	2025-02-17 15:43:05	2025-02-19 17:55:30
In (unshielded)	1824.01	2025-02-11 15:43:10	2025-02-12 15:14:51
In (shielded)	1811.21	2025-02-12 15:43:00	2025-02-13 16:04:25
In-Ni-Monitor (unsh.)	1824.01	2025-02-11 15:43:10	2025-02-12 15:33:09
Ti	24898.22	2025-02-13 15:43:00	2025-02-14 17:08:28
Ti-Ni-Monitor	24898.22	2025-02-13 15:43:00	2025-02-14 16:14:04

Table B.4.: Irradiation timing parameters for samples at the Weber tube position, showing effective irradiation time and measurement schedule.

B.2. Reactor Power Profiles and Schematics

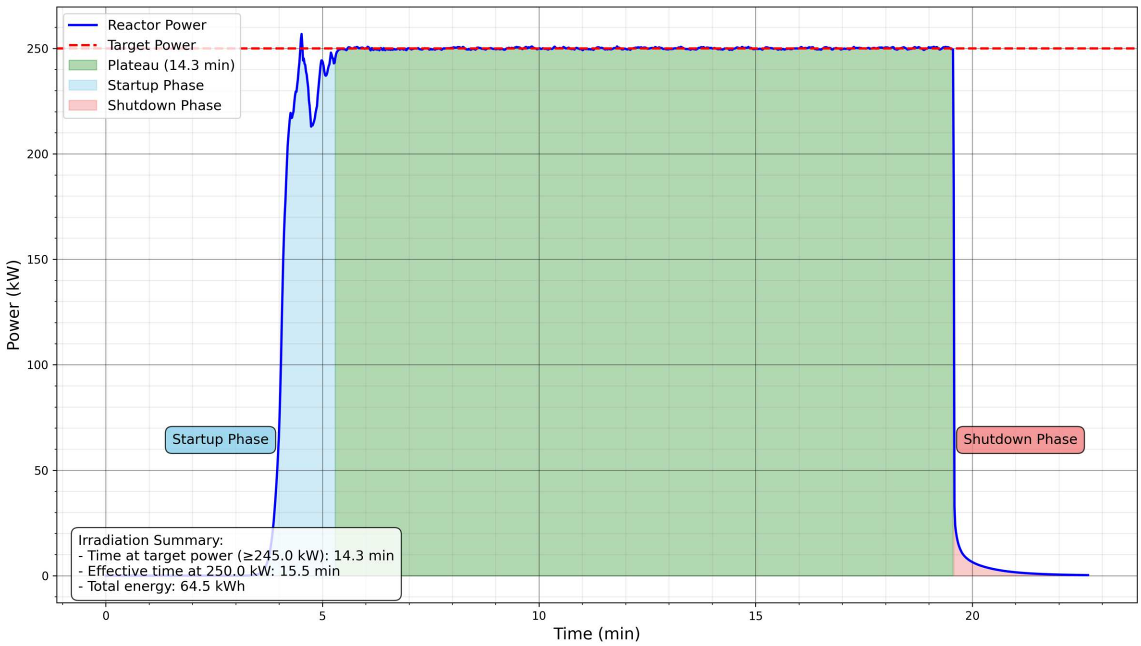


Figure B.1.: Reactor power during gold irradiation in the CIP

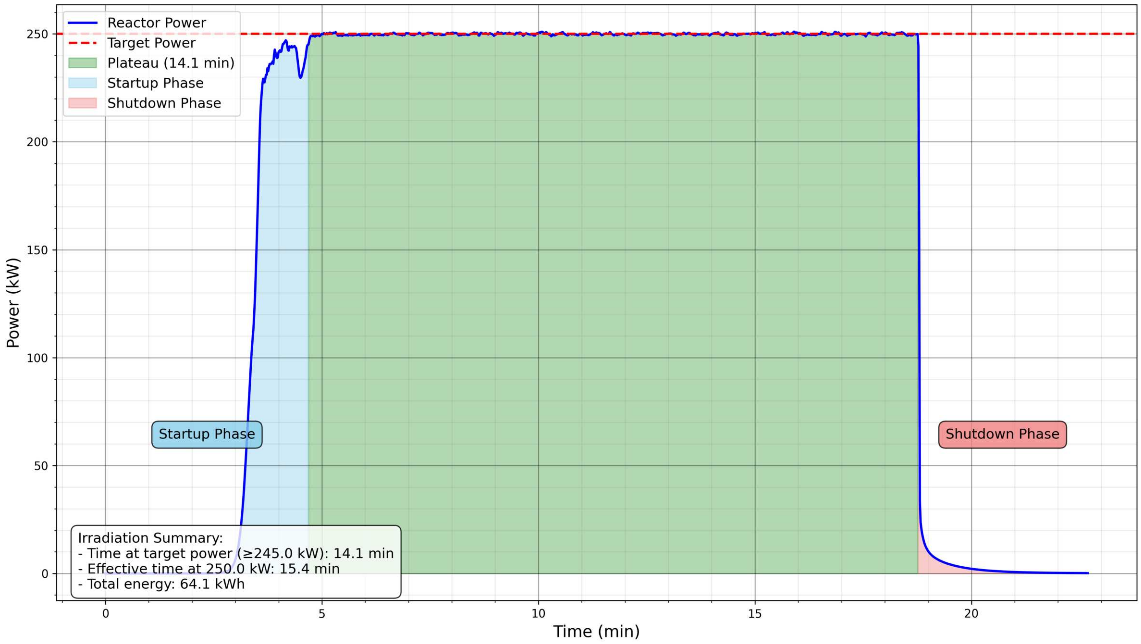


Figure B.2.: Reactor power during titanium-indium irradiation in the CIP

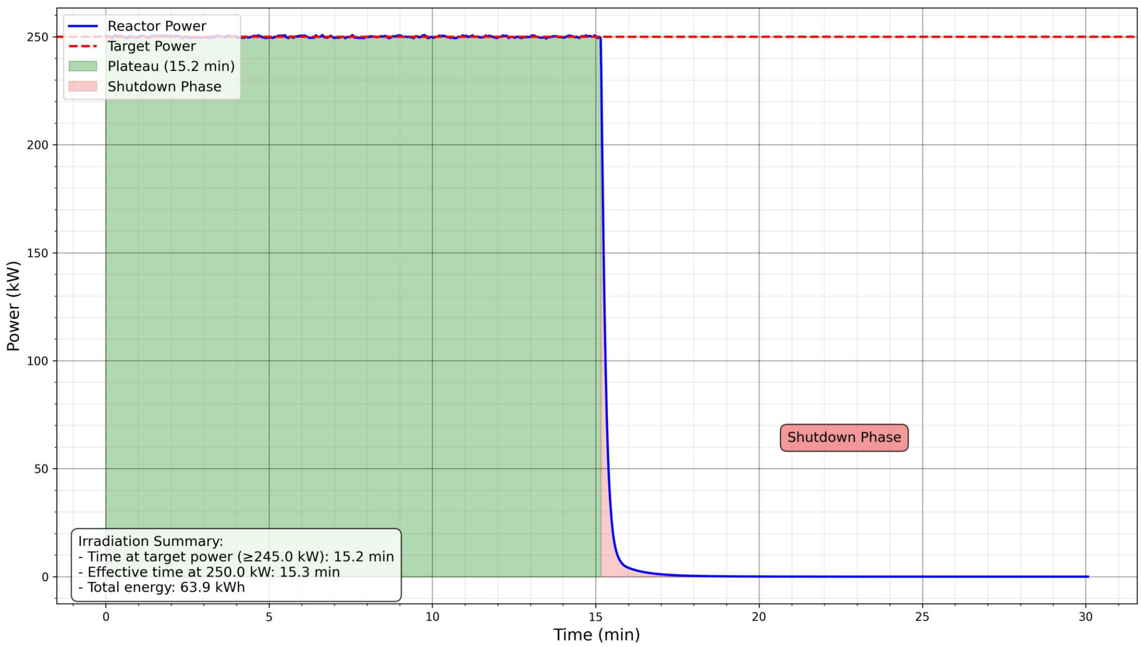


Figure B.3.: Reactor power during cadmium-shielded gold irradiation in the Weber tube

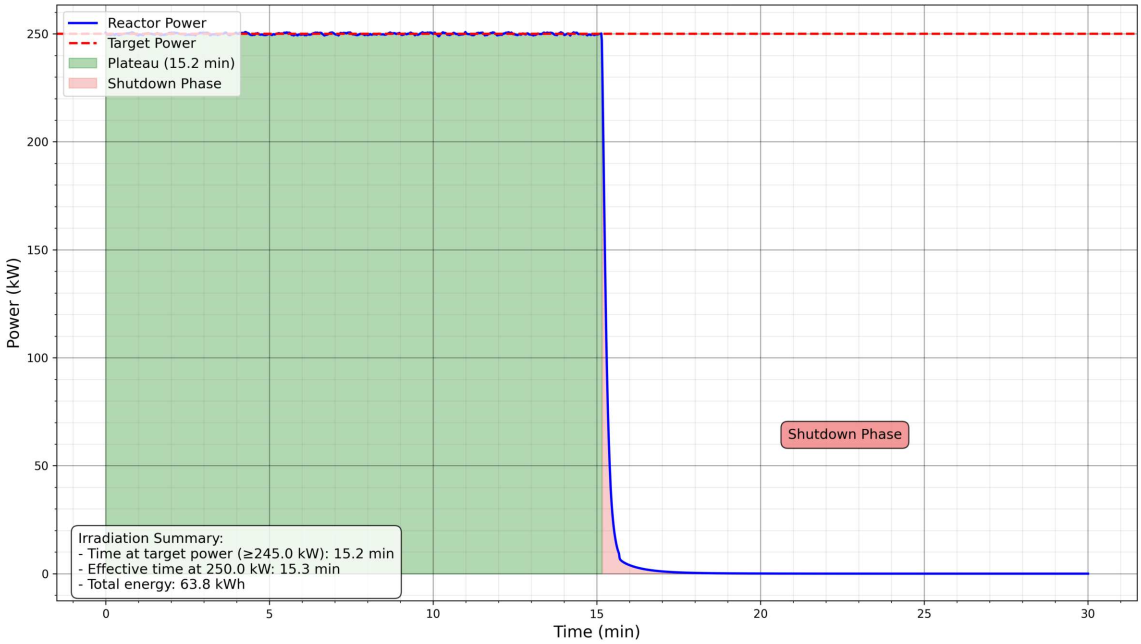


Figure B.4.: Reactor power during unshielded gold irradiation in the Weber tube

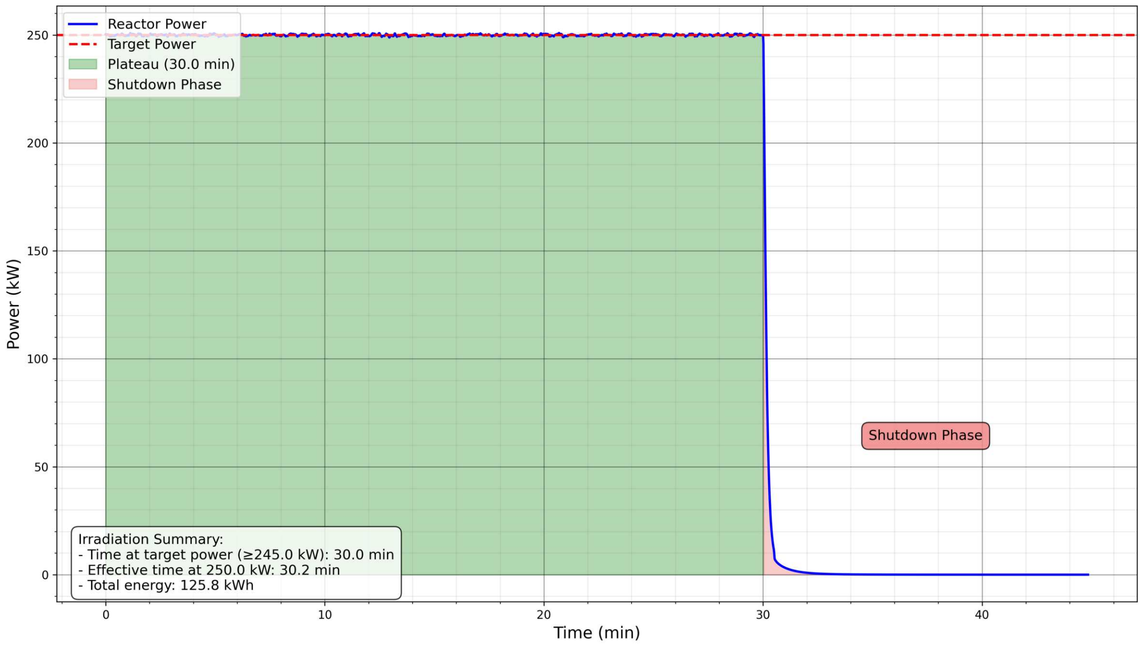


Figure B.5.: Reactor power during cadmium-shielded indium irradiation in the Weber tube

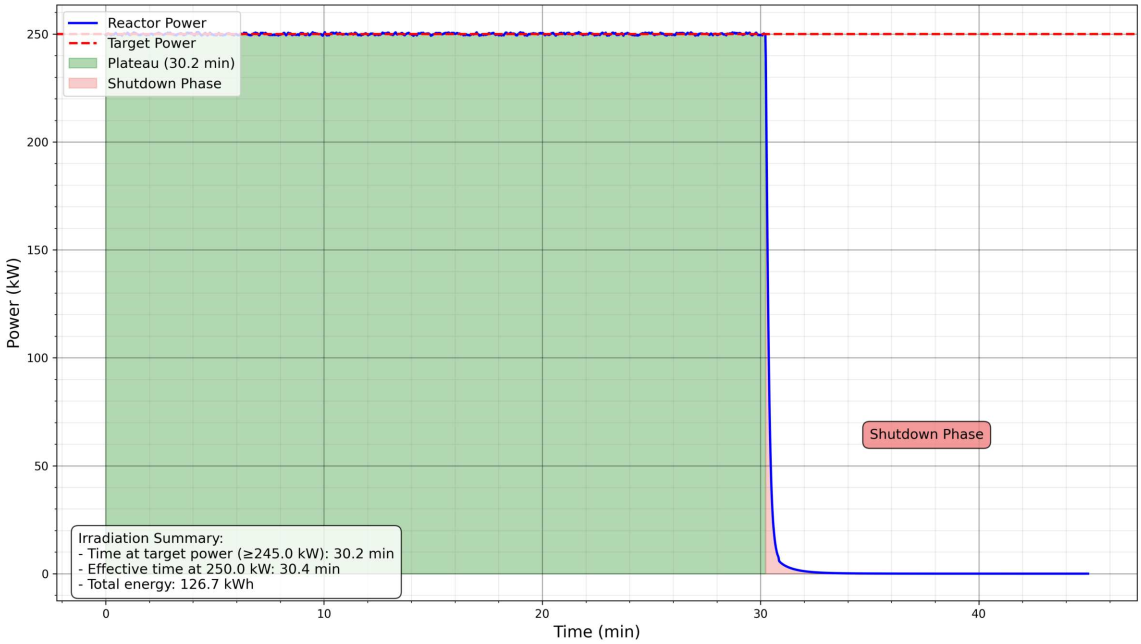


Figure B.6.: Reactor power during unshielded indium irradiation in the Weber tube

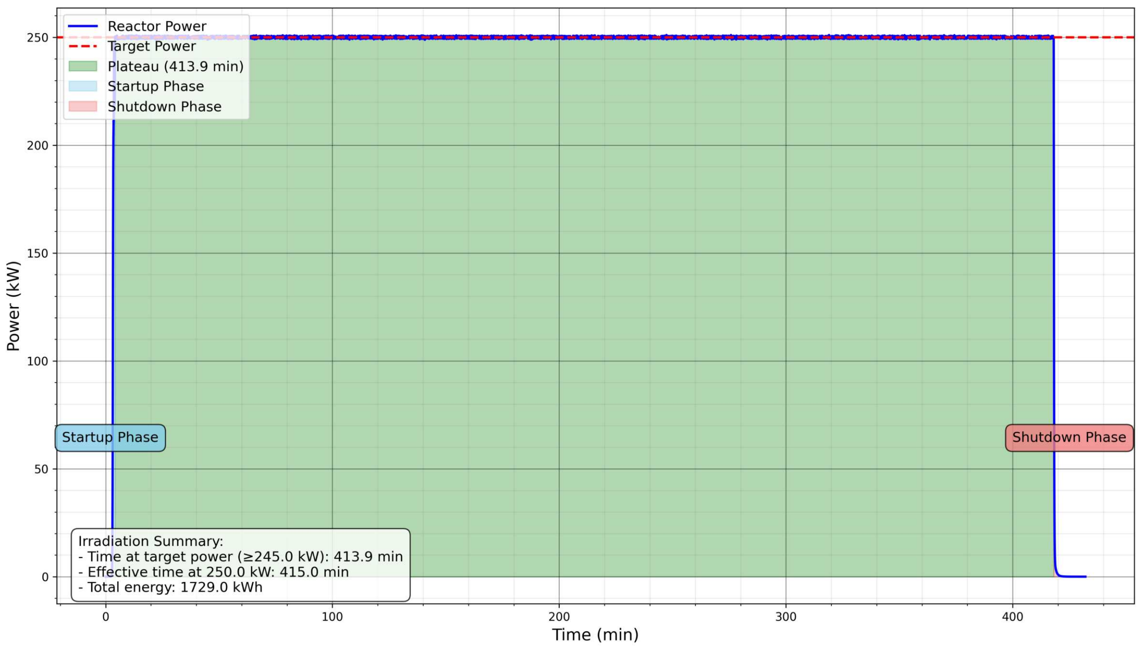


Figure B.7.: Reactor power during titanium irradiation in the Weber tube

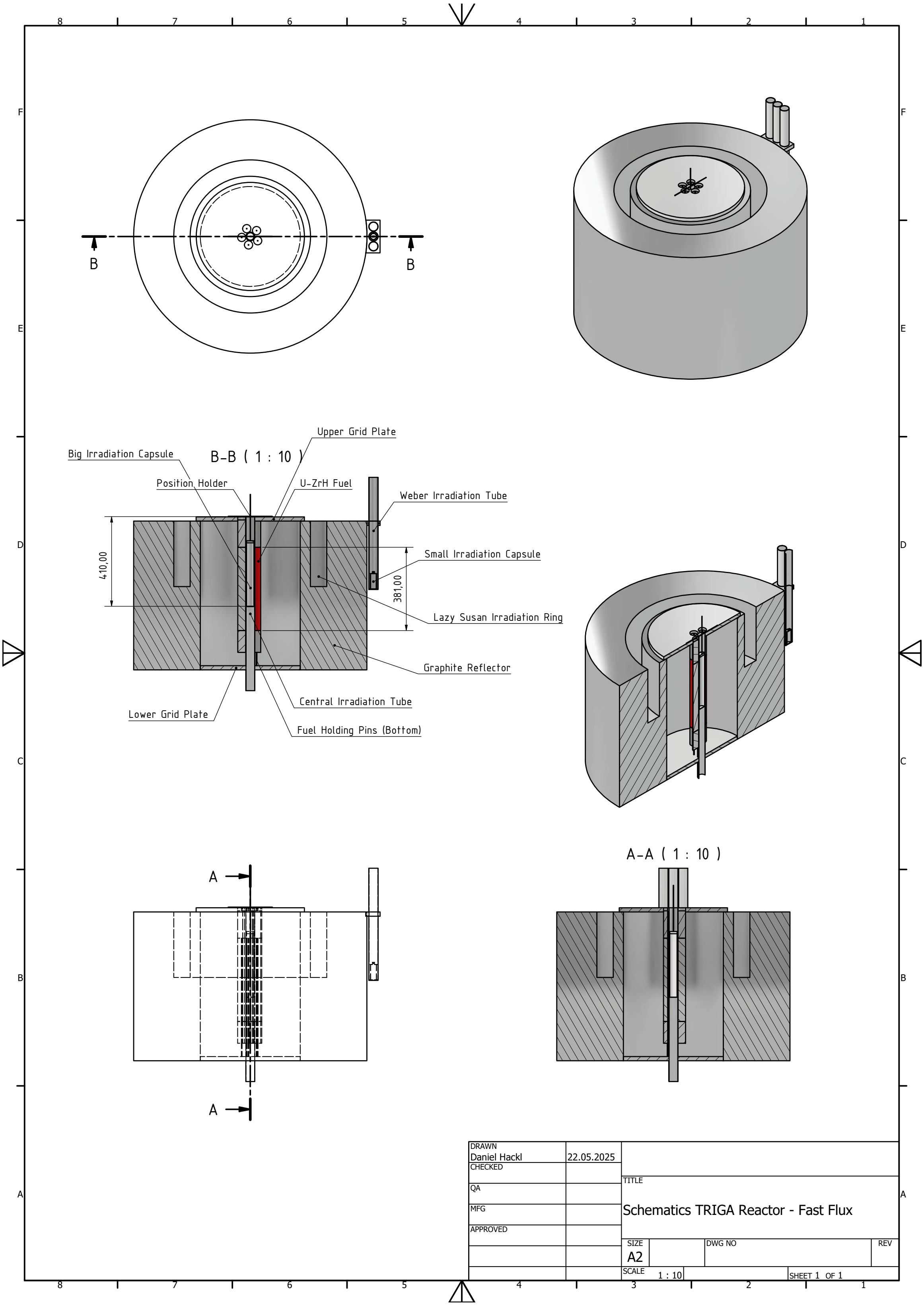


Figure B.8.: TRIGA Reactor schematics for fast flux determination

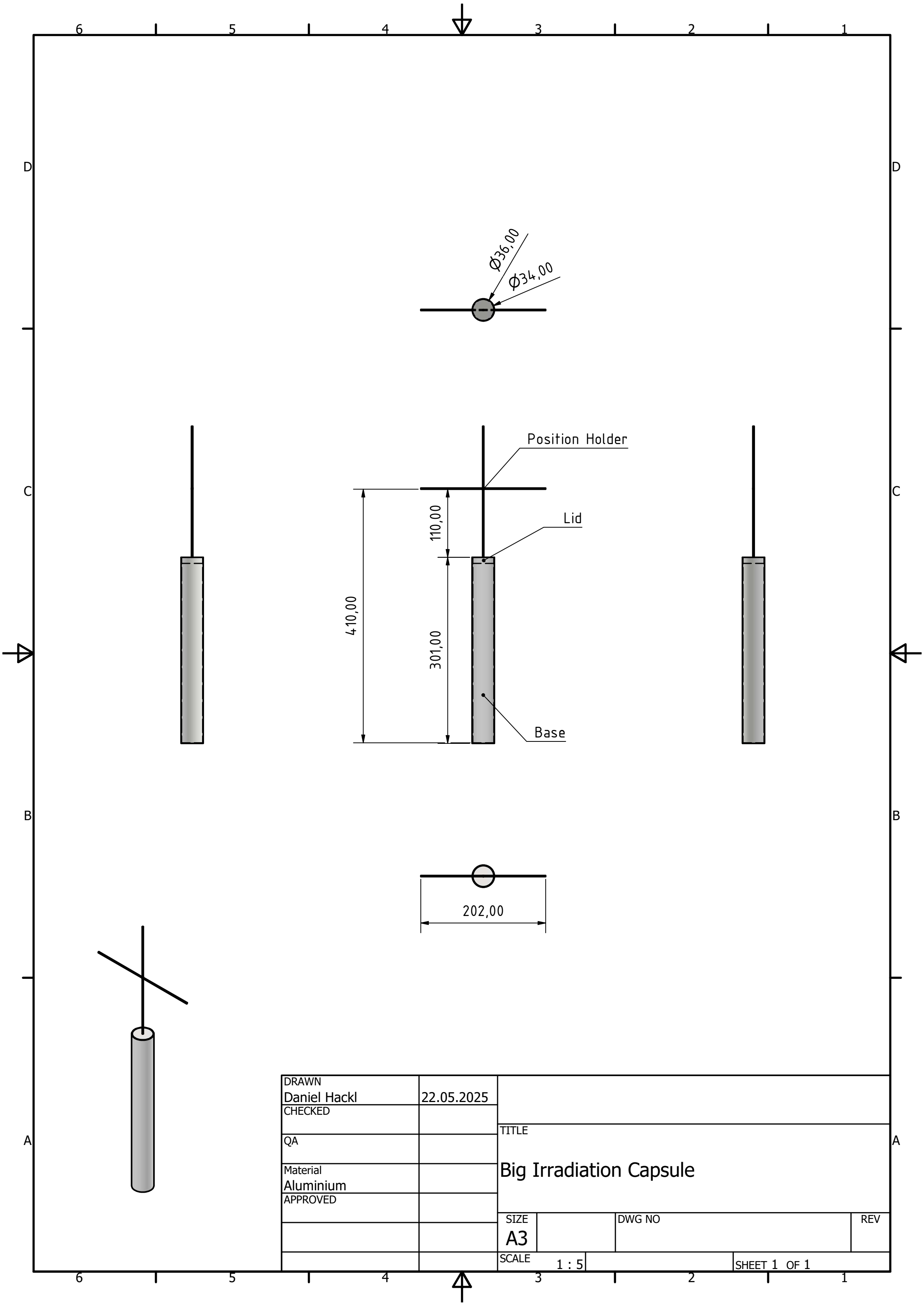


Figure B.9.: Big irradiation capsule schematic

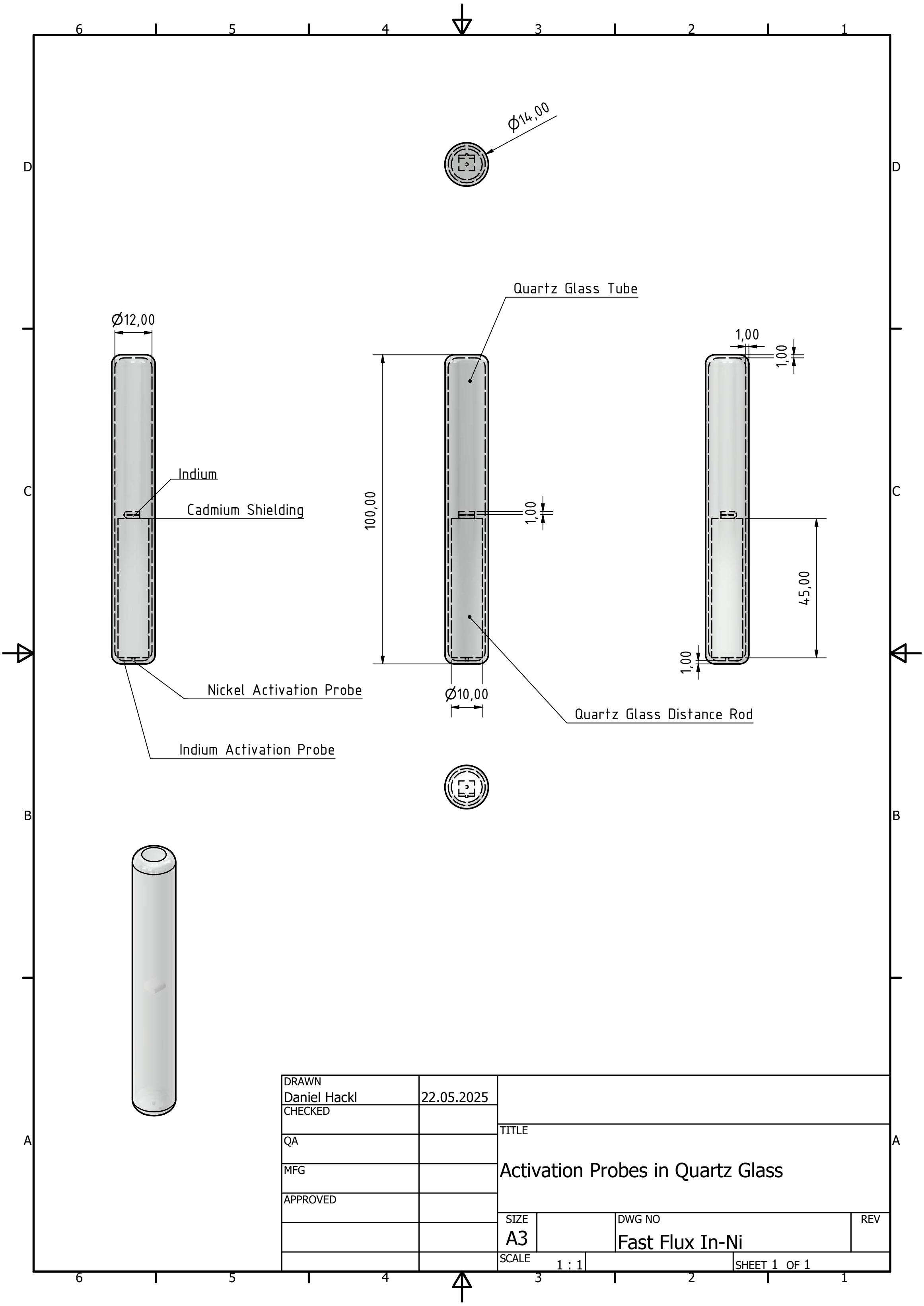


Figure B.10.: Fast flux In-Ni activation probe in quartz glass for the CIP

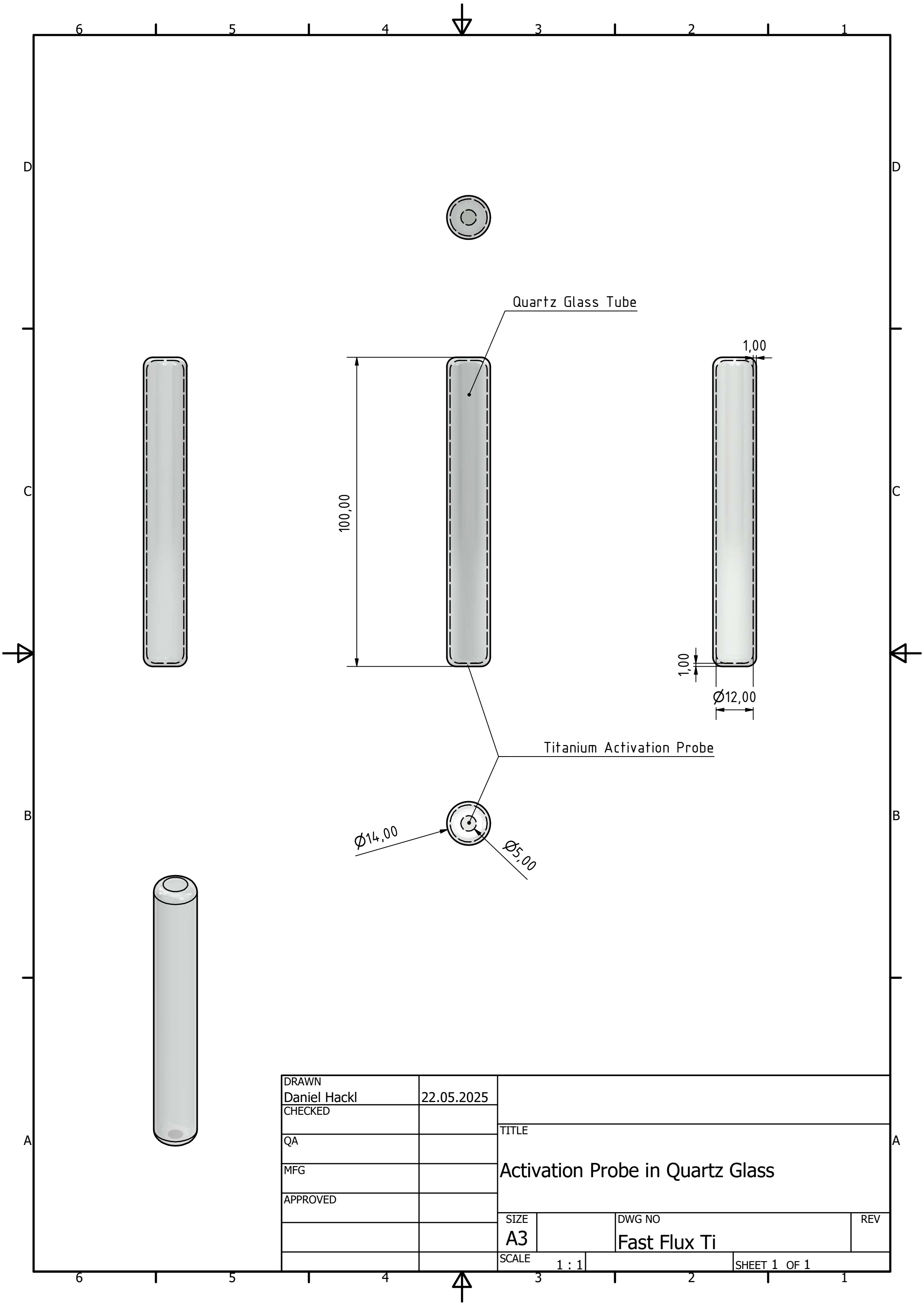


Figure B.11.: Fast flux Ti activation probe in quartz glass for the CIP

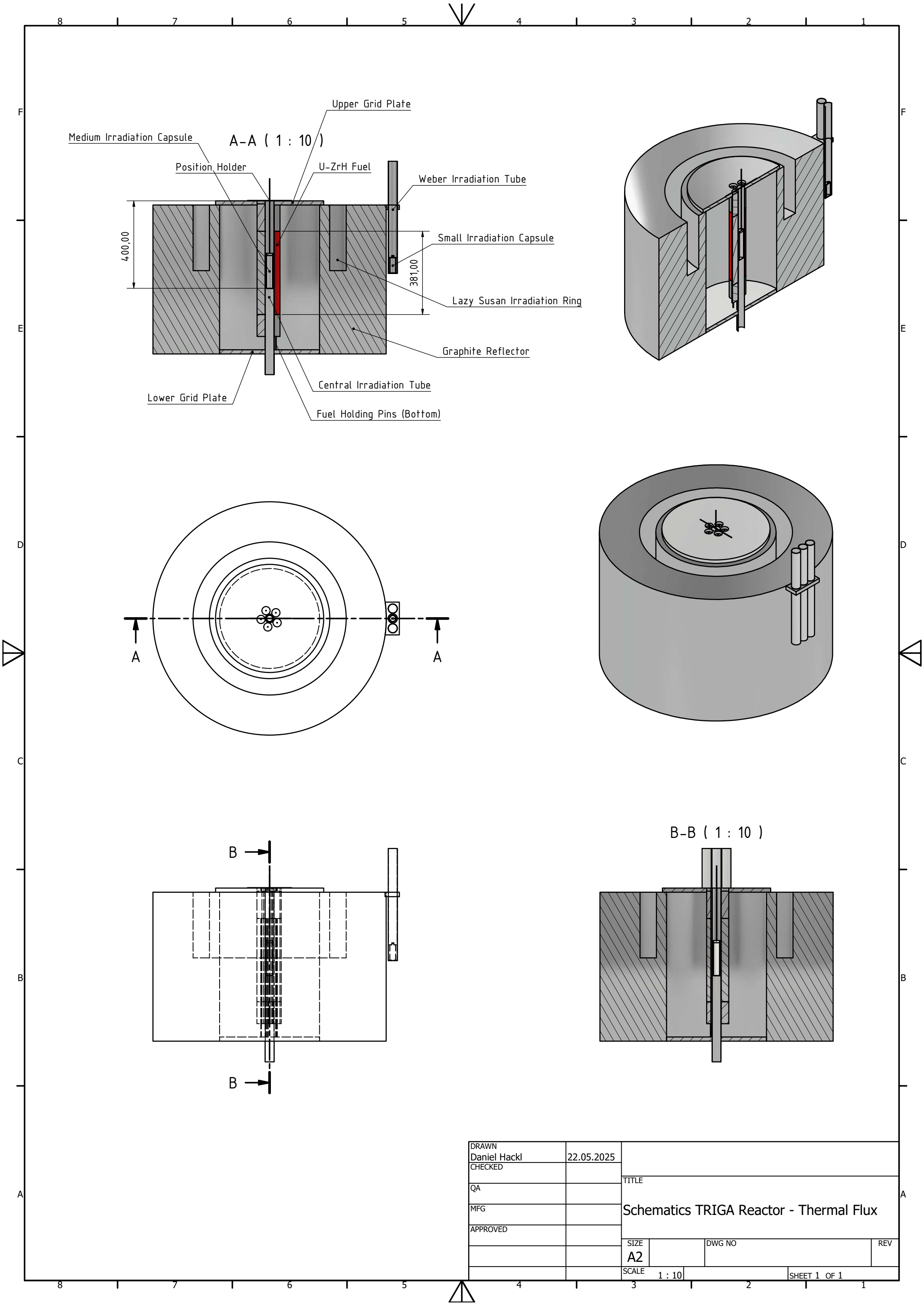


Figure B.12.: TRIGA Reactor schematics for thermal flux determination

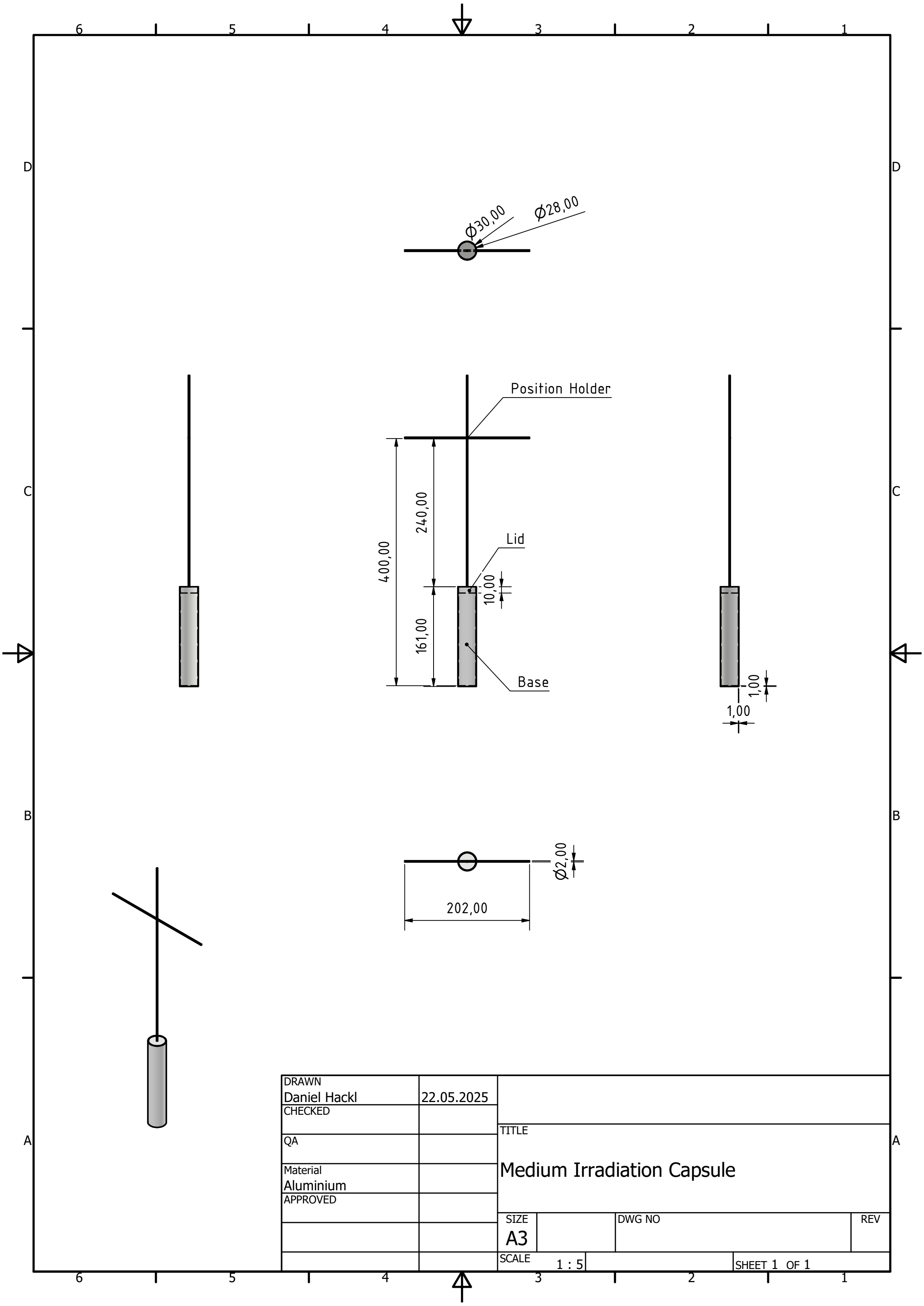


Figure B.13.: Medium irradiation capsule schematic

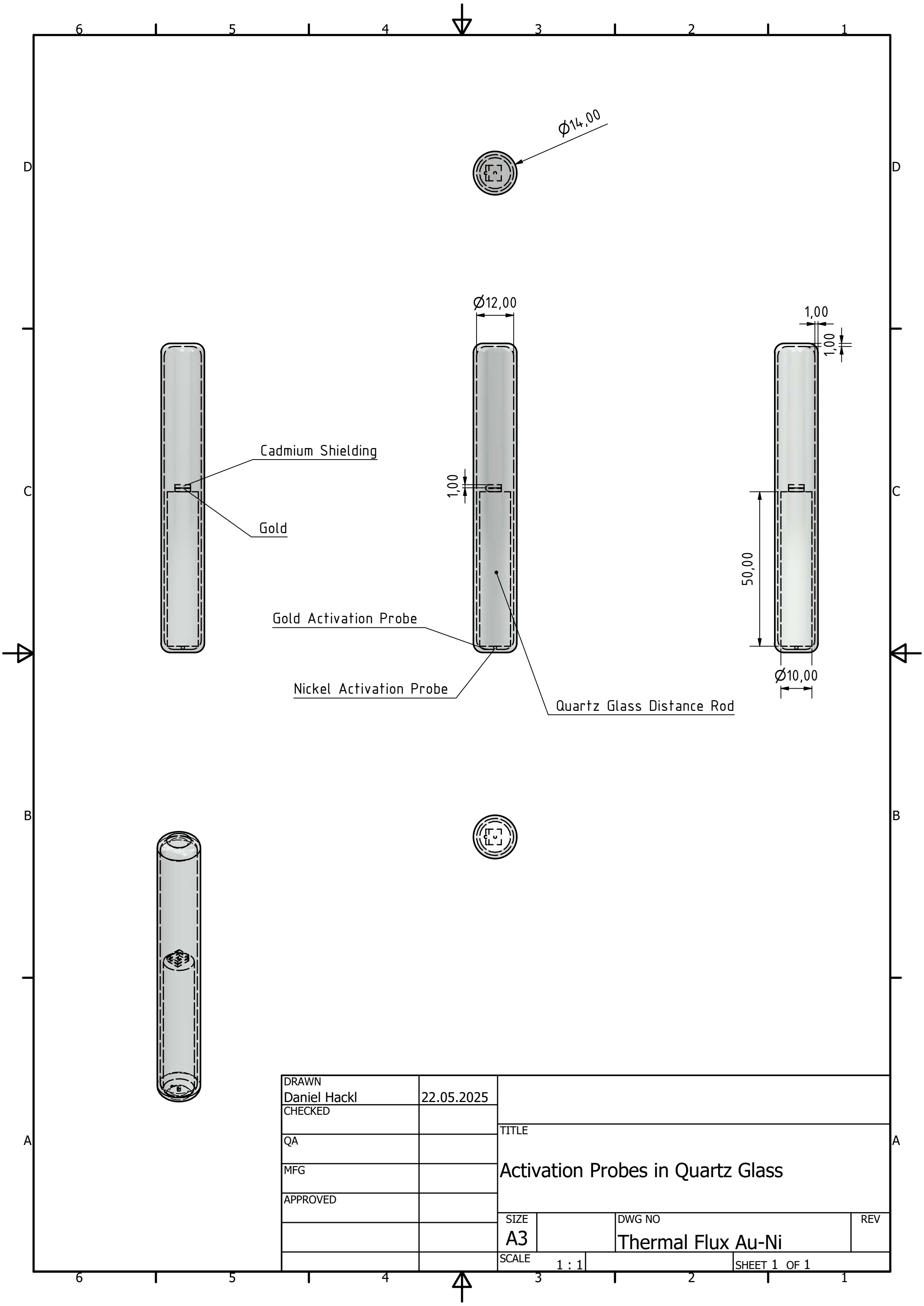


Figure B.14.: Thermal flux Au-Ni activation probe in quartz glass for the CIP

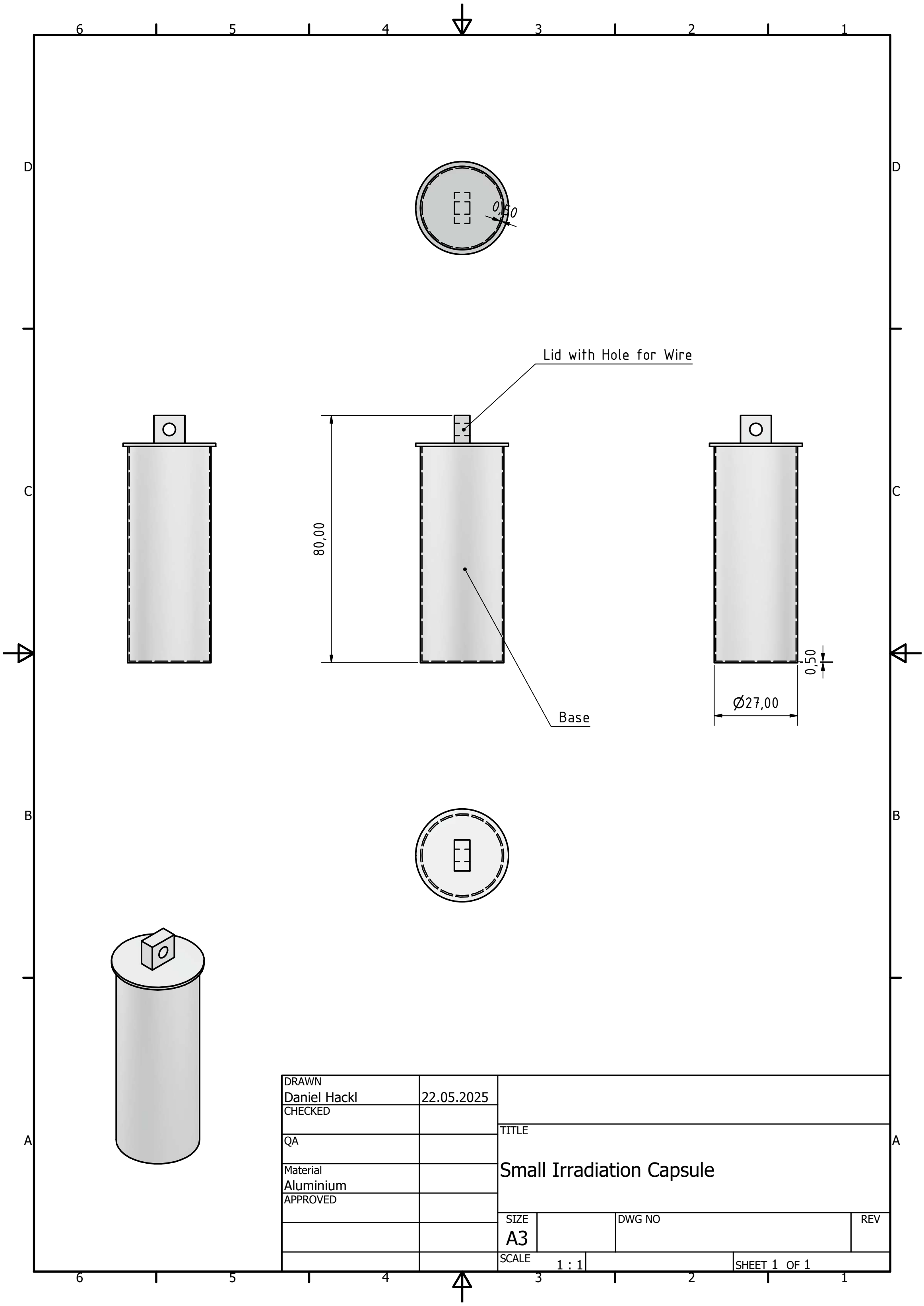


Figure B.15.: Small irradiation capsule schematic

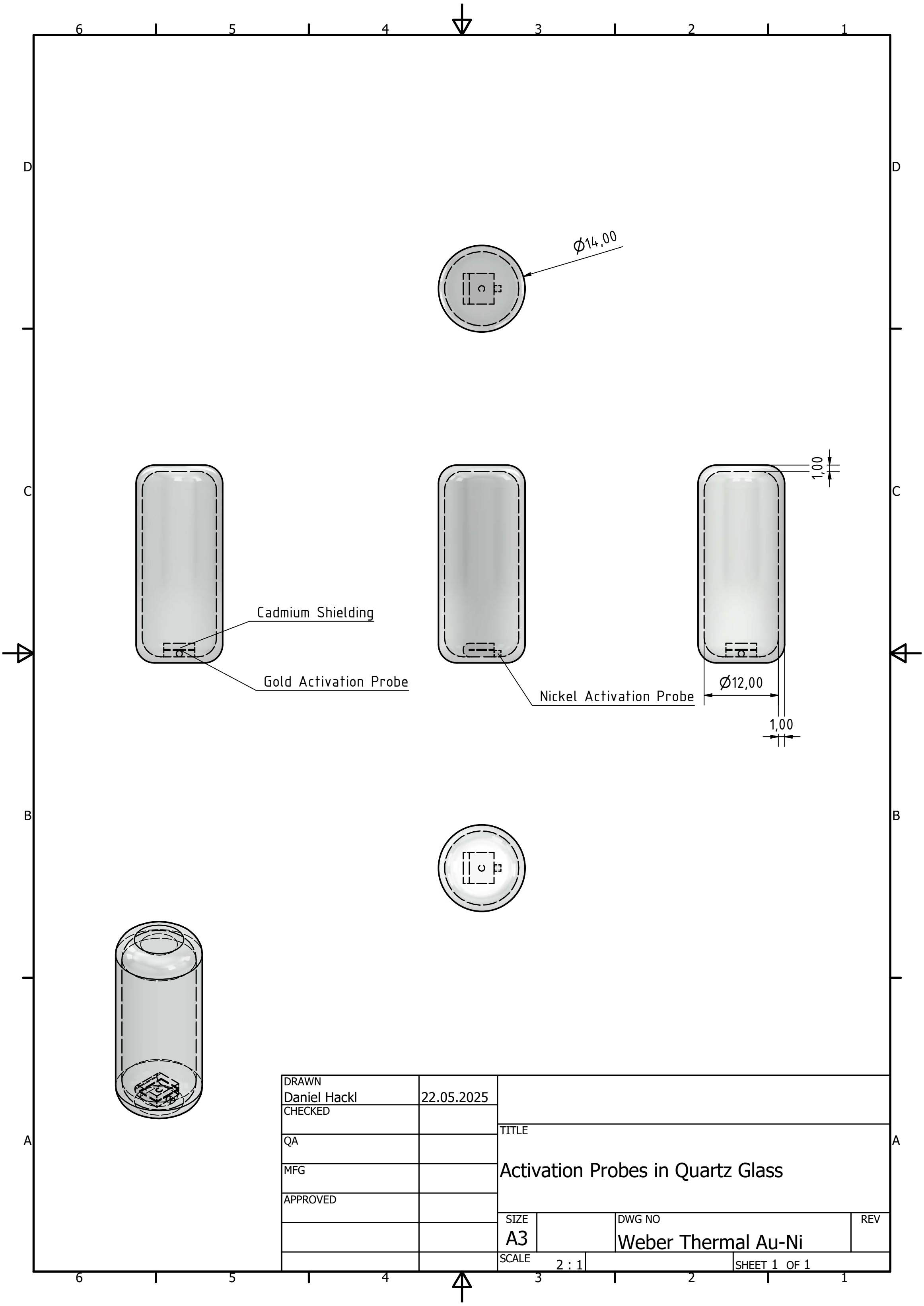


Figure B.16.: Weber thermal Au-Ni activation probe in quartz glass for the Weber Tube

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